

# TOPICS IN GEOMETRIC REPRESENTATION THEORY AND ENUMERATIVE GEOMETRY

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Abstract: Sketch notes for a course in BIMSA/Tsinghua for Spring term 2026.

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PART I

QUANTUM GROUPS AND STABLE  
ENVELOPES

1. Three ways to define the quantum group  $U_{\hbar}(\mathfrak{g})$ 

Imagine you had forgotten everything you knew, even the most basic noncommutative algebra in quantum mechanics

$$A = \mathbf{C}[[\hbar]]\langle x, p \rangle / (xp - px = \hbar).$$

How would you reconstruct it?

First, the generators and relations definition as above. Second, we could start with

$$A_0 = A/\hbar = \mathbf{C}[x, p] \simeq \mathcal{O}(T^*\mathbf{C})$$

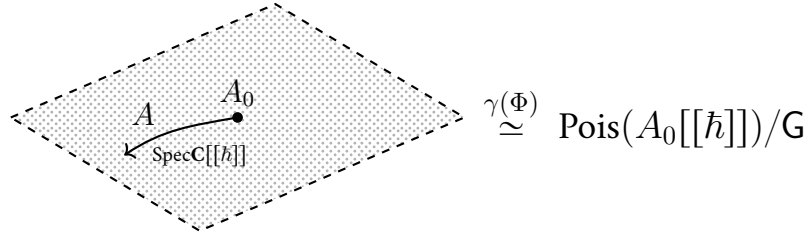
where a shadow of the product on  $A$  is remembered through the symplectic structure  $\{x, p\} = 1$  on  $T^*\mathbf{C}$ , and try to “exponentiate” this to a product on  $A$  itself, with

$$x \cdot p = xp + a_1\hbar + a_2\hbar^2 + \dots$$

such that  $[ , ] = \hbar\{ , \} + b_2\hbar^2 + \dots$ .

Kontsevich and Tamarkin worked out a more canonical way to view this:

**Theorem.** (Kontsevich–Tamarkin, [Ko03; Ta03]) *For any commutative algebra  $A_0$ , the space of its  $\mathbf{C}[[\hbar]]$ -deformations as an associative algebra is isomorphic to the space of Poisson brackets on  $A_0$  modulo the action of  $\mathbf{G}_{\{ , \}} = \text{Aut}_{\{ , \}}(A_0[[\hbar]])$ :*



Moreover, the choice of isomorphism  $\gamma(\Phi)$  is given by a choice of Drinfeld associator  $\Phi \in \text{GRT}_1 \subseteq U(\mathfrak{f}_2)^{\otimes 3}$ .

Third, Kontsevich gives an explicit formula

$$x \cdot p = \sum_{\Gamma} m_{\Gamma}(x, p)$$

as a sum over Feynman diagrams  $\Gamma$ .

1.0.1. *Motivation from physics.* Let  $\mathcal{T}_0$  be a quantum field theory. What is the space of deformations of  $\mathcal{T}$ ? The above is for one-dimensional TQFTs with algebra of operators  $\mathcal{T}_0(S^0) = A_0$ . For quantum groups we are going to be interested in three-dimensional TQFTs, where we study the deformation of the *braided monoidal category*  $\mathcal{T}_0(S^1) = U(\mathfrak{g})\text{-Mod}$ , which is braided monoidal for formal reasons.

| Deform $A_0$ as an algebra                                                 | Deform $U(\mathfrak{g})$ -Mod as a braided monoidal category             | Deforming $\mathcal{T}$ |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------|
| Poisson form $\{ , \}$                                                     | Poisson bracket $r$                                                      |                         |
| $g \in \mathbf{G} = \text{Aut}_{\{ , \}}(A_0[[\hbar]])$                    | Drinfeld twists $F \in U(\mathfrak{g}) \otimes U(\mathfrak{g})[[\hbar]]$ |                         |
| Associator $\Phi \in \text{GRT}_1 \subseteq U(\mathfrak{f}_2)^{\otimes 3}$ | ?                                                                        |                         |
| Kontsevich formality                                                       | Drinfeld's Theorem                                                       |                         |

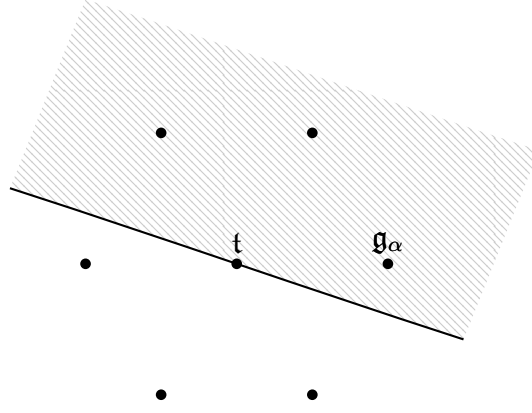
### 1.1. Semisimple Lie algebras.

1.1.1. Let  $\mathfrak{g}$  be a finite dimensional semisimple Lie algebra over  $\mathbf{C}$ , and  $\mathfrak{t} \subseteq \mathfrak{g}$  a maximal Lie subalgebra with trivial bracket.<sup>1</sup> It follows that  $\mathfrak{g}$  splits up into  $\mathfrak{t}$ -eigenspaces

$$\mathfrak{g} = \bigoplus_{\alpha \in \Lambda} \mathfrak{g}_{\alpha}$$

where the *root lattice*  $\Lambda \subseteq \mathfrak{t}^{\vee}$  is generated by the  $\alpha$ 's with  $\mathfrak{g}_{\alpha} \neq 0$  (the *roots*). By maximality  $\mathfrak{g}_0 = \mathfrak{t}$ , and by the Jacobi identity

$$[\mathfrak{g}_{\alpha}, \mathfrak{g}_{\beta}] \subseteq \mathfrak{g}_{\alpha+\beta}.$$



Choose a *stability function*  $z : \Lambda \rightarrow \mathbf{C}$ , so pulling back the upper half plane splits up  $\mathfrak{g} = \mathfrak{n}_{-} \oplus \mathfrak{t} \oplus \mathfrak{n}_{+}$ , giving a notion of *positive* and *negative* roots. A positive root is *simple* if it is not the sum of two positive roots.

Let  $\alpha_1, \dots, \alpha_r$  be the simple positive roots.

**Theorem 1.1.2.** (Killing) *A finite-dimensional complex Lie algebra  $\mathfrak{g}$  is semisimple if and only if the Killing form*

$$\kappa : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbf{C} \quad (x, y) \mapsto \mathrm{tr}_{\mathfrak{g}}([x, [y, -]])$$

*is nondegenerate. In that case,  $\kappa$  is the unique (up to scalar) symmetric bilinear form invariant under the action of  $\mathfrak{g}$ , i.e.  $\mathrm{Sym}^2(\mathfrak{g}^{\vee})^{\mathfrak{g}} \simeq \mathbf{C} \cdot \kappa$ .*

As applying  $[\mathfrak{g}_{\alpha}, -]$  enough times is the zero operation unless  $\alpha = 0$ , this implies for semisimple Lie algebras that the only nonzero parts of the Killing form are

$$\kappa : \mathfrak{g}_{\alpha} \times \mathfrak{g}_{-\alpha} \rightarrow \mathbf{C}.$$

In particular, it restricts to a nondegenerate pairing on  $\mathfrak{t}$ .

<sup>1</sup>For general  $\mathfrak{g}$  the definition of maximal torus is different.

**Theorem 1.1.3.** (Chevalley) *Apart from  $\mathfrak{g}_0 = \mathfrak{t}$ , each nonzero  $\mathfrak{g}_\alpha = \mathbf{C} \cdot e_\alpha$  is one-dimensional, and  $\mathfrak{g}$  has generators and relations definition*

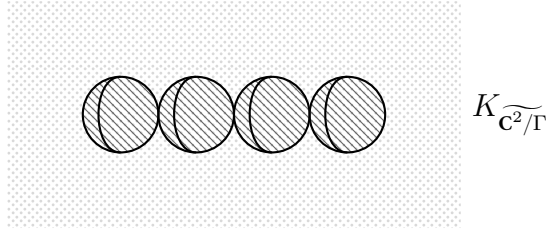
$$\mathfrak{g} = \langle e_i, h_i, f_i \rangle / \left\{ \begin{array}{lll} [h_i, h_j] = 0 & [h_i, e_j] = a_{ij}e_j & [h_i, f_j] = -a_{ij}f_j \\ [e_i, -]^{1-a_{ij}}(e_j) = 0 & [e_i, f_j] = \delta_{ij}h_i & [f_i, -]^{1-a_{ij}}(f_j) = 0 \end{array} \right\}$$

where  $e_i = e_{\alpha_i}$ ,  $f_i = e_{-\alpha_i}$ , and  $h_i = \frac{2}{\kappa(\alpha_i, \alpha_i)}\kappa(\alpha_i, -)$  are the dual basis to the simple roots with respect to the Killing form, and  $a_{ij}$  are integers.<sup>2</sup>

We represent the Cartan matrix  $a = (a_{ij})$  visually using *Dynkin diagrams*, e.g.

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix} \quad \begin{array}{cccc} 2 & -1 & 2 & -1 \\ \bullet & \bullet & \bullet & \bullet \end{array} \quad \mathfrak{sl}_5$$

1.1.4. *Remark.* In the ADE case there is a CY3 category  $\mathcal{C}(\mathfrak{g}) = D^b(\text{Coh}_T(K_{\widetilde{\mathcal{C}^2/\Gamma}}))$  with Grothendieck group  $\Lambda$ , Euler form  $\kappa$ , and elements with class  $\alpha_i$  given by  $\mathcal{O}_{\text{curve}}$  whose Euler is form given by intersection pairing.



$z$  is meant to come from a Bridgeland stability condition.

## 1.2. Explicit definition of $U_{\hbar}(\mathfrak{g})$ .

1.2.1. Let  $\mathfrak{g}$  be a finite dimensional semisimple Lie algebra over  $\mathbf{C}$ . The *Drinfeld–Jimbo quantum group*  $U_{\hbar}(\mathfrak{g})$  is the algebra over  $\mathbf{C}[[\hbar]]$  generated by elements  $e_i, f_i, h_i$  indexed by positive simple roots  $i$  with relations the same as in  $U(\mathfrak{g})$  except for

$$[e_i, f_j] = \delta_{ij}h_i \quad \rightsquigarrow \quad [e_i, f_j] = \delta_{ij} \frac{q_i^{h_i} - q_i^{-h_i}}{q_i - q_i^{-1}} = \delta_{ij} \left( h_i + \hbar^2 \frac{\kappa(\alpha_i, \alpha_i)^2}{48} (h_i^3 - h_i) + \dots \right)$$

where  $q_i = e^{\frac{(\alpha_i, \alpha_i)}{2}\hbar}$ .

**Proposition 1.2.2.** *If  $\mathfrak{g}$  is semisimple, this deformation is trivial:  $U_{\hbar}(\mathfrak{g}) \simeq U(\mathfrak{g})[[\hbar]]$  as associative algebras.*

*Proof Sketch.* Given a product

$$m : U(\mathfrak{g}) \otimes U(\mathfrak{g}) \rightarrow U(\mathfrak{g})[[\hbar]]$$

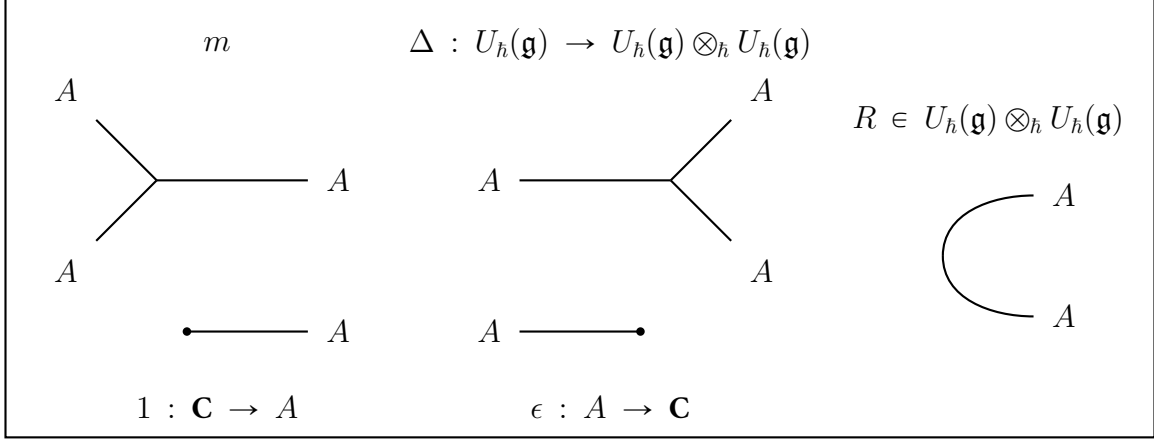
<sup>2</sup>Here,  $1 - a_{ii} = 0$ .

given by  $m = m_0 + m_1\hbar + m_2\hbar^2 + \dots$ , the associativity condition implies that  $m_1$  defines a class in

$$\mathrm{HH}^2(U(\mathfrak{g})) = \mathrm{Ext}_{U(\mathfrak{g})\text{-BiMod}}^2(U(\mathfrak{g}), U(\mathfrak{g})) \subseteq \mathrm{Hom}_{U(\mathfrak{g})\text{-BiMod}}(U(\mathfrak{g}) \otimes U(\mathfrak{g}), U(\mathfrak{g}))$$

and  $\mathrm{HH}^2(U(\mathfrak{g})) = 0$  for semisimple  $\mathfrak{g}$  (this is the hard part). Thus we can conjugate  $m$  by a formal automorphism to kill  $m_1$ . Iterating this process kills all  $m_i$  for  $i \geq 1$ , so  $m$  is conjugate to  $m_0$ .  $\square$

The interesting thing is the *coproduct* and invertible *R-matrix* on  $A = U_\hbar(\mathfrak{g})$ , which we write using strings:



where diagrams are read left to right, and parallel strings correspond to tensor products of  $A$ 's. They are defined by

$$\Delta : (e_i, h_i, f_i) \mapsto (e_i \otimes 1 + q_i^{h_i} \otimes e_i, \quad h_i \otimes 1 + 1 \otimes h_i, \quad f_i \otimes q_i^{-h_i} + 1 \otimes f_i),$$

and

$$R = R^0 \cdot \prod_{\alpha \in \Lambda^+}^{\curvearrowright} R^\alpha,$$

is a product over positive roots of (quantum) exponentials

$$R^\alpha = \exp_{q_\alpha} \left( \frac{q_\alpha - q_\alpha^{-1}}{a_\alpha} e_\alpha \otimes e_{-\alpha} \right), \quad R^0 = \exp \left( \hbar \sum (a^{-1})_{ij} h_i \otimes h_j \right),$$

where  $a_\alpha$  is the factor appearing in  $[e_\alpha, e_{-\alpha}] = a_\alpha \frac{q^{h_\alpha} - q^{-h_\alpha}}{q_\alpha - q_\alpha^{-1}}$  (which one can show always exists). We take the product in the order of increasing argument of  $z(\alpha)$ .<sup>3</sup>

**Definition 1.2.3.** A **quasitriangular (quasi)bialgebra**

$$(A, m, 1, \Delta, \epsilon, R, (\Phi))$$

is a vector space  $A$  with the structure of an associative algebra  $(m, 1)$ , a  $(\Phi)$ -coassociative coalgebra  $(\Delta, \epsilon)$ , an invertible element  $R \in A \otimes A$  (and an invertible element  $\Phi \in A \otimes A \otimes A$ ) satisfying

$$\Delta(ab) = \Delta(a)\Delta(b), \quad R \cdot \Delta(a) \cdot R^{-1} = \Delta^{\mathrm{op}}(a),$$

<sup>3</sup>By [KT91, Thm. 8.1] we may take any total order on  $\Lambda^+$  so long as if  $\alpha = \alpha_1 + \alpha_2$  then  $\alpha$  lies between  $\alpha_1, \alpha_2$  in the total order.

the *hexagon relations*

$$(\Delta \otimes \text{id})(R) = R_{13}R_{12}, \quad (\text{id} \otimes \Delta)(R) = R_{13}R_{23}$$

and  $\Delta(1) = 1 \otimes 1$ ,  $\epsilon \cdot m = \epsilon \otimes \epsilon$ . Here  $\Delta^{\text{op}} = \sigma \cdot \Delta$  is the opposite coproduct and  $R_{ij}$  means  $R$  placed in the  $i$  and  $j$  tensor factors.

(In the quasibialgebra case we also require the pentagon relations  $\Phi_{1,2,3,4}\Phi_{12,3,4} = \Phi_{2,3,4}\Phi_{1,23,4}\Phi_{1,2,3}$  and counit  $(1 \otimes \epsilon \otimes 1)\Phi = 1 \otimes 1$  relations,  $\Phi$ -coassociativity of the coproduct

$$\Phi(\Delta \otimes \text{id})\Delta(a)\Phi^{-1} = (\text{id} \otimes \Delta)\Delta(a)$$

and the  $\Phi$ -hexagon relations

$$(\text{id} \otimes \Delta)(R) = \Phi_{231}^{-1}R_{13}\Phi_{213}\Phi_{23}\Phi_{123}^{-1}, \quad (\Delta \otimes \text{id})(R) = \Phi_{312}^{-1}R_{13}\Phi_{132}R_{12}\Phi_{123}. \quad (1)$$

**Proposition 1.2.4.** (Drinfeld, Jimbo, [KT91]) *The above makes  $U_h(\mathfrak{g})$  into a quasitriangular bialgebra.*

*Proof Sketch.* To show that  $\Delta$  defines a bialgebra, we need to check that it kills relations. For instance, writing  $K_i = q^{h_i}$ ,

$$\begin{aligned} \Delta([e_i, f_j]) &= \Delta(e_i)\Delta(f_j) - \Delta(f_j)\Delta(e_i) \\ &= (e_i \otimes 1 + K_i \otimes e_i)(f_j \otimes K_j^{-1} + 1 \otimes f_j) - (f_j \otimes K_j^{-1} + 1 \otimes f_j)(e_i \otimes 1 + K_i \otimes e_i) \\ &= [e_i, f_j] \otimes K_j^{-1} + (\text{two cross terms, which cancel}) + K_i \otimes [e_i, f_j] \\ &= \delta_{ij} \frac{K_i - K_i^{-1}}{q_i - q_i^{-1}} \otimes K_i^{-1} + \delta_{ij} K_i \otimes \frac{K_i - K_i^{-1}}{q_i - q_i^{-1}} \\ &= \delta_{ij} \frac{K_i \otimes K_i - K_i^{-1} \otimes K_i^{-1}}{q_i - q_i^{-1}} \\ &= \delta_{ij} \frac{q^{\Delta(h_i)} - q^{-\Delta(h_i)}}{q_i - q_i^{-1}} \\ &= \Delta \left( \delta_{ij} \frac{q^{h_i} - q^{-h_i}}{q_i - q_i^{-1}} \right). \end{aligned}$$

The hexagon relations follow from the fact that if  $xy = qyx$ , then

$$\exp_q(x + y) = \exp_q(x) \exp_q(y).$$

For instance, since  $q^{h_\alpha}e_\alpha = q_\alpha e_\alpha q^{h_\alpha}$ , we have for  $\alpha$  a simple positive root that

$$\begin{aligned} (\Delta \otimes \text{id})(R^\alpha) &= \exp_{q_\alpha} \left( \frac{q_\alpha - q_\alpha^{-1}}{a_\alpha} \Delta(e_\alpha) \otimes e_{-\alpha} \right) \\ &= \exp_{q_\alpha} \left( \frac{q_\alpha - q_\alpha^{-1}}{a_\alpha} (e_\alpha \otimes 1 + q_\alpha^{h_\alpha} \otimes e_\alpha) \otimes e_{-\alpha} \right) \\ &= \exp_{q_\alpha} \left( \frac{q_\alpha - q_\alpha^{-1}}{a_\alpha} (e_\alpha \otimes 1 \otimes e_{-\alpha}) \right) \exp_{q_\alpha} \left( \frac{q_\alpha - q_\alpha^{-1}}{a_\alpha} (q_\alpha^{h_\alpha} \otimes e_\alpha \otimes e_{-\alpha}) \right) \end{aligned}$$

$$= R_{13}^\alpha (R_{13}^0 R_{23}^\alpha (R^0)_{13}^{-1})$$

where the last equality follows since  $R^0(1 \otimes e_{-\beta})(R^0)^{-1} = q^{h_\beta} \otimes e_\beta$  for any positive root  $\beta$ . We can prove the above for every positive root  $\alpha$ , which one can show implies the hexagon relations.

□

1.2.5. *Structure on  $U_h(\mathfrak{g})$ -LMod.* A *braided monoidal category with fibre functor* is a  $\mathbf{C}$ -linear category  $\mathcal{C}$  is the data of functors and natural isomorphism

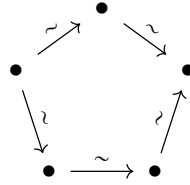
$$\begin{array}{ccc} \mathcal{C} & \otimes_{\mathcal{C}} : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C} & \begin{array}{c} \mathcal{C} \times \mathcal{C} \xrightarrow{\otimes_{\mathcal{C}}} \mathcal{C} \\ \wr \Downarrow \beta \\ \mathcal{C} \times \mathcal{C} \xrightarrow{\otimes_{\mathcal{C}}^{\text{op}}} \mathcal{C} \end{array} \\ F : \mathcal{C} \rightarrow \text{Vect} & 1_{\mathcal{C}} : \text{Vect} \rightarrow \mathcal{C} & \end{array}$$

as well as associativity isomorphism data

$$\alpha_{c,c',c''} : (c' \otimes_{\mathcal{C}} c'') \otimes_{\mathcal{C}} c''' \simeq c' \otimes_{\mathcal{C}} (c'' \otimes_{\mathcal{C}} c''')$$

$$\begin{array}{ccc} \mathcal{C} \times \mathcal{C} \times \mathcal{C} & \xrightarrow{\otimes_{\mathcal{C}} \cdot (\otimes_{\mathcal{C}} \times \text{id})} & \mathcal{C} \\ \wr \Downarrow \alpha & & \\ \mathcal{C} \times \mathcal{C} \times \mathcal{C} & \xrightarrow{\otimes_{\mathcal{C}} \cdot (\text{id} \times \otimes_{\mathcal{C}})} & \mathcal{C} \end{array}$$

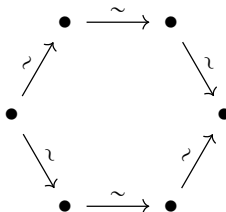
and the following relations: a *pentagon* compatibility relation for products of four objects



and the relations that (suppressing associator isomorphisms)

$$\beta_{c \otimes_{\mathcal{C}} c', c''} = \beta_{c, c''} \beta_{c', c''}, \quad \beta_{c, c' \otimes_{\mathcal{C}} c''} = \beta_{c, c''} \beta_{c, c'} \quad (2)$$

which are called the *hexagon* relations since (including the associator isomorphisms) they have the shape



and compatibility with the unit  $1_{\mathcal{C}}$  and fibre functor  $F$  isomorphisms

$$J_{c,c'} : F(c \otimes_{\mathcal{C}} c') \simeq F(c) \otimes_{\mathcal{C}} F(c'), \quad 1_{\mathcal{C}} \otimes_{\mathcal{C}} c \xrightarrow{u_L} c \xrightarrow{u_R} c \otimes_{\mathcal{C}} 1_{\mathcal{C}},$$

both compatible with the associator in the obvious way, and  $u_L, u_R$  compatible with the braiding  $\beta$ .

1.2.6. *Warning.* Note that this means  $F$  is a monoidal functor but not braided monoidal:  $F(\beta)$  is definitely not the usual braiding on  $\text{Vect}$ , this is the interesting part of the structure.

**Proposition 1.2.7.** *If we assume all categories  $\mathcal{C}$  are  $\mathbf{C}$ -linear cocomplete and the fibre functor  $F$  is cocontinuous, we get a bijection<sup>4</sup>*

$$\begin{aligned} \{\text{quasitriangular quasibialgebras } A\} &\simeq \{\text{braided monoidal categories } \mathcal{C} \text{ with a fibre functor}\} \\ A &\mapsto A\text{-LMod}. \end{aligned}$$

*Proof.* Given any category  $\mathcal{C}$  with functor  $F$  to  $\text{Vect}$ , we let  $A = \text{End} F$  be the natural endomorphisms of  $F$ , which is an algebra with respect to composition. The conditions on  $\mathcal{C}$  are sufficient that the tautological map

$$\mathcal{C} \rightarrow \text{End} F\text{-LMod}(\text{Vect}),$$

which always exists, is an equivalence by Barr–Beck. The other structures on  $A = \text{End} F$  are induced as follows:

- As  $\mathcal{C}$  is monoidal,  $A \otimes A$  has an  $A$ -module structure, and so we get  $\Delta : A \rightarrow A \otimes A$  by acting on  $1 \otimes 1$ . The associator  $\Phi$  is the image of  $1 \otimes 1 \otimes 1$  under

$$\alpha_{A,A,A} : (A \otimes A) \otimes A \xrightarrow{\sim} A \otimes (A \otimes A).$$

- As  $\mathcal{C}$  is braided monoidal, we get an equivalence

$$\beta_{A,A} : A \otimes A \xrightarrow{\sim} A \otimes A,$$

and  $R$  is the image of the identity under this equivalence. The hexagon relations (2) for  $\beta$  induce the hexagon relations (1) for  $R$ . The  $A$ -linearity of  $\beta$  implies the relation  $R \cdot \Delta(a) \cdot R^{-1} = \Delta^{\text{op}}(a)$ .

- $1_{\mathcal{C}}$  induces the counit.

□

**Corollary 1.2.8.**  *$U_h(\mathfrak{g})\text{-LMod}$  is a braided monoidal category.*

<sup>4</sup>In general, we just get an adjunction between the categories whose objects are on the left and right.

1.2.9. *Remark: a geometric definition.* A braided monoidal category  $\mathcal{C}$  is equivalent to a monoidal category with two monoidal structures  $\otimes, \boxtimes$  which are compatible:

$$(A \otimes B) \boxtimes (C \otimes D) \simeq (A \boxtimes C) \otimes (B \boxtimes D)$$

The same as two coproducts

1.2.10. *Construction via doubling.*

Interlude: monodromy, exodromy and the Riemann–Hilbert correspondence

Recall that a *connection* on a vector bundle on smooth scheme or complex manifold  $X$  is a map of vector bundles, or equivalently of  $\mathcal{O}$ -modules,

$$\nabla : T_X \otimes_{\mathbb{C}} V \rightarrow V \quad \Leftrightarrow \quad \nabla : \mathcal{T}_X \otimes_{\mathcal{O}} \mathbf{V} \rightarrow \mathbf{V} \quad \Leftrightarrow \quad \nabla : \mathbf{V} \rightarrow \mathbf{V} \otimes_{\mathcal{O}} \Omega_X^1$$

satisfying the Leibniz rule  $\nabla(fv) = f\nabla(v) + df \otimes v$ . A connection is *flat* if  $\nabla^2 = 0$  as a map  $T_X \otimes T_X \otimes V \rightarrow V$ .

Two connections are *equivalent* if there is an isomorphism of vector bundles  $\varphi : \mathbf{V} \simeq \mathbf{V}'$  intertwining the connections:

$$\begin{array}{ccc} \mathbf{V} & \xrightarrow{\nabla} & \mathbf{V} \otimes_{\mathcal{O}} \Omega_X^1 \\ \wr \downarrow \varphi & & \wr \downarrow \varphi \otimes \text{id} + \varphi^{-1} d\varphi \\ \mathbf{V}' & \xrightarrow{\nabla'} & \mathbf{V}' \otimes_{\mathcal{O}} \Omega_X^1 \end{array}$$

For instance, if  $\mathbf{V} = W \otimes \mathcal{O}$  then a connection is the same thing as a  $\mathfrak{gl}(W)$ -valued 1-form  $A$ , where  $\nabla = d + A$ , and a gauge transformation is the same thing as a  $\text{GL}(W)$ -valued function  $g$ , acting on the connection by  $A \mapsto gAg^{-1} + gdg^{-1}$ . Flatness is equivalent to  $dA + A \wedge A = 0$ .

1.2.11. A *meromorphic connection* on a vector bundle  $\mathbf{V}$  is a map  $\nabla$  as above defined on an open dense subscheme, for instance

$$\nabla : \mathbf{V} \rightarrow \mathbf{V} \otimes_{\mathcal{O}} \Omega_X^1(D)$$

for some divisor  $D$ .

1.2.12. These are all examples of D-module. A *D-module* is a sheaf  $\mathcal{M}$  of  $\mathcal{O}_X$ -modules with an action

$$\nabla : \mathcal{T}_X \otimes_{\mathcal{O}} \mathcal{M} \rightarrow \mathcal{M}$$

of the sheaf of Lie algebras  $\mathcal{T}_X$  (which is equivalent to flatness) satisfying Leibniz. The above are both examples of D-modules.

1.2.13. *Local systems.* A *local system*  $\mathcal{L}$  is a sheaf of  $\mathbb{C}$ -vector spaces such that there is an open cover  $\{U_i\}$  of  $X$  such that  $\mathcal{L}|_{U_i} \simeq \underline{W_i}$  is the trivial local system on finite dimensional vector space  $W_i$ .

The data of  $\mathcal{L}$  is equivalent to its *monodromy*, a functor

$$\Pi_1(X) \rightarrow \text{Vect}$$

sending a point  $x \in X$  to the stalk  $\mathcal{L}_x$  and a path  $\gamma : x \rightarrow y$  to the *parallel transport* map  $\mathcal{L}_x \simeq \mathcal{L}_y$ . If  $X$  is connected this is the same thing as a homomorphism

$$\Pi_1(X, x) \rightarrow \text{GL}(\mathcal{L}_x).$$

**Proposition 1.2.14.** (Riemann–Hilbert equivalence, easy case) *Let  $X$  be a smooth complex manifold. There is an equivalence of categories*

$$\{\text{local systems on } X\} \simeq \{\text{vector bundles with flat connections on } X\}$$

$$\begin{aligned}\mathcal{L} &\mapsto \mathcal{L} \otimes_{\mathbf{C}} \mathcal{O}_X \\ \mathbf{V}^\nabla &\leftrightarrow (\mathbf{V}, \nabla)\end{aligned}$$

1.2.15. *Local systems and monodromy.*

1.2.16. *Constructible sheaves.*

1.2.17. *Six functors.*  $Rf_*$  and  $Lf_!$ ; define their adjoints.

1.2.18. *Exodromy.*

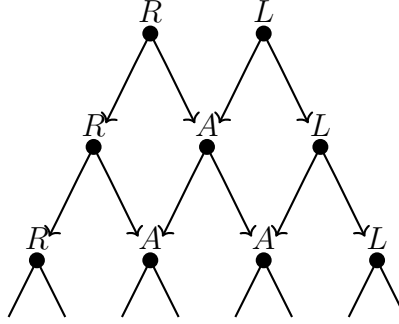
1.2.19. *Vector bundles with meromorphic connection.*

1.2.20. *D-modules.*

1.2.21. *Riemann–Hilbert correspondence.*

## Interlude: cobar complex

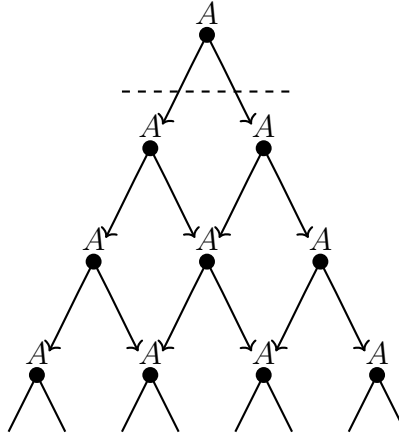
1.2.22. If  $A$  is a coalgebra, and  $L, R$  are left and right comodules, the *cobar complex*  $\Omega(R, A, L)$



is the complex on  $R \otimes A^{\otimes n} \otimes L$  with differential given by

$$d(r \otimes a_1 \otimes \cdots \otimes a_n \otimes l) = \Delta(r) \otimes a_1 \otimes \cdots \otimes a_n \otimes l - r \otimes \Delta(a_1) \otimes a_2 \otimes \cdots \otimes a_n \otimes l + \cdots \pm r \otimes a_1 \otimes \cdots \otimes a_n \otimes \Delta(l).$$

Note that  $A \xrightarrow{\sim} \Omega(A, A, A)$  gives a resolution of  $A$  as a  $A$  bicomodule:



That resolution is cofree, therefore injective, and so we have the *Hochschild cochain complex*

$$\mathrm{HC}^\bullet(A) = \mathrm{LHom}_{A\text{-BiCoMod}}(A, A) \simeq \mathrm{Hom}_{A\text{-BiCoMod}}(A, \Omega(A, A, A)).$$

If  $A = \mathrm{Sym}(V)$  then

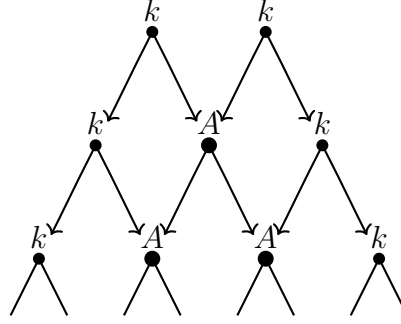
$$\mathrm{HH}^\bullet(A) \simeq \mathrm{Sym}(V) \otimes \mathrm{Sym}(V[-1]).$$

If  $A$  is a “smooth” coalgebra, then the HKR Theorem says that  $\mathrm{HH}^\bullet(A) \simeq \bigwedge_A^\bullet \mathrm{CoDer}(A, A)$  where the right hand side is also  $\mathrm{Sym}_A(\mathrm{CoDer}(A, A)[-1])$ .

1.2.23. If  $A$  is given a comodule  $k \rightarrow A$ , we will instead be considering the complex

$$\Omega(k, A, k) \simeq k \otimes_A^\vee \Omega(A, A, k) = k \otimes_A^{\mathrm{R}\vee} k$$

giving a free resolution of  $k$  as a left  $A$ -comodule.



given by the cohomology of the **co-Hochschild complex** for a coalgebra  $A$

$$\begin{aligned}
 C^n(A) &= (A^{\otimes n}, d), \\
 d(a_1 \otimes \cdots \otimes a_n) &= 1 \otimes a_1 \otimes \cdots \otimes a_n \\
 &\quad + \sum_{i=1}^n (-1)^i a_1 \otimes \cdots \otimes \Delta(a_i) \otimes \cdots \otimes a_n \\
 &\quad + (-1)^{n+1} a_1 \otimes \cdots \otimes a_n \otimes 1,
 \end{aligned}$$

**Lemma 1.2.24.** *When  $A = \text{Sym}V$ , the cohomology of  $C^\bullet(A)$  is*

$$k, V, \wedge^2 V, \wedge^3 V, \dots$$

*the maps induced by  $\wedge^n V \rightarrow A^{\otimes n}$  and with inverse defined on boundaries given by the alternating sum.*

*Proof.*

□

Thus, we get a *canonical* element of  $\wedge^n \mathfrak{g}$  for each cohomology class in  $H^n(C^\bullet(U(\mathfrak{g})))$ !

1.2.25. We consider the swap involution

$$\sigma : a_1 \otimes \cdots \otimes a_n \mapsto (-1)^{\frac{n(n+1)}{2}} a_n \otimes \cdots \otimes a_1$$

**Lemma 1.2.26.** *The subspace of  $\Omega(k, A, k)^-$  of  $\sigma$ -antiinvariant elements is a subcomplex, with cohomology*

$$0, \mathfrak{g}, 0, \wedge^3 \mathfrak{g}, \dots$$

*Moreover, if a coboundary here is  $\mathfrak{g}$ -invariant, it is the coboundary of a  $\mathfrak{g}$ -invariant element. is additionally a*

## 2. Drinfeld's Theorem: constructing $U_{\hbar}(\mathfrak{g})$ using differential equations and deformation theory

Let  $\mathfrak{g}$  be a finite dimensional semisimple Lie algebra over a field  $k$  of characteristic zero. We now study  $k[[\hbar]]$ -deformations of  $U(\mathfrak{g})$  as a quasitriangular quasibialgebra, up to a notion of *twist equivalence*. This captures the deformations up to equivalence of  $\text{Rep}U(\mathfrak{g})$  as a braided monoidal category with fibre functor.

The headline result is that the deformation theory is close to being rigid, i.e. the deformations are more or less unique up to equivalence. See [Va95; SV91a; SV91b] for other (more geometric) proofs of the Drinfeld–Kohno Theorem.

**Theorem 2.0.1.** [Dr90; Dr89] *Let  $t$  be any  $\mathfrak{g}$ -invariant element of  $\text{Sym}^2 \mathfrak{g}$ .*

- (1) *There is a deformation over  $k[[\hbar]]$ , moreover there is one of the form*

$$R = 1 \otimes 1 + \hbar t/2 + \mathcal{O}(\hbar^2), \quad \Phi = 1 \otimes 1 \otimes 1 + \mathcal{O}(\hbar^2)$$

*and  $\Delta = \Delta_0$ . There are  $\mathfrak{g}$ -invariant  $k$ -valued power series in variable  $\tau$*

$$R_k(\tau) = 1 \otimes 1 + \tau/2 + \mathcal{O}(\tau^2), \quad \Phi_k(\tau) := \Phi_k(\tau_{12}, \tau_{23}) = 1 \otimes 1 \otimes 1 + \mathcal{O}(\tau^2),$$

*the first symmetric, such that specialising  $\tau = \hbar t$  for any  $t$  gives a  $k$ -deformation. We call such a pair a **universal  $k$ -deformation**.*

- (2) *There is a universal  $\mathbf{C}$ -deformation with  $\Delta = \Delta_0$ , and  $R_{\text{KZ}} = \exp(\hbar t/2)$  and  $\Phi_{\text{KZ}}$  both given by the exodromy of the KZ equations:*

- (3) *Every universal  $k$ -deformation is of the form*

$$R_k(\tau) = \exp(f(\tau)/2)$$

*where  $f(\tau) = \tau + \mathcal{O}(\tau^2)$  is a  $k$ -valued power series in  $\tau$ .*

*Uniqueness (i.e. rigidity) results:*

- (4) *Every deformation is twist-equivalent to one of the form*

$$(U(\mathfrak{g}_{\hbar})[[\hbar]], \Delta_0, \exp(\hbar t/2), \Phi)$$

*which we call **standard**. Here  $\mathfrak{g}_{\hbar}$  is a deformation of  $\mathfrak{g}$  as a Lie algebra.*

- (5) *Any deformation  $(A, \Delta_0, R, \Phi)$  is equivalent by a twist  $F \in (\text{Sym}^2 \mathfrak{g})^{\mathfrak{g}}[[\hbar]]$  to*

$$(R_{\mathbf{Q}}(\hbar\theta), \Phi_{\mathbf{Q}}(\hbar\theta))$$

*for a unique  $\theta \in (U(\mathfrak{g}) \otimes U(\mathfrak{g}))[[\hbar]]$ . The twist is unique up to multiplication by  $(u^{-1} \otimes u^{-1})\Delta(u)$  for a unique  $u \in 1 + \mathfrak{g}[[\hbar]]$ .*

*Stronger still, any such deformation is equivalent by a twist  $F \in (\text{Sym}^2 \mathfrak{g})^{\mathfrak{g}}[[\hbar]]$  to a unique universal deformation*

$$(\exp(\hbar\theta/2), \Phi_k(\hbar\theta))$$

where now  $\theta \in (\text{Sym}^2 \mathfrak{g})^{\mathfrak{g}}$  (no  $\hbar$ !) and  $F$  is unique as before.

(6) Any two  $k$ -universal deformations  $R_i(\tau), \Phi_i(\tau)$  are twist-equivalent after a change of power series

$$F(\tau) : (R_1(\tau), \Phi_1(\tau)) \mapsto (R_2(f(\tau)), \Phi_2(f(\tau)))$$

where the  $k$ -power series  $f(\tau) = \tau + \mathcal{O}(\tau^2)$  and  $F(\tau)$  are unique and unique up to multiplication by  $(u^{-1} \otimes u^{-1})\Delta(u)$ , respectively.

We prove (1) in §2.1, then (6) in §2.3, (2) in §2.4, and (4) in §2.5.

We will *not* prove:

(1) There is a universal standard deformation of the form

$$\Phi_{\text{KZ}}(\hbar\Omega) = \exp(P_{\text{KZ}}(\hbar\Omega_{12}, \hbar\Omega_{23})) \in \mathbf{Q}\langle\langle\hbar\Omega_{12}, \hbar\Omega_{23}\rangle\rangle,$$

where  $P_{\text{KZ}}$  is an element in the free completed Lie algebra on two generators.

(2)  $\Phi_{\text{KZ}}(\hbar\Omega)$  is unique up to an element  $F(\hbar\Omega) \in \mathbf{Q}\langle\langle\hbar\Omega\rangle\rangle$ .

(3) **GT action? zeta functions?**

## 2.1. Universal deformations.

2.1.1. We construct a  $\mathbf{Q}$ -universal deformation, which proves all of (1) simultaneously. To do this, we construct truncations  $R_n(\tau), \Phi_n(\tau)$  of the power series modulo  $\hbar^n$ , and use Hochschild cohomology computations to construct the next term of the power series:

$$\Phi_{n+1} = \Phi_n + \hbar^n \varphi, \quad R_{n+1} = R_n + \hbar^n r$$

where  $\varphi, r$  are  $\mathfrak{g}$ -invariant,  $r$  is symmetric and  $\varphi$  is antisymmetric, and both are defined as a  $\mathbf{Q}$ -linear word in lower order terms and  $\tau$ .

Consider the failure of the pentagon relation

$$\Phi_{n,1,2,34}\Phi_{n,12,3,4} = \Phi_{n,2,3,4}\Phi_{n,1,23,4}\Phi_{n,1,2,3} + \hbar^n \mu \mod \hbar^{n+1} \quad (3)$$

and failure of the hexagon relation (from which failure of the other hexagon relation may be read off)

$$(\Delta \otimes \text{id})R_n = \Phi_n^{312}R_n^{13}(\Phi_n^{132})^{-1}R_n^{23}\Phi_n + \hbar^n \nu \mod \hbar^{n+1} \quad (4)$$

where  $\mu \in U(\mathfrak{g})^{\otimes 4}$  and  $\nu \in U(\mathfrak{g})^{\otimes 3}$ .

**Lemma 2.1.2.** *Viewed as an element of the  $n = 4$  part of the cobar complex*

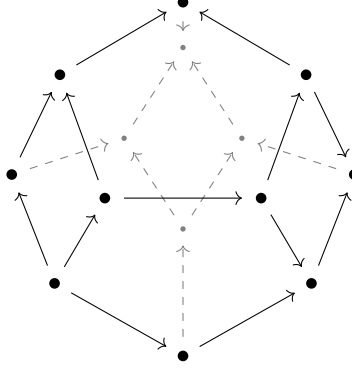
$$\Omega(k, A, k) = (A^{\otimes n}, d)$$

*we have that  $d\mu = 0$ . Likewise, viewed as an element of  $n = 2$  part of the complex*

$$\Omega(k, A, k) = (A^{\otimes n}, d) \otimes A,$$

*and assuming we have already constructed  $\Phi$ ,  $d\nu = 0$ .*

*Proof.* We take a sum of (3) over all six pentagons in the associahedron  $K_5$  whose points correspond to possible bracketings of a five-letter word,



to give that

$$\mu_{1,2,3,4} - \mu_{1,2,34,5} + \mu_{12,3,4,5} = \mu_{2,3,4,5} - \mu_{1,23,4,5} + \mu_{1,2,3,45}$$

which is exactly  $d\mu = 0$ .

Likewise, if  $\Phi$  is constructed, we use  $\Phi$ -coassociativity of  $\Delta$  to give that

$$\Phi^{1,2,3}(\Delta \otimes \text{id} \otimes \text{id})(\Delta \otimes \text{id})(R_n) = (\text{id} \otimes \Delta \otimes \text{id})(\Delta \otimes \text{id})(R_n)\Phi^{1,2,3}$$

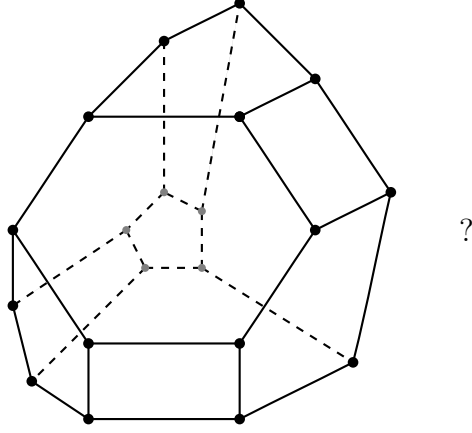
and inserting (4) in place of  $(\Delta \otimes \text{id})(R_n)$  gives that modulo  $\hbar^{n+1}$ ,

$$\Phi^{1,2,3}(\Delta \otimes \text{id} \otimes \text{id})(\Phi^{312}R_n^{13}(\Phi^{132})^{-1}R_n^{23}\Phi + \hbar^n\nu) = (\text{id} \otimes \Delta \otimes \text{id})(\Phi^{312}R_n^{13}(\Phi^{132})^{-1}R_n^{23}\Phi + \hbar^n\nu)\Phi^{1,2,3}$$

gives  $\nu^{12,3,4}, \nu^{1,23,4}$  in the below, and applying (4) in place of  $(\Delta \otimes \text{id})(R_n)$  in the 123 and the 234 components again to give  $\nu^{123}, \nu^{234}$ , we can rearrange (using the pentagon relation and quasitriangularity of  $R_n$ ) to give that modulo  $\hbar^{n+1}$ ,

$$d\nu = \nu^{234} - \nu^{12,3,4} + \nu^{1,23,4} - \nu^{124} = 0.$$

*n.b.* I expect this is interpretable in terms of another polytope, e.g. the cyclohedron  $W_4$  whose points correspond to possible bracketings of a four-letter word up to cyclic rotation, but I haven't checked this:



□

**Lemma 2.1.3.** *There is a skew-symmetric element  $\varphi$  of degree  $n = 3$  such that*

$$d\varphi = \varphi_{2,3,4} - \varphi_{12,3,4} + \varphi_{1,23,4} - \varphi_{1,2,34} + \varphi_{1,2,3} = \mu$$

*and, assuming  $\Phi$  is constructed, there is a symmetric element  $r$  of degree  $n = 2$  such that*

$$dr = r_{13} - r_{12,3} + r_{23} = \nu - 3\varphi.$$

*Proof.* Note that these equalities are the leading order terms of (3) and (4) respectively.

We prove the above by applying Lemma 1.2.26, firstly to the  $\sigma$ -antisymmetric element  $\mu$  in degree  $n = 4$ :

$$\sigma_{2341} \cdot \mu = -\mu = -(-1)^{\frac{n(n+1)}{2}} \mu,$$

then to the  $\sigma$ -antisymmetrisation of  $\nu$ , which is

$$\bar{\nu} = \nu + \sigma_{21} \cdot \nu = \nu + \nu^{213}.$$

By Lemma 1.2.26, both are cocycles and hence coboundaries of elements  $\varphi$  and  $r$  of degree  $n = 3$  and  $n = 2$  respectively. By examining (4) one can show that  $\nu - \nu^{213} = 6\varphi$ , giving the second equality. This also shows that not just that  $\varphi = -\varphi_{312}$ , but  $\varphi \in \bigwedge^3 U(\mathfrak{g})$ ; moreover  $\varphi$  is  $\mathfrak{g}$ -invariant since  $\mu$  is.

It remains to show that  $r$  may be taken to be  $\mathfrak{g}$ -invariant and symmetric.

□

At this point, we have constructed  $\Phi_{n+1}$  and, assuming that  $\Phi$  is constructed,  $R_{n+1}$ , satisfying the pentagon and hexagon relations. The other relations are easier.

We want

$$(\text{id} \otimes \epsilon \otimes \text{id})(\Phi) = 1, \quad \Phi^{-1} = \Phi^{312}$$

as well as

$$(\epsilon \otimes \text{id})(R) = 1, \quad (\text{id} \otimes \epsilon)(R) = 1,$$

as well as  $\Phi$ -coassociativity and  $R$ -quasitriangularity of  $\Delta$ . **prove this?**

This constructs a  $\mathbf{Q}$ -universal deformation, and thus proves (1) and (2) of Theorem 2.0.1.

## 2.2. Twists.

2.2.1. *Twists.* We note that two braided monoidal categories with fibre functors are equivalent if there is an equivalence of categories fitting into a commutative diagram

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow[\sim]{\gamma} & \mathcal{C}' \\ F \searrow & & \swarrow F' \\ & \text{Vect} & \end{array} \quad (5)$$

with a natural transformations

$$\mathcal{J}_{V,V'} : \gamma(V \otimes V') \xrightarrow{\sim} \gamma(V) \otimes' \gamma(V') \quad (6)$$

such that  $J'_{\gamma(V),\gamma(V')} \cdot F'(\mathcal{J}_{V,V'}) = J_{V,V'}$  as natural transformations, which respects associators and braidings

$$\begin{array}{ccc} \gamma((V_1 \otimes V_2) \otimes V_3) \xrightarrow[\sim]{\mathcal{J}_{1,2}\mathcal{J}_{12,3}} (\gamma(V_1) \otimes \gamma(V_2)) \otimes \gamma(V_3) & \xrightarrow[\sim]{\mathcal{J}} & \gamma(V_1) \otimes' \gamma(V_2) \\ \gamma(\alpha_{V_1,V_2,V_3}) \downarrow \wr & \wr \downarrow \alpha'_{\gamma(V_1),\gamma(V_2),\gamma(V_3)} & \gamma(\beta_{V_1,V_2}) \downarrow \wr \\ \gamma(V_1 \otimes (V_2 \otimes V_3)) \xrightarrow[\sim]{\mathcal{J}_{1,23}\mathcal{J}_{2,3}} \gamma(V_1) \otimes (\gamma(V_2) \otimes \gamma(V_3)) & \xrightarrow[\sim]{\mathcal{J}} & \gamma(V_2 \otimes V_1) \otimes' \gamma(V_1) \\ & \wr \downarrow \beta'_{\gamma(V_1),\gamma(V_2)} & \end{array}$$

as well a unit isomorphism  $\gamma(1) \simeq 1$  satisfying various compatibilities, such that under this identification  $\mathcal{J}_{1,V'}$  and  $\mathcal{J}_{V,1}$  are the identity.

**Definition 2.2.2.** If  $A$  is an algebra, its group  $G_A$  of twists are the invertible elements of  $F \in A \otimes A$  satisfying the counit conditions  $(\epsilon \otimes \text{id})F = (\text{id} \otimes \epsilon)F = 1$ .

**Proposition 2.2.3.**  $G_A$  acts on the set of quasitriangular quasibialgebra structures on  $A$  by

$$F \cdot (A, \Delta, \Phi, R) = (A, F^{-1}\Delta F, F_{2,3}^{-1}F_{1,23}^{-1}\Phi F_{1,23}F_{2,3}, F_{21}^{-1}RF).$$

*We need some extra conditions so that the RHS actually defines a braided monoidal category with fibre functor*

Since the right hand side arises from a braided monoidal category with fibre functor, the right hand side defines a quasitriangular quasibialgebra structure on  $A$  by Proposition 1.2.7. See [ES, §13.3.3. §14.2.1] for an interpretation in terms of nonabelian cohomology.

**Proposition 2.2.4.**  $A\text{-Mod}$  and  $A'\text{-Mod}$  are equivalent as braided monoidal categories with fibre functors if and only if  $A \simeq A'$  as algebras and  $A'$  is a twist of  $A$ .

*Proof.* The commutative diagram (5) implies that  $A \simeq A'$  as algebras, and we will identify them from now on. We now set

$$F = \mathcal{J}_{A,A} \cdot 1 \otimes 1.$$

Since the map  $\mathcal{J}_{V,V'}$  is a map of  $A$ -modules, this implies

$$\begin{array}{ccc} \gamma(V_1 \otimes V_2) & \xrightarrow[\sim]{\mathcal{J}_{V_1,V_2}} & \gamma(V_1) \otimes' \gamma(V_2) \\ \downarrow \Delta(a) \cdot & & \downarrow \Delta'(a) \cdot \\ \gamma(V_1 \otimes V_2) & \xrightarrow[\sim]{\mathcal{J}_{V_1,V_2}} & \gamma(V_1) \otimes' \gamma(V_2) \end{array}$$

commutes. Together with the above two commuting diagrams, this shows that  $A' = F \cdot A$ .  $\square$

The main example is if  $R = R^0 \cdot \prod_{\alpha \in \Lambda^+} R^\alpha$ , then the product  $F = \prod_{\alpha \in S} R^\alpha$  over any sector  $S$  of the complex plane gives a twist. [find reference and make precise](#)

### 2.3. Uniqueness of deformations.

2.3.1. We prove the first part of (5), whose proof also works for (6). To begin with, by twisting we may assume that  $R$  is symmetric. Indeed, writing out the condition that  $F' R F'^{-1}$  be symmetric, we may use any  $F' = F_0 \sqrt{R \sqrt{R^{21}} R}^{-1}$  where  $F_0$  is symmetric works.

We prove (5) by constructing the truncations

$$F_n, R(\theta_n), \Phi(\theta_n)$$

twist  $F \in (U(\mathfrak{g}) \otimes U(\mathfrak{g}))[[\hbar]]$  and element  $\theta$  term-by-term (n.b.  $\Phi(-)$  and  $R(-)$  are invertible power series so the above determines  $\theta_n$ ), as in the proof of (1). It boils down to showing that

**Lemma 2.3.2.** *If  $(\Phi_i, R_i)$  are two deformations with  $R_i$  symmetric, which both agree modulo  $\hbar^n$ , and*

$$\Phi_1 - \Phi_2 = \varphi \hbar^n \mod \hbar^{n+1}, \quad R_1 - R_2 = r \hbar^n \mod \hbar^{n+1},$$

*then we have that*

$$\varphi = f^{23} - (\Delta \otimes \text{id})(f) + (\text{id} \otimes \Delta)(f) - f^{12}$$

*where  $f$  is a symmetric element in  $(U(\mathfrak{g}) \otimes U(\mathfrak{g}))^\mathfrak{g}$  with  $(\epsilon \otimes \text{id})f = (\text{id} \otimes \epsilon)f = 0$ , and unique up to addition by  $v \otimes 1 - \Delta(v) + 1 \otimes v$  where  $v$  is an element of the centre  $U(\mathfrak{g})^\mathfrak{g}$  with  $\epsilon(v) = 0$ .*

*Proof.* Similar.  $\square$

Using (3) we prove the second part of (5) similarly.

### 2.4. The KZ equations.

2.4.1. Let  $\mathfrak{g}$  be a finite dimensional semisimple Lie algebra over  $\mathbf{C}$  and  $t \in (\text{Sym}^2 \mathfrak{g})^{\mathfrak{g}}$ . For finite dimensional representations  $V_1, \dots, V_n$  we write

$$\mathbf{V} = V_1 \otimes \dots \otimes V_n \otimes \mathcal{O}_{\mathbf{C}^n}[[\hbar]]$$

for the trivial vector bundle on  $\mathbf{C}^n$ .

**Theorem 2.4.2.** (Canonical definition of  $\nabla_{\text{KZ}}$ ) *There is a unique (up to isomorphism) meromorphic flat connection  $\nabla_{\text{KZ}}$  on  $\mathbf{V}$  with such that:*

- (1) *It is equivariant<sup>5</sup> for the translation action  $\mathbf{G}_a \times \mathbf{C}^n \rightarrow \mathbf{C}^n$  sending  $t \cdot (z_1, \dots, z_n) = (z_1 + t, \dots, z_n + t)$ , and for action of the symmetric group  $\mathfrak{S}_n$ .*
- (2) *It has regular singularities along the diagonals  $\Delta_{ij} = \{z_i = z_j\}$  and is regular elsewhere.*
- (3) *The exodromy along a path  $\gamma$  onto a point of  $\Delta_{jk}$  avoiding the other diagonals is  $\hbar J_{V_j, V_k} : \otimes V_i \xrightarrow{\sim} \otimes V_i$ .*
- (4) *The monodromy around  $\Delta_{jk}$  clockwise is  $e^{\hbar 2t_{jk}} : \otimes V_i \xrightarrow{\sim} \otimes V_i$  acting on the  $j$  and  $k$  tensor factors.*

*Proof Sketch.* The KZ connection

$$\nabla_{\text{KZ}, i} = \partial_{z_i} + \hbar \sum_{j \neq i} \frac{t_{ij}}{z_i - z_j}$$

has these properties.

Conversely, because  $\mathbf{C}^n$  is affine, any meromorphic connection on  $\mathbf{C}^n$  takes the form  $\nabla_i = \partial_{z_i} + \sum_j A_{ij}(z_1, \dots, z_n)$ , or equivalently as a map

$$\mathbf{V} \rightarrow \mathbf{V} \otimes_{\mathcal{O}} \Omega^1$$

given by  $d + \sum_{i,j} A_{ij}(z_1, \dots, z_n, \hbar) dz_j$ . The second summand, let us refer to it as  $A$ , is a  $\mathfrak{gl}(V)$ -valued meromorphic section of  $\Omega^1$ .

Note that we have by Saito and Arnold that

$$\Gamma(\text{Conf}_n \mathbf{C}, \Omega^1) = \mathbf{C}[z_i, (z_i - z_j)^{-1}] \{d \log(z_i - z_j)\}$$

where  $\text{Conf}_n \mathbf{C} = \mathbf{C}^n \setminus \{z_i = z_j\}$  is the ordered configuration space of  $n$  points in  $\mathbf{C}$ . Thus by the conditions on the singularities,  $A$  is an element of this vector space, such that:

- By translation equivariance,  $A$  is only a function of  $(z_i - z_j)^{\pm}$ .
- By monodromy and  $\mathfrak{S}_n$ -equivariance, the constant coefficient in front of  $d \log(z_i - z_j)$  is  $\pm \hbar t_{ij}$ , the sign positive or negative whether  $i > j$  or  $i < j$ .

---

<sup>5</sup>Strongly equivariant in the  $\mathcal{D}$ -module language.

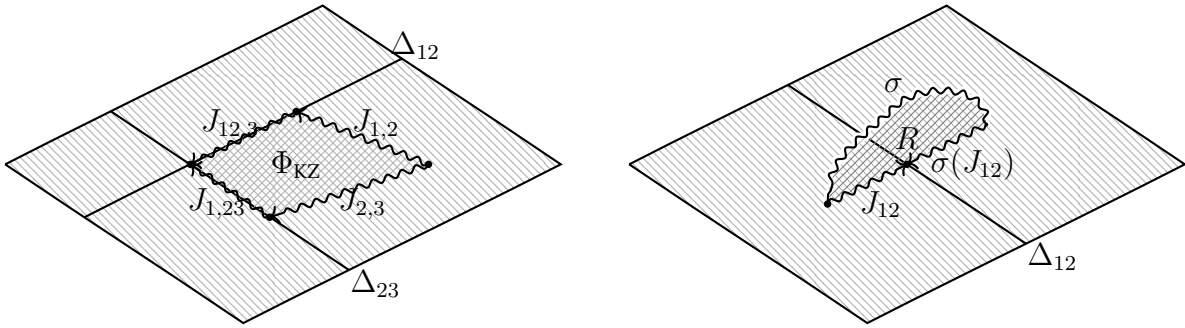
This shows that

$$\nabla_i = \partial_{z_i} + \hbar \sum_{j \neq i} \frac{t_{ij}}{z_i - z_j} + \mathcal{O}(1)$$

where the term  $\mathcal{O}(1)$  is further constrained by the flatness condition. To finish, one needs to show that up to equivalence of connections, one may kill the  $\mathcal{O}(1)$  term.

□

This Theorem now implies (2) of Theorem 2.0.1. We can *define*  $\Phi_{\text{KZ}}$  and  $R_{\text{KZ}}$  by means of exodromy off the diagonals onto  $\Delta_{12} \cap \Delta_{23}$ , and around  $\Delta_{12}$ , respectively: they are the *ratios* of the exodromy on either side:



The pentagon and hexagon relation follow by building a contractible two-cell out of five and six face two-cells corresponding to  $\Phi_{\text{KZ}}$  and  $R_{\text{KZ}}$ 's, which implies that the corresponding products of  $\Phi_{\text{KZ}}$  and  $R_{\text{KZ}}$  are trivial. The two-face is obvious to construct in the pentagon case by working inside the real slice  $\mathbf{R}^4 \subseteq \mathbf{C}^4$ .

Explicitly, following [ES], for three points 0,  $z$ , 1 we consider

$$\frac{dF}{dz} = \hbar \left( \frac{t_{12}}{z} + \frac{t_{23}}{z-1} \right) F$$

and consider the unique solutions of the form

$$F_0(z) = (1 + z f_0(z)) z^{\hbar t_{13}}, \quad F_1(z) = (1 + (1-z) f_1(1-z)) (1-z)^{\hbar t_{23}}$$

for power series  $f_i$ , then define  $\Phi_{\text{KZ}} = F_0(z)/F_1(z)$ , which is a constant because both the above are solutions to the KZ equations, and  $\mathfrak{g}$ -invariant because the equations are  $\mathfrak{g}$ -invariant. Similarly, we define  $R_{\text{KZ}} = e^{\hbar t/2}$  by considering the monodromy around  $z = 0$ .

The  $\Phi$ -coassociativity of  $\Delta$  follows from  $\mathfrak{g}$ -invariance, and the counit axiom since applying  $\text{id} \otimes \epsilon \otimes \text{id}$  to the right hand side of the above gives zero.

More interesting relations follow from three-cells. For the pentagon relation, let us consider the four point KZ equations

$$\frac{dF}{dz} = \hbar \left( \frac{t_{12}}{z} + \frac{t_{23}}{z-u} + \frac{t_{24}}{z-1} \right) F$$

$$\frac{dF}{du} = \hbar \left( \frac{t_{13}}{u} + \frac{t_{23}}{u-z} + \frac{t_{34}}{u-1} \right) F$$

We view the points  $0zu1$  as on a line, and the bracketing  $((\bullet\bullet)\bullet)\bullet$  corresponds to the solution

$$F(u, z) = (1 + f(u, z)) z^{\hbar t_{12}} u^{\hbar(t_{13}+t_{23})}$$

where  $f$  is analytic in  $u, z$  and vanishes at  $z = u = 0$ .

## 2.5. Standard form.

2.5.1. We now prove (3) using the KZ equations, which proves it for  $k = \mathbf{C}$ .

In general, if we have that

$$F(\tau) \cdot (\exp(\tau/2), \Phi(\tau)) = (R'(\tau'), \Phi'(\tau'))$$

as in (4), then it follows that

$$R'(\tau') = F(\tau) \exp(\tau/2) F(\tau)^{-1} = \exp(\tau/2)$$

By (6), we have that

2.5.2. Important will be a special class of deformations called **standard**:

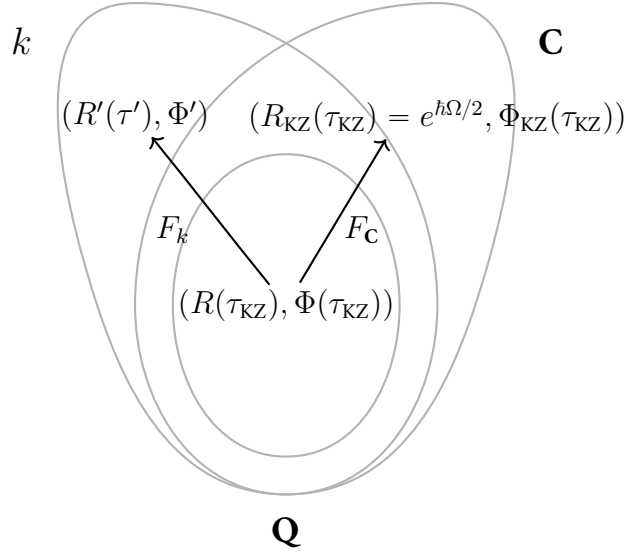
$$A = (U(\mathfrak{g})[[\hbar]], \Delta_0, R = e^{\hbar t/2}, \Phi_{\text{KZ}})$$

where  $t \in \mathfrak{g} \otimes \mathfrak{g}$  is symmetric and  $\mathfrak{g}$ -invariant, and  $\Delta_0(x) = x \otimes 1 + 1 \otimes x$  is the trivial coproduct. A **universal  $k$ -deformation** is one of the form

$$(U(\mathfrak{g})[[\hbar]], \Delta_0, R(\hbar\Omega), \Phi(\hbar\Omega)) \tag{7}$$

with  $R(\hbar\Omega) = 1 + \hbar\Omega/2 + \mathcal{O}(\hbar^2)$  and  $\Phi(\hbar\Omega) = 1 \otimes 1 \otimes 1 + \mathcal{O}(\hbar^2)$  power series in  $k$ .

2.5.3. We then note that the KZ equations give a universal  $\mathbf{C}$ -deformation, and by **what?** the twist equivalence with the universal rational deformation preserves the  $R$ -matrix and involves no power series change in  $\tau$ .

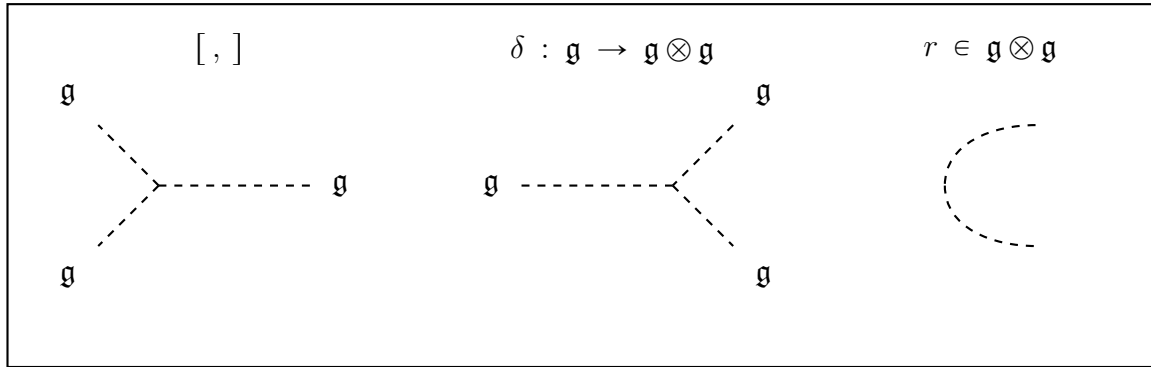


It follows that there is a universal standard **Q**-deformation, and therefore by (6) any **Q**-deformation is twist equivalent to a standard one.

## 2.6. Afterwords.

### 2.6.1. Relation to formality.

### 2.6.2. Fibre functor point of view. More generally, let $\mathfrak{g} = (\mathfrak{g}, \delta)$ be a Lie bialgebra



Corresponding to Manin triple

$$\mathfrak{g} = \mathfrak{g}_+ \oplus \mathfrak{g}_-$$

we have a decomposition as  $\mathfrak{g}$ -modules

$$U(\mathfrak{g}) = (\text{Ind}_{\mathfrak{g}_+}^{\mathfrak{g}} \mathbf{C}) \otimes (\text{Ind}_{\mathfrak{g}_-}^{\mathfrak{g}} \mathbf{C})$$

which we write as  $M_{\pm} \simeq U(\mathfrak{g}_{\pm}) \cdot 1_{\pm}$ .

**Theorem 2.6.3.** (Etingof–Kazhdan) Under *refequivalence*, a standard bialgebra  $A$  corresponds to the braided monoidal category

$$\mathcal{C} = (U(\mathfrak{g})[[\hbar]]\text{-Mod}, \otimes_0, \sigma \cdot R^{\hbar\Omega/2}, \Phi_{\text{KZ}}, J_{\text{KZ}})$$

and forgetful monoidal fibre functor (for the KZ associator on  $\mathcal{C}$  and the trivial associators on  $\text{Vect}[[\hbar]]!$ )

$$F : \mathcal{C} \rightarrow (\text{Vect}[[\hbar]], \otimes_0, \Phi_0)$$

$$V \mapsto V = \text{Hom}_{U(\mathfrak{g})[[\hbar]]}(U(\mathfrak{g})[[\hbar]], V) = \text{Hom}_{\mathfrak{g}}(U(\mathfrak{g}), V)[[\hbar]]$$

where the isomorphism exhibiting  $F$  as a monoidal functor

$$J_{V,W} : F(V) \otimes F(W) \simeq F(V \otimes W)$$

is induced by

$$\begin{aligned} M_+ \otimes M_-[[\hbar]] &\xrightarrow{\Delta} (M_+ \otimes M_+) \otimes (M_- \otimes M_-)[[\hbar]] \\ &\xrightarrow[\sim]{\Phi_{\text{KZ}}} (M_+ \otimes (M_+ \otimes M_-)) \otimes M_-[[\hbar]] \\ &\xrightarrow{\beta_{23}} (M_+ \otimes (M_- \otimes M_+)) \otimes M_-[[\hbar]] \\ &\xrightarrow[\sim]{\Phi_{\text{KZ}}} (M_+ \otimes M_-) \otimes (M_+ \otimes M_-)[[\hbar]] \end{aligned}$$

where  $\Phi_{\text{KZ}}$  denote appropriate compositions of associators.

2.6.4. *Afterword: Physics and the Casimir–KZ equations.*

2.6.5.

2.6.6. *The KZ–Casimir connection [AT24].* Let  $\mathfrak{h} \subseteq \mathfrak{g}$  denote the maximal torus. For every  $\mathfrak{g}$ -representations  $V_1, \dots, V_n$ , we have a connection on

$$\mathfrak{t}_{\text{reg}} \times \mathfrak{h}_{\text{reg}}$$

### 3. Nakajima quiver varieties

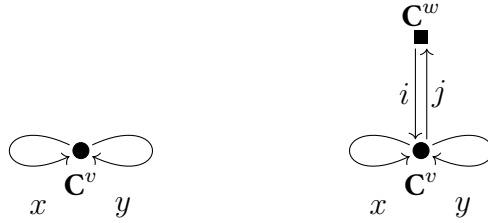
#### 3.1. Motivation: the ADHM construction.

3.1.1. Let us consider two types of coherent sheaves

- *Torsion*: zero-dimensional coherent sheaves  $\mathcal{A}$  on  $\mathbf{C}^2$ ,
- *Torsion free*: coherent sheaves  $\mathcal{B}$  on  $\mathbf{P}^2$  with a *framing*  $\mathcal{B}|_{\mathbf{P}^1} \simeq \mathcal{O}^{\oplus w}$  at infinity.

In both cases, let  $v$  denote the second Chern class and in the second case let  $w$  denote the rank. Both's first chern classes vanish.

We understand the first using the vector space  $\Gamma(\mathcal{A}) = \mathbf{C}^v$ , which is a module over the ring of functions  $\mathbf{C}[x, y]$ .



The data of  $\mathcal{B}$  is more complicated. By a spectral sequence argument [Na99],  $\mathcal{B}$  is the cohomology of a complex

$$\mathcal{O}_{\mathbf{P}^2}(-1) \otimes V \xrightarrow{\begin{pmatrix} z_0 j \\ z_0 B_1 - z_1 \\ z_0 B_2 - z_2 \end{pmatrix}} \mathcal{O}_{\mathbf{P}^2} \otimes (W \oplus V \oplus V) \xrightarrow{\begin{pmatrix} z_0 i & -(z_0 B_2 - z_2) & z_0 B_1 - z_1 \end{pmatrix}} \mathcal{O}_{\mathbf{P}^1}(1) \otimes V$$

for the vector spaces  $V = H^1(\mathbf{P}^2, \mathcal{B}(-2))$ , and  $W \oplus V \oplus V = H^1(\mathbf{P}^2, \mathcal{B} \otimes \mathcal{T}_{\mathbf{P}^2}^*)$ . The conditions on the maps are  $[x, y] + ij = 0$ , and the *stability* condition that  $\text{im}(i)$  generates  $\mathbf{C}^v$  as a  $\mathbf{C}\langle x, y \rangle$ -module.

The group  $\text{GL}_v$  (and  $\text{GL}_w$ ) acts on the space of such data, and we define

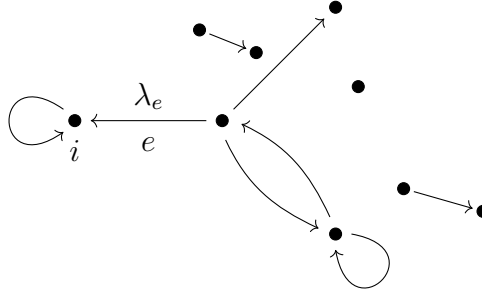
$$\mathcal{M}_{\mathcal{A}} = \{(x, y) : [x, y] = 0\} / \text{GL}_v, \quad \mathcal{M}_{\mathcal{B}} = \{(x, y, i, j) : [x, y] + ij = 0, \text{ stability}\} / \text{GL}_v.$$

Points in  $\mathcal{M}_{\mathcal{A}}$  can have nontrivial stabilisers so in general the quotient is a stack, but points in  $\mathcal{M}_{\mathcal{B}}$  have trivial stabilisers so the quotient is a variety, the first example of a *Nakajima quiver variety*.

#### 3.2. Quivers.

3.2.1. The above are examples of moduli of quiver representations, and of framed quiver representations, respectively.

### 3.2.2. Quivers.



A **quiver** is a directed graph  $Q = (Q_0, Q_1)$  whose sets  $Q_0, Q_1$  of vertices  $i$  and edges  $e$  are finite. It is **graded** with respect to a lattice  $N \simeq \mathbf{Z}^r$  if we have a weight function  $\text{wt}: Q_1 \rightarrow N$  assigning an element of the lattice  $\text{wt}(e) \in N$  to each edge. Write  $s(e), t(e)$  for the source and target of the edge.

The opposite quiver  $Q^{\text{op}}$  is formed by reversing the direction of all arrows, and negating all gradings:  $\text{wt}^{\text{op}}(e^*) = -\text{wt}(e)$  where  $e^*$  is the reversed version of  $e$ . We call a quiver **symmetric** if  $Q = Q^{\text{op}}$  and graded quiver **graded symmetric** if  $(Q, \text{wt}) \simeq (Q^{\text{op}}, \text{wt}^{\text{op}})$ .

3.2.3. *Representations.* A **representation** of  $Q$  is an assignment of a finite dimensional vector space  $V_i$  to each vertex and a linear map  $\rho_e$  to each edge

$$\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} V_i \xleftarrow{\rho_e} V_j \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \dots$$

This is equivalent to a finite dimensional left module over the *path algebra*  $\mathbf{C}Q$  of  $Q$ , whose basis is the set of paths in  $Q$  and multiplication is given by concatenation of paths. The *dimension* vector of a representation is the element  $d = (\dim V_i)_{i \in Q_0} \in \mathbf{N}^{Q_0}$ , writing  $d_i$  for the dimension at the  $i$ th vertex. We write  $\delta_i$  for the dimension vector which is 1 at vertex  $i$  and zero elsewhere.

The category  $\text{Rep}Q$  of representations of  $Q$  is an abelian category, and the rank of the derived Hom spaces

$$\chi(\rho, \rho') = \dim(\text{Hom}(\rho, \rho')) - \dim(\text{Ext}^1(\rho, \rho')),$$

is called the *Euler form*, which only depends on the dimension vectors  $d, d'$  of  $\rho, \rho'$ , and is

$$\chi(d, d') = \sum_i d_i d'_i - \sum_{e: i \rightarrow j} d_i d'_j.$$

3.2.4. *Potential.* A **potential** of  $Q$  is an element  $W \in \mathbf{C}Q/[\mathbf{C}Q, \mathbf{C}Q]$ .

A potential is given by a linear combination of cyclic words in  $Q$ , where two cyclic words are considered to be the same if one can be cyclically permuted to be the other. If  $W$  is a single cyclic word and  $e \in Q_1$ , then we define

$$\frac{\partial W}{\partial e} = \sum_{W=cec'} c'c$$

and we extend this definition linearly to general  $W$ . The *Jacobi algebra* associated to  $(Q, W)$  is defined to be

$$\text{Jac}(Q, W) := \mathbf{C}Q / \left\langle \frac{\partial W}{\partial e} : e \in Q_1 \right\rangle.$$

The direct sum and tensor product of representations are defined vertex-wise.

### 3.3. Moduli stacks of quiver representations.

3.3.1. The **moduli stack of representations** of  $Q$  of dimension  $d$  is the quotient stack

$$\mathcal{M}_{Q,d} := \text{Rep}_d(Q)/\text{GL}_d$$

of the following vector space by the following group:

$$\begin{aligned} \text{Rep}_d(Q) &:= \prod_{e:i \rightarrow j} \text{Hom}(\mathbf{C}^{d_i}, \mathbf{C}^{d_j}) \\ \text{GL}_d &:= \prod_{i \in Q_0} \text{GL}_{d_i} \end{aligned}$$

where the group  $\text{GL}_d$  acts on  $\text{Rep}_d(Q)$  by conjugation. The union over all dimension vectors is the moduli stack of objects of the abelian category  $\text{Rep}Q$  in the sense of [TV]. For instance, its  $\mathbf{C}$ -points are precisely the groupoid of finite dimensional representations of  $Q$ .

3.3.2. *Torus equivariance.* Let  $Q$  be a quiver graded by the lattice  $N$  of a torus  $T = \text{Hom}(N, \mathbf{G}_m)$ , with its grading denoted  $\text{wt} : Q_1 \rightarrow N$  as before. Thus we have an action of the torus on the representation space

$$T \times \text{Rep}_d(Q) \rightarrow \text{Rep}_d(Q), \quad t \cdot \rho_e = t^{\text{wt}(e)} \rho_e$$

where  $t^{\text{wt}(e)} \in \mathbf{G}_m$  is the image of an element of the torus under the character  $\lambda_e : T \rightarrow \mathbf{G}_m$  associated to  $\text{wt}(e) \in N$ , which acts by multiplication on the element of the vector space  $\rho_e \in \text{Hom}(\mathbf{C}^{d_i}, \mathbf{C}^{d_j})$ .

The homomorphism  $\lambda_e$  gives a morphism of stacks  $\lambda_e : \text{BT} \rightarrow \text{BG}_m$ . Taking pullback, this defines a cohomology class  $\mathbf{t}(e) \in \text{HH}^2(\text{BT}, \mathbf{Q})$  given by  $\mathbf{t}(e) = \lambda_e^*(u)$  where  $u$  is the generator of  $\text{HH}^2(\text{BG}_m, \mathbf{Q}) \simeq \mathbf{Q}[u]$ . Note that each weight  $\text{wt}(e)$  also defines a line bundle  $\mathcal{L}_e$  on  $\text{BT}$ . We will sometimes abuse notation and write

$$\text{wt}(e) = c_1(\mathcal{L}_e) = \mathbf{t}(e) \in \text{HH}^2(\text{BT}, \mathbf{Q}). \quad (8)$$

3.3.3. *Tautological bundles and  $T$ -equivariant moduli space.* We then define

$$\mathcal{M}_{Q,d}^T = \text{Rep}_d(Q)/(\text{GL}_d \times T).$$

The **tautological bundle**  $\mathcal{E}_{i,d}$  for vertex  $i$  is the pullback of the universal vector bundle along

$$\mathcal{M}_{Q,d}^T \rightarrow \text{BGL}_d \times \text{BT} \rightarrow \text{BGL}_{d_i} \times \text{BT} \rightarrow \text{BGL}_{d_i}.$$

Its fibre over a point corresponding to a representation  $\rho$  is the vector space  $V_i$ . The *tautological map* of vector bundles  $\rho_e : \mathcal{E}_i \rightarrow \mathcal{E}_j$  attached to any edge  $e$  is induced by the map

$$\mathcal{M}_{Q,d}^T \rightarrow \text{Hom}(\mathbf{C}^{d_i}, \mathbf{C}^{d_j})/\text{GL}_{d_i} \times \text{GL}_{d_j}.$$

Its fibre over a point is the linear map  $\rho_e : V_i \rightarrow V_j$ .

3.3.4. *Ext complex.* The **Ext complex** is the two-term complex of vector bundles on  $\mathcal{M}_{Q,d}^T \times_{\text{BT}} \mathcal{M}_{Q,d'}^T$ , given by

$$\text{Ext}_{d,d'} = \left( \bigoplus_i \mathcal{E}_{i,d}^\vee \boxtimes \mathcal{E}_{i,d'} \rightarrow \bigoplus_{e:i \rightarrow j} \mathcal{E}_{i,d}^\vee \boxtimes \mathcal{E}_{j,d'} \otimes \mathcal{L}_e \right). \quad (9)$$

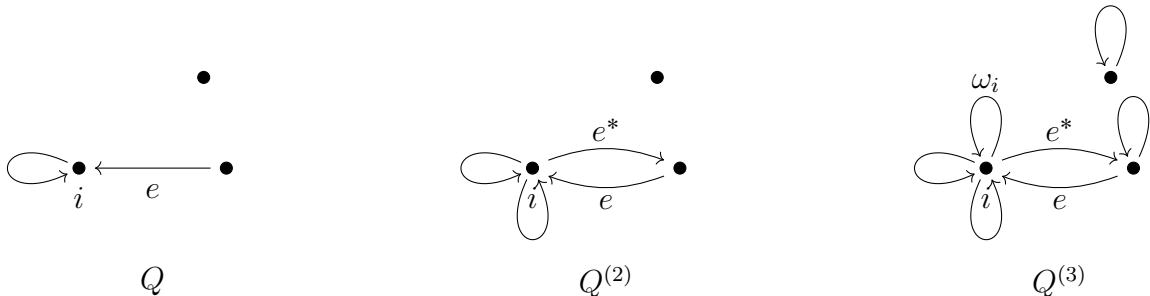
We denote the zeroth and first factors by  $\text{Ext}_0$  and  $\text{Ext}_1$ . The differential is given by viewing the left summands as the vector bundle of maps from  $\mathcal{E}_i \boxtimes 1$  to  $1 \boxtimes \mathcal{E}_i$ , and composing with the tautological maps  $\rho_e$  for each edge.

We will often have to dualise and also have to pull back the Ext-complex by the swap map  $\sigma : \mathcal{M}_{Q,d}^T \times_{\text{BT}} \mathcal{M}_{Q,d'}^T \rightarrow \mathcal{M}_{Q,d'}^T \times_{\text{BT}} \mathcal{M}_{Q,d}^T$ ; for this we note that  $\mathcal{L}_e^\vee$  corresponds to the character  $-\text{wt}(e)$ , and  $\sigma^* \mathcal{L}_e = \mathcal{L}_e$ .

3.3.5. *Relation to the potential.* The weight function induces a  $N$ -grading on  $\mathbf{C}Q$ , sending a path to the sum of the weights of its edges. We say the potential is *invariant* for the  $T$ -action, if  $W$  has weight zero under this grading:  $\text{wt}(W) = 0$ .

### 3.4. Doubled and tripled quivers.

3.4.1. *Preprojective and Jacobi algebras.* The *double*  $Q^{(2)}$  and *triple*  $Q^{(3)}$  of a quiver  $Q$  are quivers formed by adding a copy  $e^* : j \rightarrow i$  of each edge  $e : i \rightarrow j$  in the opposite direction, and then adding a loop  $\omega_i$  to each vertex, respectively.



We note that

$$\mathrm{Rep}(Q^{(2)}, v) \simeq T^*\mathrm{Rep}(Q, v)$$

is the cotangent bundle of the representation space of  $Q$ . The action of  $\mathrm{GL}_v$  on  $\mathrm{Rep}(Q^{(2)}, v)$  is Hamiltonian with respect to the natural symplectic structure, and the moment map is given by

$$\mu : \mathrm{Rep}(Q^{(2)}, v) \rightarrow \mathfrak{gl}_v^* \simeq \mathfrak{gl}_v, \quad (x, x^*) \mapsto \sum_e [x_e, x_{e^*}].$$

We define the moduli stack of *preprojective representations* as

$$\Pi_Q\text{-Mod} = \mu^{-1}(0)/\mathrm{GL}_v \simeq H^0 T^* \mathcal{M}_Q$$

where  $\Pi_Q = \mathbb{C}Q^{(2)}/\langle \sum_e [e, e^*] \rangle$ .

We identify this with the zeroth homology of the cotangent complex of  $T^* \mathcal{M}_Q$ :

**Lemma 3.4.2.** *If  $Y/\mathrm{G}$  is a quotient stack by a smooth scheme  $Y$ , its cotangent complex is the complex in degrees 0, 1*

$$T^*(Y/\mathrm{G}) \simeq (T^*Y \rightarrow \mathfrak{g}^* \otimes \mathcal{O}_Y)/\mathrm{G}.$$

*Proof.* Follows from the distinguished triangle of the cotangent complex of a composition, applied to  $Y/\mathrm{G} \rightarrow \mathrm{BG}$ . □

**Corollary 3.4.3.** *The above quotient has a Poisson structure.*

### 3.5. Calabi–Yau structures.

3.5.1. The *canonical tripled potential* on  $Q^{(3)}$  is

$$\widetilde{W} = \left( \sum_i \omega_i \right) \left( \sum_e [e, e^*] \right).$$

We have an isomorphism

$$\Pi_Q[\omega] \simeq \mathrm{Jac}(Q^{(3)}, \widetilde{W})$$

for the polynomial ring  $\Pi_Q[\omega]$ , sending  $\omega \mapsto \sum \omega_i$ .

3.5.2.

### 3.6. Framed quivers.

3.6.1. If  $Q$  is a quiver, the associated *framed quiver*  $Q^{\text{fr}}$  is formed by adding a new copy  $\hat{q}$  of each vertex  $q$  of  $Q$ , and add an edge  $q \rightarrow \hat{q}$ .



We write

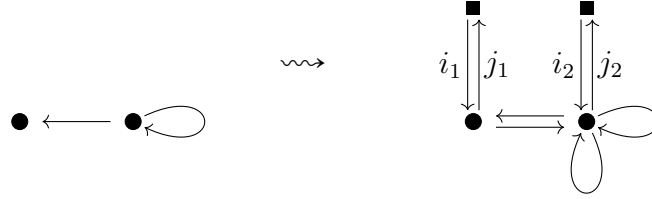
$$\text{Rep}(Q, v, w) = \text{Rep}(Q^{\text{fr}}, (v, w))$$

for the representation space with dimension  $v$  at the original vertices and dimension  $w$  at the new *framing* vertices.

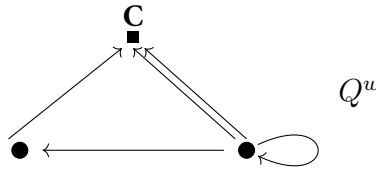
3.6.2. We will be considering the double of the framed quiver,

$$T^*\text{Rep}(Q, v, w) = \text{Rep}((Q^{\text{fr}})^{(2)}, (v, w))$$

where



3.6.3. *Remark: Crawley–Boevey quivers.* There is another point of view of framed quiver representations, where instead we consider a single node at infinity but  $w_i$  edges  $i \rightarrow \infty$ :



We then consider representations of dimension one at the infinite node, giving a dimension vector  $\hat{v} = (v, 1)$ . There is a  $\text{GL}_v$ -equivariant vector space isomorphism

$$\text{Rep}(Q^w, \hat{v}) \xrightarrow{\sim} \text{Rep}(Q, v, w) \tag{10}$$

given by taking the  $w_i$  many maps  $V_i \rightarrow \mathbf{C}$  and arranging them into a single map  $V_i \rightarrow \mathbf{C}^{w_i}$ .

### 3.7. Stability conditions.

3.7.1. A **(King) stability condition** on a quiver  $Q$  is a map

$$\theta : K_0(\text{Rep} Q) \simeq \mathbf{Z}^{Q_0} \rightarrow \mathbf{R},$$

viewed as an element  $\theta \in \mathbf{R}^{Q_0}$ . The *slope* of a representation is

$$\mu_\theta(V) = \frac{\theta \cdot \dim_{Q_0} V}{\dim V},$$

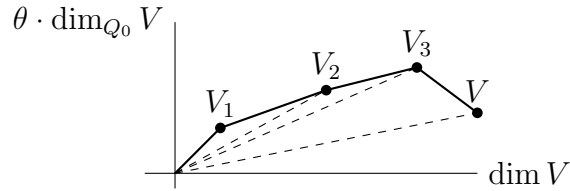
A representation is **(semi)stable** if it has no proper subrepresentations of greater (or equal) slope. When  $\theta = 0$  or  $\theta = \theta^+ = (1, \dots, 1)$ , the slope takes the constant value 0 or 1, and all representations are semistable, and are stable if and only if it has no proper subrepresentations.

**Theorem 3.7.2.** *This notion of stability agrees with GIT stability for  $X = \text{Rep}(Q, v)$  and*

$$\chi_\theta = \prod \det(-)^{-\theta_i} : \text{GL}_w \rightarrow \mathbf{G}_m.$$

**Corollary 3.7.3.** (King)  $\text{PGL}_v$  acts freely on the stable representations  $\text{Rep}_\theta^s(Q, v)$ .

**Proposition 3.7.4.** (Harder–Narasimhan, Jordan–Holder) *Every representation  $V$  has a unique filtration by subrepresentations  $0 = V_0 \subseteq V_1 \subseteq V_2 \subseteq \dots \subseteq V_n = V$*



such that each  $V_{k+1}/V_k$  is  $\theta$ -semistable. If  $W$  is semistable of slope  $\mu$ , there is a unique filtration  $0 = W_0 \subseteq W_1 \subseteq \dots \subseteq W_m = W$  such that each  $W_{k+1}/W_k$  is stable of slope  $\mu$ .

*Proof.* We pick a subrepresentation  $V_{n-1}$  of maximal slope, and among these maximal under inclusion, and repeat to get the Harder–Narasimhan filtration. Note that such a maximum exists as the slope only depends on the dimension vector

$$\mu_\theta : K_0(\text{Rep} Q) \simeq \mathbf{Z}^I \rightarrow \mathbf{Q}$$

and the set of dimension vectors  $0 \leq d \leq \dim_{Q_0} V$  which could arise from subrepresentations of  $V$  is finite.

Uniqueness follows as [continue](#)

For Jordan–Holder, it is enough to show that kernels and cokernels of maps between semistable representations of the same slope are also semistable of the same slope. [write details](#) □

3.7.5. *Framed stability conditions.* Given a stability condition  $\theta$  on  $Q$ , we form a stability condition on  $Q^w$  with

$$\hat{\theta} = \left( \theta, - \sum w_i \theta_i \right).$$

Given a King stability condition on  $Q$ , we say that a framed representation  $(V, W)$  is **(semi)stable** if the representation under (10) is  $\hat{\theta}$ -semistable.

For instance,

**Lemma 3.7.6.** *A representation  $\rho = (V, W)$  is  $\theta^+$ -semistable if and only if there is no proper subrepresentation  $V' \subseteq V$  contained in  $\ker j$ .*

*Proof.* Follows by applying Theorem 3.7.2 to the isomorphism (10). □

**Corollary 3.7.7.** (King) *For  $\theta = \theta^+$ , stability and semistability on  $\text{Rep}(Q, v, w)$  coincide.*

### 3.8. Quotients.

3.8.1. Let  $G$  act on a scheme  $X$  and let  $\chi : G \rightarrow \mathbf{G}_m$  be a character. The **GIT** and **stack** quotients are

$$X//_{\chi}G = \text{Proj} \left( \bigoplus_{n \geq 0} \Gamma(X, \mathcal{O}_X)^{G, \chi^n} \right), \quad X/G = \lim(G \times X \rightrightarrows X)$$

where  $\Gamma(X, \mathcal{O}_X)^{G, \chi^n}$  is the space of  $G$ -invariant sections of  $\mathcal{O}_X$  twisted by the character  $\chi^n$ .

3.8.2. *Example.* When  $\chi = 1$  is trivial this is just the affinisation  $X//G = \text{Spec} \Gamma(X, \mathcal{O}_X)^G$ . For any  $\chi$  there is a canonical map

$$\pi : X//_{\chi}G \rightarrow X//G$$

induced by the inclusion of invariant functions.

3.8.3. *Example.*  $\mathbf{C}^{n+1} \setminus 0$  are the (semi)stable points for the  $\mathbf{G}_m$ -action with weight one on  $\mathbf{C}^{n+1}$ , and the GIT quotient and stack quotients are both  $\mathbf{P}^n$ .

3.8.4. The  $\chi$ -**semistable points**  $X_{\chi}^{ss}$  are those given by the nonvanishing locus of any  $f \in \Gamma(X, \mathcal{O}_X)^{G, \chi^n}$  for some  $n > 0$ . It is **stable** if in addition the  $G$ -action on that locus is closed and that point has finite stabilisers.

There is always a map

$$X_{\chi}^{ss}/G \rightarrow X//_{\chi}G.$$

Let  $G$  be reductive. This map is an isomorphism if  $G$  acts freely on  $X_{\chi}^{ss}$ , or if  $X_{\chi}^s = X_{\chi}^{ss}$  and  $G$  acts with finite stabilisers on  $X_{\chi}^s$ .

**Theorem 3.8.5.**  $X_{\chi}^{ss} \rightarrow X//_{\chi}G$  is the universal (initial)  $G$ -invariant map from  $X_{\chi}^{ss}$  into a scheme.

**Theorem 3.8.6.**  $X \rightarrow X/G$  is the universal  $G$ -torsor  $P$  over a stack with a  $G$ -equivariant map to  $X$ .

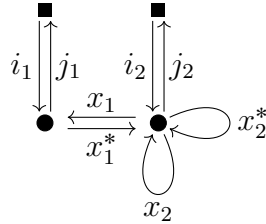
### 3.9. Quiver varieties.

**Definition 3.9.1.** Let  $Q$  be a quiver and  $\theta$  a King stability condition for  $Q$  (which we view as a King stability condition on  $(Q^f)^{(2)}$ ). The **Nakajima quiver variety** is

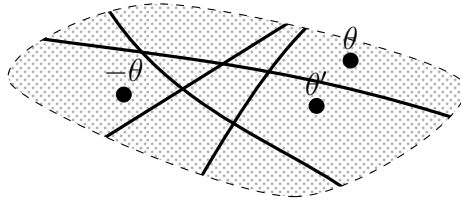
$$\mathcal{M}_\theta(v, w) = \mu^{-1}(0)^{\text{ss}} //_{\theta} \text{GL}_v \subseteq T^* \text{Rep}(Q, v, w) / \text{GL}_v.$$

Here, the moment map  $\mu$  is explicitly

$$\begin{aligned} \mu : \text{Hom}(Q^{(2)}, v) \times \text{Hom}(V, W) \times \text{Hom}(W, V) &\rightarrow \mathfrak{gl}_v^* \\ (x, x^*, i_k, j_k) &\mapsto \sum [x, x^*] + \sum i_k j_k. \end{aligned}$$

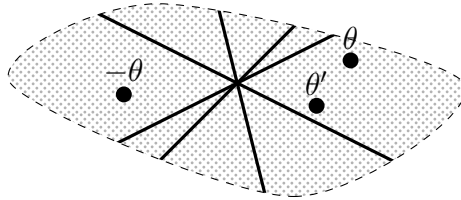


Thus we have a quotient stack  $\mu^{-1}(0)_{\theta}^{\text{ss}} / \text{GL}_v$  for each stability condition  $\theta$ , which is locally constant as  $\theta$  varies but can jump at certain walls where stability and semistability fail to coincide. Along these walls, the stack and GIT quotient can differ.



$$\text{Stab} = \text{Hom}(K_0(\text{Rep} Q), \mathbf{R})$$

By [Na96], the walls are of the form hyperplanes  $\theta \cdot \alpha_i = 0$  for finitely many dimension vectors  $\alpha_i \in \mathbf{R}^{Q_0}$ . This is because by [Na96] away from these walls there are no strictly semistable points, and the picture actually looks like



$$\text{Stab} = \text{Hom}(K_0(\text{Rep} Q), \mathbf{R})$$

The following is taken from [Gi09, Prop. 5.1.5]

**Lemma 3.9.2.** A representation  $\rho = (x, x^*, i, j)$  is  $\theta^+$ -semistable if and only if for any subrepresentation  $V'$  of  $Q^{(2)}$ , if

$$V' \subseteq \ker j \quad \Rightarrow \quad \theta(\dim_{Q_0} V') \leq 0,$$

$$V' \supseteq \text{im } i \quad \Rightarrow \quad \theta(\dim_{Q_0} V') \leq \theta(\dim_{Q_0} V).$$

When  $\theta = 0$  these conditions hold for all  $\theta$ . When  $\theta = \theta^+$ , these two conditions become

$$V' \subseteq \ker j \quad \Rightarrow \quad V' = 0,$$

and when  $\theta = -\theta^+$ , the second condition becomes

$$V' \supseteq \operatorname{im} i \quad \Rightarrow \quad V' = V.$$

3.9.3. Let us consider  $\theta = \theta^+$ .

Here we take the GIT quotient, but for  $\theta = \theta^+$  or whenever stability and semistability coincide (as for generic  $\theta$ ), this agrees with the stack quotient by Corollaries 3.7.3 and 3.7.7.

**Proposition 3.9.4.** *When  $\theta = \theta^+$ ,  $\mathcal{M}_\theta(v, w)$  is a smooth quasiprojective variety.*

*Proof.* We prove this for the stack quotient, which by Corollary 3.7.7 agrees with the GIT quotient.  $T^*\operatorname{Rep}(Q, v, w)/\operatorname{GL}_v$  being smooth, it is enough to show that the map  $\mu : T^*\operatorname{Rep}(Q, v, w) \rightarrow \mathfrak{gl}_v^*$  is regular at semistable points  $\rho$ . This follows from  $\operatorname{GL}_v$  acting freely on  $T^*\operatorname{Rep}(Q, v, w)^{\text{ss}}$ , which is a consequence of the stability condition.  $\square$

3.9.5. *Example.* For  $\theta = \theta^+$  and the quiver

$$\begin{array}{c} W \simeq \mathbf{C}^n \\ \blacksquare \\ \begin{array}{c} \downarrow i \\ \uparrow j \end{array} \\ \bullet \\ V \simeq \mathbf{C}^k \end{array}$$

The moment map condition becomes  $\mu = ij = 0$ , i.e.  $\ker j \supseteq \operatorname{im} i$ , and the stability condition for  $\theta_+$  is that  $i$  is surjective, so we have **maybe  $i, j$  need to be swapped**

$$0 \rightarrow K \rightarrow W \rightarrow V \rightarrow 0.$$

The inclusion  $K \hookrightarrow W$  defines a point in  $\operatorname{Gr}(n - k, n)$  and the map  $W \rightarrow W/K$  defines a point in the cotangent fibre, and so

$$\mathcal{M}_{\theta^+}(k, n) \simeq T^*\operatorname{Gr}(n - k, n).$$

3.9.6. *Example.* For  $\theta = 0$ , **or  $\theta^+$ ?**

$$\begin{array}{c} \mathbf{C}^w \\ \blacksquare \\ \begin{array}{c} \downarrow i \\ \uparrow j \end{array} \\ \bullet \\ \mathbf{C}^v \\ \begin{array}{cc} x & y \end{array} \end{array}$$

is stable if and only if the image of  $i$  generates  $\mathbf{C}^v$  as a  $\mathbf{C}\langle x, y \rangle$ -module. **not quite, work out in class** For instance,

$$\mathcal{M}(v, 1) \simeq \text{Hilb}_v(\mathbf{C}^2).$$

### 3.10. Cohomology of quiver varieties.

3.10.1. To begin with we notice that if we ignored the semistable condition,

$$\iota : \mathcal{M}_\theta(v, w) = \mu^{-1}(0)^{\text{ss}}/\text{GL}_v \hookrightarrow \mu^{-1}(0)/\text{GL}_v = \mathcal{M}(v, w),$$

the  $\text{GL}_w$ -equivariant cohomology of the right hand side is very easy to work out:

**Lemma 3.10.2.** *We have that*

$$H_{\text{GL}_w}^\bullet(\mathcal{M}(v, w)) \simeq H^\bullet(\text{BGL}_v) \otimes H^\bullet(\text{BGL}_w)$$

*is generated freely by the nonzero chern classes  $c_k(\mathcal{V}_i)$  and  $c_k(\mathcal{W}_i)$  of the tautological bundles.*

*Proof.*  $\mu^{-1}(0)$  is  $\text{GL}_v \times \text{GL}_w$ -equivariantly contractible. □

**Theorem 3.10.3.** [MN18] *Whenever  $\mathcal{M}_\theta(v, w)$  is smooth, the “Kirwan” pullback map*

$$\iota^* : H_{\text{GL}_w}^\bullet(\mathcal{M}(v, w)) \rightarrow H_{\text{GL}_w}^\bullet(\mathcal{M}_\theta(v, w))$$

*is surjective.*

### 3.11. The structure on quiver varieties.

3.11.1. We have the following Theorem:

**Theorem 3.11.2.** [MO12] *For generic  $\theta$ , the collection of spaces  $\mathcal{X}_w = \bigsqcup_v \mathcal{M}_\theta(v, w)$  satisfies the following properties:*

- (1) *There is an action of  $\mathbf{G} = \text{PGL}_w$ .*
- (2)  *$\mathcal{X}_w$  is a smooth symplectic variety, with a proper  $\mathbf{G}$ -equivariant map  $\pi : \mathcal{X}_w \rightarrow \mathcal{X}_w^0$  to an affine variety  $\mathcal{X}_w^0$ . The symplectic form  $\omega$  is an eigenvector for  $\mathbf{G}$  and fixed by the maximal torus  $T \subseteq \mathbf{G}$ .  $\mathcal{X}_w$  is formal as a  $T$ -variety.*
- (3) (Factorisation) *Considering the rank  $r$  subtorus induced by decomposition*

$$\mathbf{A} \subseteq \mathbf{T} \quad \Leftrightarrow \quad w = w_1 + \cdots + w_r,$$

*we have*

$$\mathcal{X}_w^{\mathbf{A}} \simeq \mathcal{X}_{w_1} \times \cdots \times \mathcal{X}_{w_r}. \tag{11}$$

*Proof.* We let  $\mathcal{X}_w^0 = \text{Spec} \mathcal{O}(\mathcal{X}_w)$  be the affinisation of  $\mathcal{X}_w$ , which carries a canonical map from  $\mathcal{X}_w$ .

equivalently as in MO it is a map to  $\mathcal{M}_{0,0}$

We prove factorisation (11) assuming that  $\mathfrak{s} = \mathfrak{ss}$ . There is an inclusion

$$\text{Rep}(Q, v_1, w_1) \times \cdots \times \text{Rep}(Q, v_r, w_r) \hookrightarrow \text{Rep}(Q, v, w) \quad (12)$$

for every decomposition  $v = v_1 + \cdots + v_r$  and  $w = w_1 + \cdots + w_r$ , intertwining the actions of  $\text{GL}_{v_1} \times \cdots \times \text{GL}_{v_r} \rightarrow \text{GL}_v$ , and hence inducing a map on quiver varieties

$$\iota : \mathcal{M}_\theta(v_1, w_1) \times \cdots \times \mathcal{M}_\theta(v_r, w_r) \hookrightarrow \mathcal{M}_\theta(v, w).$$

We now consider the  $\mathbf{A}$ -fixed points of the right hand side. Recall from [Ro05] that the category of stacks forms a 2-category, and, extending the notion of fixed points of a variety, the  $\mathbf{A}$ -fixed points for a stack  $\mathcal{X}$  with  $\mathbf{A}$ -action is defined by

$$\text{Hom}_{\mathbf{A}\text{-Stk}}(\mathcal{Y}, \mathcal{X}) = \text{Hom}_{\text{Stk}}(\mathcal{Y}, \mathcal{X}^{\mathbf{A}}).$$

In particular, if  $\mathbf{A}$  acts trivially on  $\text{BG}$  then [reference](#)

$$(\text{BG})^{\mathbf{A}} = \bigsqcup_{\phi : \mathbf{A} \rightarrow \mathbf{G}} \text{BC}_{\phi(\mathbf{A})}(\mathbf{G})$$

is the union over homomorphisms  $\phi : \mathbf{A} \rightarrow \mathbf{G}$  of the classifying stacks of the centralisers of the images of  $\phi$ . For instance, if  $\mathbf{G} = \text{GL}_v$  and  $\mathbf{A}$  is as above then

$$C_{\mathbf{A}}(\mathbf{G}) = \text{GL}_{v_1} \times \cdots \times \text{GL}_{v_r}.$$

Likewise, we have for a quotient stack that

$$(Y/\mathbf{G})^{\mathbf{A}} = \bigsqcup_{\phi : \mathbf{A} \rightarrow \mathbf{G}} Y^{\phi(\mathbf{A})}/C_{\mathbf{A}}(\mathbf{G}).$$

Given the homomorphism  $\phi : \mathbf{A} \rightarrow \text{GL}_v$  induced by the decomposition  $v = v_1 + \cdots + v_r$ , we have that  $\text{Rep}(Q, v, w)^{\phi(\mathbf{A})}$  is precisely the image of the inclusion (12). This proves the factorisation result.

□

#### 4. Stable envelopes and Yangians

So far we have studied quantum groups  $U_q(\mathfrak{g})$  attached to *finite* dimensional Lie algebras like  $\mathfrak{sl}_n$ . From the physics point of view its categories of modules these are expected to be categories of line operators of a three dimensional *topological* QFT  $\mathcal{T}$  called 3d Chern–Simons:

$$\mathrm{Rep}^{\mathrm{f.d.}} U_q(\mathfrak{g}) \simeq \mathcal{T}(S^1)$$

where  $S^1 \subseteq \mathbf{R}^2$ .

In what follows we will study quantum groups of the best behaved infinite dimensional Lie algebras: *affine* Lie algebras  $\widehat{\mathfrak{g}}$ , which arise as extensions of the Lie algebra  $L\mathfrak{g} = \mathfrak{g}[t^{\pm}]$  of loops into  $\mathfrak{g}$ :

$$0 \rightarrow \mathbf{C} \cdot c \rightarrow \widehat{\mathfrak{g}} \rightarrow L\mathfrak{g} \rightarrow 0$$

in which the correction to the Lie bracket in  $L\mathfrak{g}$  is given by the *residue pairing*

$$\mathrm{Res} : L\mathfrak{g} \otimes L\mathfrak{g} \rightarrow \mathbf{C} \cdot c, \quad (x(t), y(t)) \mapsto c \cdot \mathrm{Res}_{t=0} \kappa(x(t), y(t)) dt.$$

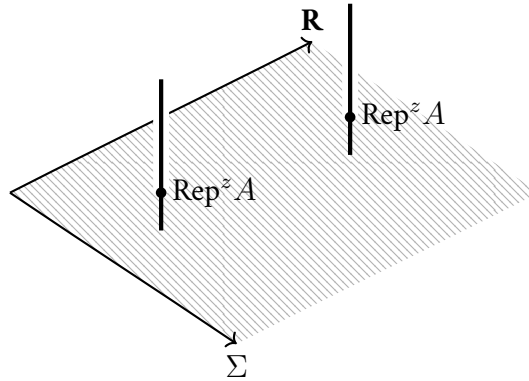
The relevant physics object is a four dimensional *holomorphic-topological* QFT  $\mathcal{T}$  called 4d Chern–Simons [CWY17]. Now the local structure of spacetime is not  $\mathbf{R}^3$  but  $\mathbf{R}^2 \times \Sigma$ , where  $\Sigma$  is a Riemann surface with nonvanishing one form. There are three options,

$$\Sigma \in \{\mathbf{C}, \mathbf{C}^{\times}, E\}$$

where  $E$  is an elliptic curve. Therefore we get three different types of algebras, with line operator categories expected to be given by a certain category of modules

$$\mathrm{Rep}^z Y_h(\widehat{\mathfrak{g}}) \simeq \mathcal{T}(S_z^2), \quad \mathrm{Rep}^z U_q(\widehat{\mathfrak{g}}) \simeq \mathcal{T}(S_z^2), \quad \mathrm{Rep}^z E_{h,\tau}(\widehat{\mathfrak{g}}) \simeq \mathcal{T}(S_z^2)$$

where  $S_z^2 \subseteq \mathbf{R} \times \Sigma$  is a small sphere around  $(0, z)$ .



As far as I am aware there is not an exact analogue of Drinfeld’s big Theorem for these quantum groups—this is a question you should think about if interested. Instead, we use a different construction of quantum groups, the *RTT construction*.

| 3d QFT                         | 4d QFT                         |                                       |                            | 5d       |
|--------------------------------|--------------------------------|---------------------------------------|----------------------------|----------|
| $\mathbf{R} \times \mathbf{R}$ | $\mathbf{C} \times \mathbf{R}$ | $\mathbf{C}^\times \times \mathbf{R}$ | $E_\tau \times \mathbf{R}$ | $\cdots$ |
| $U_h(\mathfrak{g})$            | $Y_h(\mathfrak{g})$            | $U_h(\widehat{\mathfrak{g}})$         | $E_{h,\tau}(\mathfrak{g})$ | $\cdots$ |
| $\text{Fun}(-; \mathbf{F}_q)$  | $H^\bullet(-)$                 | $K(-)$                                | $\text{Ell}(-)$            | $\cdots$ |

#### 4.1. RTT construction.

4.1.1. Let  $V$  be a finite dimensional vector space and  $R \in \text{GL}_{\mathbf{C}}(V \otimes V)$  be an element satisfying the *Yang–Baxter equation*

$$R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$$

in  $\text{GL}_{\mathbf{C}}(V \otimes V \otimes V)$ , where the subscripts indicate which factors of  $V \otimes V \otimes V$  the operator  $R$  acts on.

**Definition 4.1.2.** Let  $A(R)$  be the algebra with generators  $t_{ij}$ , where  $1 \leq i, j \leq \dim V$ , and  $(\dim V)^2$  many relations given by the RTT relation

$$R_{12}T_1T_2 = T_2T_1R_{12}$$

inside  $\mathfrak{gl}_{\mathbf{C}[t_{ij}]}(V \otimes V[t_{ij}])$ , where  $T$  is the matrix with entries  $t_{ij}$ .

**Lemma 4.1.3.**  $A(R)$  is a Hopf algebra with coproduct

$$\Delta(t_{ij}) = \sum_{k=1}^{\dim V} t_{ik} \otimes t_{kj}.$$

There is a filtration on  $A(R)$  by degree.

**Definition 4.1.4.** Let  $U(R) \subseteq A(R)^\vee$  be the subalgebra generated by the coefficients  $\ell_{ij}^\pm$  of  $L^\pm$ , with action on degree  $k$  elements as

$$L^+(T_1 \cdots T_k) = R_1^+ \cdots R_k^+,$$

$$L^-(T_1 \cdots T_k) = R_1^- \cdots R_k^-,$$

where  $R_i^+ = \sigma_{0i} R_{0i} \sigma_{0i}$  and  $R_i^- = R_{0i}^{-1}$ .

4.1.5. *Example.* When  $V = \mathbf{C}^2$  and

$$R = \left( \begin{array}{c|c} q & \\ \hline 1 & \\ q - q^{-1} & 1 \end{array} \right)$$

the RTT relations reduce to

$$t_{11}t_{12} = qt_{12}t_{11}$$

$$t_{11}t_{21} = qt_{21}t_{11}$$

$$t_{12}t_{21} = t_{21}t_{12}$$

$$t_{12}t_{22} = qt_{22}t_{12}$$

$$t_{21}t_{22} = qt_{22}t_{21}$$

$$t_{11}t_{22} - t_{22}t_{11} = (q - q^{-1})t_{12}t_{21}$$

we have that  $A(R) \simeq U_q(\mathfrak{sl}_2)$ . Moreover,

$$L^+ = \sqrt{q} \begin{pmatrix} e^{-\hbar H/2} & \hbar X \\ & e^{\hbar H/2} \end{pmatrix}, \quad L^- = \frac{1}{\sqrt{q}} \begin{pmatrix} e^{\hbar H/2} & \\ \hbar Y & e^{-\hbar H/2} \end{pmatrix}$$

where we have that

$$\begin{aligned} e^{\pm \hbar H/2} X &= q^{\pm 1} X e^{\pm \hbar H/2}, & e^{\pm \hbar H/2} Y &= q^{\mp 1} Y e^{\pm \hbar H/2}, \\ XY - YX &= \frac{q - q^{-1}}{\hbar^2} (e^{\hbar H/2} - e^{-\hbar H/2}). \end{aligned}$$

4.1.6. *Example.* More generally,

**Theorem 4.1.7.** *When*

$$R = \sum_{i \neq j} e_{ii} \otimes e_{jj} + q \sum_i e_{ii} \otimes e_{ii} + (q - q^{-1}) \sum_{i < j} e_{ij} \otimes e_{ji}$$

we have  $U(R) \simeq U_q(\mathfrak{sl}_n)$ .

## 4.2. What type of thing should the RTT construction be?

4.2.1. Let's say we have a braided monoidal category with fibre functor  $\mathcal{C} = A\text{-Mod}$ . As we have seen,  $A$  has to be a quasitriangular bialgebra with quasitriangular element  $R \in A \otimes A$ , which we can reconstruct  $A = \text{End} F$  as the algebra of endomorphisms of the forgetful functor  $F : \mathcal{C} \rightarrow \text{Vect}_{\mathbb{C}}$ .

However, imagine we were in the following more restrictive setting:

- We did not know any information about  $\mathcal{C}$  itself,
- We only have a collection  $V_i = F(c_i)$  of vector spaces, closed under  $\otimes$  and including  $\mathbf{C} = V_0$ , and linear maps  $R_{V_i V_j} : V_i \otimes V_j \rightarrow V_i \otimes V_j$  satisfying the Yang–Baxter equation,

$$R_{12} R_{13} R_{23} = R_{23} R_{13} R_{12}$$

as endomorphisms of  $V_i \otimes V_j \otimes V_k$  for all  $i, j, k$ , and  $R_{V_0 V_i} = R_{V_i V_0} = \text{id}_{V_i}$  for all  $i$ .

**Question 4.2.2.** How could we reconstruct  $(\mathcal{C}, F)$  from this data?

To begin with, we should have distinct objects  $c_i$ , so as a first approximation we can take  $\mathcal{C}_0 = \bigsqcup_i \text{Vect} \cdot c_i$ , which has no morphisms except multiples of the identity. Which morphisms should we add? The  $A$ -linear ones, which should include at least  $R_{V_i, V_j}$ .

At this point it becomes easier to reconstruct the algebra  $A$  instead. We know that

$$R_{V_i, V_j} : V_i \otimes V_j \rightarrow V_i \otimes V_j$$

should be action by an element in  $A \otimes A$ :

$$R_{V_i, V_j} = \sum \rho_{V_i}(a^{(1)}) \otimes \rho_{V_j}(a^{(2)})$$

and we can take the trace of the first (or second) factor to get the action of a single element in  $A$  on  $V_j$ . More generally, we can precompose with the action of any endomorphism  $m_i : V_i \rightarrow V_i$  to get an element

$$\alpha_{m_i, j} = \sum \text{tr}(m_i \cdot \rho_{V_i}(a^{(1)})) \cdot \rho_{V_j}(a^{(2)}) : V_j \rightarrow V_j.$$

We call this a **coefficient** of the  $R$ -matrix.

**Theorem 4.2.3.** *There is a (universal?) braided monoidal category with fibre functor  $(\mathcal{C}, F)$  answering Question 4.2.2: the module category of the algebra*

$$\mathbf{Y} = \langle \alpha_{m_j} \rangle \subseteq \prod_i \text{End}(V_i)$$

generated by the coefficients  $\alpha_{m_j}$  of the  $R$ -matrix with respect to  $m_j : V_j \rightarrow V_j$ , which as an endomorphism in  $\text{End}(V_i)$  is given by

$$\alpha_{m_j}|_{V_i} = \text{tr}_{V_j} [(m_j \otimes \text{id}) R_{V_j, V_i}] : V_i \rightarrow V_i.$$

*Proof Sketch.* By the explanation above, any algebra  $\mathbf{Y}'$  with the above properties □

### 4.3. Cohomology, K-theory and elliptic cohomology.

4.3.1. Equivariant cohomology, K-theory and elliptic cohomology. Where the equivariant parameters  $a$  live.

A **cohomology theory** is (define it).

In (special case), the cohomology theory arises from a non-formal algebraic group (justify). These are cohomology,  $K$ -theory, or Grojnowski's equivariant elliptic cohomology (justify this part), by:

**Theorem 4.3.2.** *Let  $\mathbf{G}$  be a one-dimensional algebraic group over  $\mathbf{C}$ . Then*

$$\mathbf{G} \in \{\mathbf{C}, \mathbf{C}^\times, E\}$$

where  $E$  is an elliptic curve.

*Proof.* (Maybe Serre) □

**Theorem 4.3.3.** (Belavin–Drinfeld)

### 4.4. Spectral parameter RTT construction.

#### 4.4.1. Spectral parameter RTT construction: cohomology, K-theory and elliptic cohomology.

We will state it for cohomology, where  $H^*(\mathbf{BG}_m) = k[a]$ , however it should be clear that the same construction works for  $K$ -theory and elliptic cohomology.

**Theorem 4.4.2.** *Given a collection of free  $k(a)$ -modules  $V_i(a)$  and  $k[a_1, a_2]$ -linear maps*

$$R_{ij}(a_1, a_2) = R_{ij}(a_1 - a_2) : V_i(a_1) \otimes V_j(a_2) \rightarrow V_i(a_1) \otimes V_j(a_2)$$

*with  $R_{ij}(\infty) = 1$ , and satisfying the spectral Yang–Baxter equation*

$$R_{12}(a_1, a_2)R_{13}(a_1, a_3)R_{23}(a_2, a_3) = R_{23}(a_2, a_3)R_{13}(a_1, a_3)R_{12}(a_1, a_2)$$

*then*

$$\mathbf{Y} = \langle \alpha_{m_{i_j}}(n) \rangle \subseteq \prod_{i_1, \dots, i_k} \text{End}(V_{i_1}(a_1) \otimes \dots \otimes V_{i_k}(a_k))$$

*given by (residue of bla/ p30 [OS16]) is an associative bialgebra over  $k[a]$  with a natural action on  $V_i(a)$ , and the  $R_{ij}(a) : V_i \otimes V_j \xrightarrow{\sim} V_j \otimes V_i(a)$  are intertwiners of  $\mathbf{Y}$ -modules.*

*Moreover,  $\mathbf{Y}$  is the universal such algebra.*

#### 4.4.3. Sketch-define meromorphic braided monoidal categories.

#### 4.5. Recollections on Borel–Moore homology.

4.5.1. Reminder about the six functors, cohomology, Borel–Moore homology, compactly supported cohomology, and homology. Identifications if  $X$  is smooth.

Virtual classes.

4.5.2. Localisation sequence and regular closed embeddings (locally  $A/f_1 \cdots f_r$  where  $f_{k+1}$  is not a zero divisor in  $A/f_1 \cdots f_k$ ). Normal bundles and Euler classes for regular closed embeddings.

4.5.3. Proper pushforward on BM homology. Exceptional pullback for regular closed embeddings, maps between smooths, and open embeddings.

#### 4.6. Recollections on symplectic geometry.

4.6.1. Recall that a manifold or smooth scheme  $X$  is *symplectic* if there is a two-form  $\omega \in \Omega^2(X)$  which is closed and nondegenerate:

$$d\omega = 0, \quad \Omega_X^n = \mathcal{O} \cdot \wedge^n \omega.$$

A subvariety  $\iota : Z \hookrightarrow X$  is called *isotropic* if  $\iota^* \omega = 0$ , and *Lagrangian* if in addition  $\dim Z = \frac{1}{2} \dim X$ .

Assume that  $\omega$  is fixed by the action of a torus  $\mathbb{T}$ , and  $\mathbf{G}_{m, \hbar} \subseteq \mathbb{T}$  is multiples of the identity. Given a component  $Z \subseteq X^\hbar$  we have

$$N_Z = N^- \oplus N^+$$

and the symplectic form  $\omega$  identifies  $(N^+)^{\vee} \simeq N^- \otimes \hbar$ . As  $\mathbf{G}_{m,\hbar}$  commutes with  $\mathbf{A}$ , the class

$$\epsilon^2 = (-1)^{\text{codim} Z/2} e(N_Z)|_{H_{\mathbf{A}}^{\bullet}(\text{pt})} = \prod \alpha_i^2$$

is a perfect square for any point  $\text{pt} \in Z$ . A *polarisation* is a choice of square root, and the sign  $\pm e(N_Z)$  agrees with the polarisation if it restricts to  $\epsilon$  on  $H_{\mathbf{A}}^{\bullet}(\text{pt})$ .

An *isotropic cycle* (sloppily referred to as a Lagrangian cycle in [MO12]) inside a symplectic manifold  $(X, \omega)$  with an action of torus  $\mathbf{T}$  is a homology class

$$Z = \sum_i n_i [Z_i] \in H_{\mathbf{T}}^{\bullet}(X).$$

where  $Z_i \subseteq X$  are  $\mathbf{T}$ -invariant subvarieties with  $Z_i^{sm}|_{\omega} = 0$ , with  $\dim Z_i = \frac{1}{2} \dim X$  and  $n_i \in \mathbf{Z}$ .

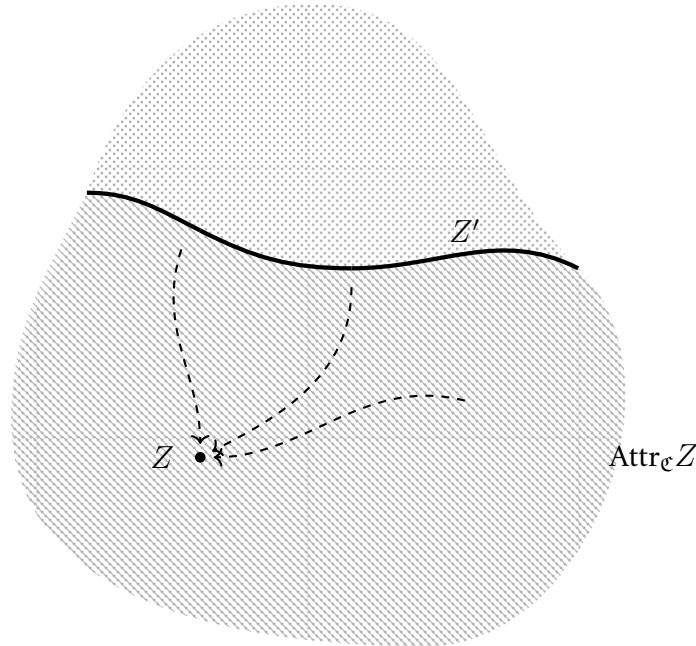
A homology class is *supported* on a  $\mathbf{T}$ -invariant closed subvariety  $\iota : Z \hookrightarrow X$  if it is in the image of the pushforward map

$$\iota_* : H_{\mathbf{T}}^{\bullet}(Z) \rightarrow H_{\mathbf{T}}^{\bullet}(X).$$

4.6.2. If  $X$  is a variety with an action by torus  $\mathbf{T}$  and  $\mathbf{A} \subseteq \mathbf{T}$ , then

$$\text{gr}H_{\mathbf{T}}^{\bullet}(X) \simeq H_{\mathbf{T}/\mathbf{A}}^{\bullet}(X) \otimes H_{\mathbf{A}}^{\bullet}(\text{pt}).$$

Thus, we define the **A-degree**  $d \in \mathbf{N}^r$  of an element  $f(a) \in H_{\mathbf{T}}^{\bullet}(X)$  its degree as a polynomial in  $H_{\mathbf{A}}^{\bullet}(\text{pt}) \simeq \mathbf{C}[a_1, \dots, a_r]$ .



For  $Z \subseteq X$  an  $\mathbf{A}$ -invariant subvariety of  $X$ , and  $\sigma : \mathbf{C}^{\times} \rightarrow \mathbf{A}$  a cocharacter then we have the **attracting set**

$$\text{Attr}_{\sigma} Z = \{x \in X : \lim_{t \rightarrow 0} \sigma(t) \cdot x \in Z\}.$$

Note that the limit, if it exists, will always lie in  $Z \cap X^A$ . This defines a partial order on the components of  $X^A$  by setting  $Z \geq Z'$  if  $Z'$  is upstream of  $Z$ , i.e. if  $Z \cap \text{Attr}_\sigma Z' \neq \emptyset$ .

The **full attracting set** is

$$\text{Attr}_\sigma^f Z = \overline{\text{Attr}_\sigma \text{Attr}_\sigma \cdots \text{Attr}_\sigma(Z)},$$

i.e. the union of attracting sets of all components  $Z' \subseteq X^A$  upstream of  $Z$ . The above iteration terminates for Noetherian  $X$ .

**Lemma 4.6.3.** *For  $X$  as below, if  $Z \subseteq X^A$  is a component of the fixed locus, we have that*

$$Z \xrightarrow{\iota_0} \text{Attr}_\sigma Z \xrightarrow{j} \text{Attr}_\sigma^f Z \xrightarrow{\iota_1} X$$

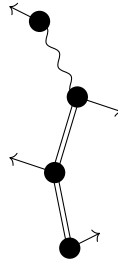
where  $\iota_0$  is a regular closed immersion,  $j$  is open and  $\iota_1$  is closed. If we split

$$N_{Z/X} = N_{Z/X}^+ \oplus N_{Z/X}^-$$

into the attracting, fixed and repelling parts, then the normal bundle of  $\iota_0$  is  $N_{Z/X}^-$ .

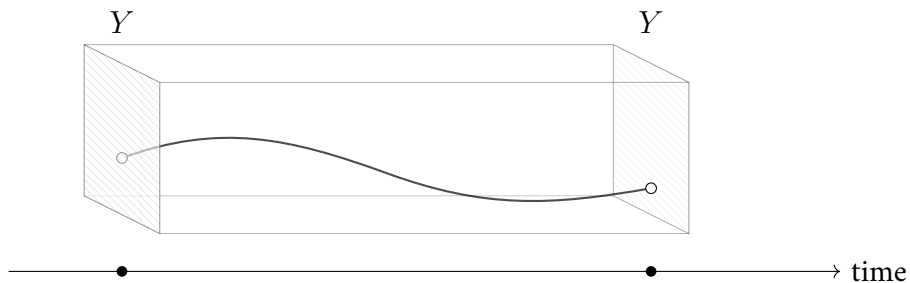
#### 4.7. Motivation II: quantisation of Lagrangians.

4.7.1. Given a system of balls attached with various rods and springs



all information of the system is captured by a point in the symplectic submanifold  $Y \subseteq T^*(\mathbf{R}^3)^4$  parametrising positions and velocities, with conditions forced by the rods.

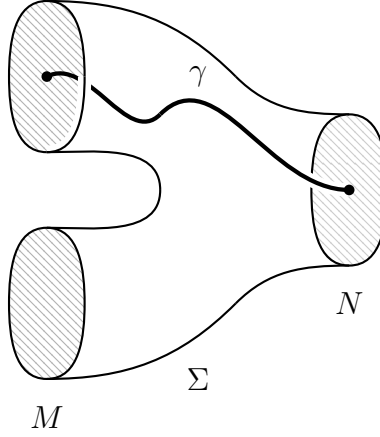
By Newton's law for springs and rods, knowing the positions and velocities of the balls lets you know the positions and velocities after an arbitrary time  $t$ :



We can package this information into the set of *allowed paths*  $Y \simeq \text{Paths}_{t_1, t_2} \subseteq \text{Maps}([t_1, t_2], Y)$  for every time interval: these are paths satisfying certain differential equations. Asking a path what its start- and endpoint is gives maps

$$\begin{array}{ccc} & \text{Paths}_{t_1, t_2} & \\ \swarrow & & \searrow \\ Y & & Y \end{array}$$

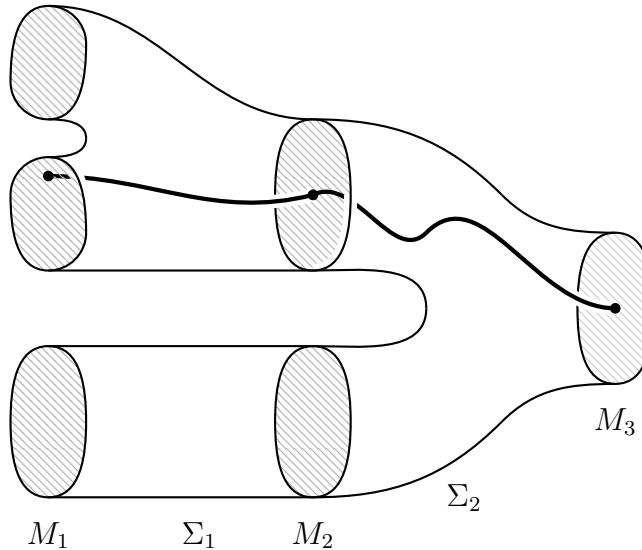
defining a *Lagrangian* submanifold of  $Y \times Y$ .



Likewise, given a more complicated cobordism from symplectic manifold  $X$  to  $Y$ , we get a Lagrangian submanifold of allowed paths  $\gamma$ :

$$\begin{array}{ccc} & \Sigma & \\ M \nearrow & & \nwarrow N \end{array} \xrightarrow{\sim} \begin{array}{ccc} & \text{Paths}_{\Sigma} & \\ X \xleftarrow{q} & & \xrightarrow{p} Y \end{array}$$

Moreover, given two composable cobordisms,



we have

$$\begin{array}{c}
 \text{Paths}_{\Sigma_1 \cdot \Sigma_2} \simeq \text{Paths}_{\Sigma_1} \times_{X_2} \text{Paths}_{\Sigma_2} \\
 \swarrow \quad \searrow \\
 \text{Paths}_{\Sigma_1} \quad \text{Paths}_{\Sigma_2} \\
 \swarrow \quad \searrow \quad \swarrow \quad \searrow \\
 X_1 \quad X_2 \quad X_3
 \end{array} \tag{13}$$

We call such a functor  $\mathcal{Z} : \text{Cob} \rightarrow \text{Lag}$  into the category of symplectic manifolds and Lagrangian correspondences a **classical QFT**, where now  $\text{Paths}_{\Sigma}$  is an arbitrary Lagrangian in  $\mathcal{Z}(M_1) \times \mathcal{Z}(M_2)$  not necessarily having to do with solutions to differential equations.

A *quantisation* of a classical QFT is loosely speaking an assignment of a vector space to each relevant symplectic manifold  $X$ —for instance  $\mathcal{O}(X)$  or  $H_{\bullet}(X)$  or  $H^{\bullet}(X)$ —and a linear map to each cobordism  $\Sigma$ . A natural candidate is

$$\mathcal{Z}_{\Sigma} = p_* q^* : H^{\bullet}(X) \rightarrow H^{\bullet}(Y)$$

given by taking a differential form  $\alpha$  on  $X$ , pulling it back to  $\text{Paths}_{\Sigma}$  and then integrating along the fibres of  $p$  to get a differential form on  $Y$ . In other words,

$$\alpha \mapsto \pi_{2*}(\pi_1^* \alpha \cap [\text{Paths}_{\Sigma}]) \tag{14}$$

where we intersect with the class of  $\text{Paths}_{\Sigma} \subseteq X \times Y$  and  $\pi_1, \pi_2$  are the two projection maps.

**4.7.2. Stable envelopes as a quantisation.** We are now going to execute this kind of process, where  $X = X^A, Y = X$ , and our main object  $[\text{Paths}_{\Sigma}] \in H_{\bullet}(X \times Y)$  will be the stable envelope, (13) will correspond to the hexagon relation and  $\mathcal{Z}_{\Sigma} \cdot \mathcal{Z}_{\Sigma^{-1}}$  will be the  $R$ -matrix defining the Yangian.

We will have a cobordism  $\Sigma = \mathbf{P}^1$  from marked points  $M = \text{pt} \sqcup \text{pt}$  to  $N = \text{pt}$ , as we will see later when we study quasimaps.

#### 4.8. Definition of the Yangian from stable envelopes.

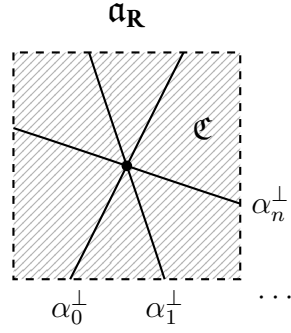
4.8.1. Given a collection of spaces  $X = X_w$  satisfying the following:

- $(X, \omega)$  is nonsingular variety and symplectic, with an action of a reductive group  $G = \text{PGL}(n)$ ,
- there is a  $G$ -equivariant proper map  $\pi : X \rightarrow X_0$  to an affine variety  $X_0$ ,
- given a maximal torus  $T$  and  $A \subseteq T \subseteq G$  such that  $\omega$  is a  $G$ -eigenvector fixed by  $A$ ,
- $X$  is formal as a  $T$ -variety, and  $H_T^{\bullet}(X)$  is free as a  $H_T^{\bullet}(\text{pt})$ -module.

We also have a splitting

$$N_{X^A/X} = N_{X^A/X}^+ \oplus N_{X^A/X}^-$$

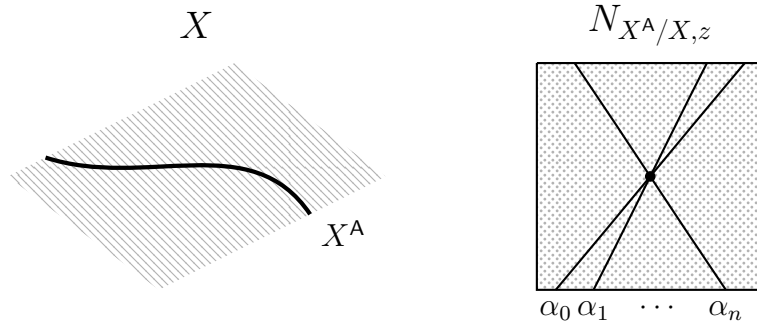
into positive and negative part. We also get a decomposition of  $\mathfrak{a}_R$  into **chambers** given by the roots appearing in  $N_{X^A/X}$ , and we fix a chamber  $\mathfrak{C}$ .



Note that the (full) attracting set only depends in the chamber, and so we write  $\text{Attr}_{\mathfrak{C}}^{(f)} Z$ .

We further assume that for each  $A$ ,

- (1)  $X^A \simeq \sum_{w=w_1+\dots+w_r} X_{w_1} \times \dots \times X_{w_r}$  if  $A$  has rank  $r$ ,
- (2) consider the  $A$ -weights of the normal bundle to the fixed points



these weights  $\chi \in \mathfrak{a}_Q^*$  are the roots  $\alpha_{ij}$  of  $G = \text{PGL}(n)$ .

In [MO12] it's noted that this definition is a bit ad-hoc.

**Theorem 4.8.2.** (Stable envelopes) *There is a Lagrangian cycle for each chamber  $\mathfrak{C}$*

$$\begin{array}{ccc} & [\text{Stab}_{\mathfrak{C}}] & \\ \swarrow & & \searrow \\ X^A & & X \end{array}$$

unique such that for any component  $Z \subseteq X^A$ ,

- (1) The retraction to  $X^A \times Z$ :

$$\begin{array}{ccc} & [\text{Stab}_{\mathfrak{C}}]|_{Z \times X^A} & \\ \swarrow & & \searrow \\ Z & & X \end{array}$$

is supported on  $Z \times \text{Attr}_{\mathfrak{C}}^f Z$ ,

- (2) the restriction to  $Z \times Z$  is

$$\begin{array}{ccc} & [\text{Stab}_{\mathfrak{C}}]|_{Z \times Z} & \\ \swarrow & & \searrow \\ Z & & Z \end{array}$$

is  $\in e(N_{Z, \mathfrak{C}}^-) \cap \Delta$ ,

(3) for any downstream  $Z' < Z$ , the restriction

$$\begin{array}{ccc} & [\text{Stab}_{\mathfrak{C}}]|_{Z' \times Z} & \\ \swarrow & & \searrow \\ Z' & & Z \end{array}$$

has A-degree less than  $\frac{1}{2} \text{codim} Z'$ .

(write down K theoretic version)

There is a map

$$\text{Stab}_{\mathfrak{C}} : H_{\top}^*(X^A) \rightarrow H_{\top}^*(X),$$

unique such that for any component  $Z \subseteq X^A$  and  $\gamma \in H_{\top}^*(Z)$ ,

- (1)  $\text{Stab}_{\mathfrak{C}}(\gamma)$  is supported on  $\text{Attr}_{\mathfrak{C}}^f Z$ ,
- (2)  $\text{Stab}_{\mathfrak{C}}(\gamma)|_Z = \pm e(N_{Z, \mathfrak{C}}^-) \cup \gamma$ ,
- (3)  $\deg_A \text{Stab}_{\mathfrak{C}}(\gamma)|_{Z'} < \frac{1}{2} \text{codim} Z'$  for all downstream  $Z' < Z$ .

From uniqueness of stable envelopes, it follows that

**Corollary 4.8.3.** *If  $A' \subseteq A \subseteq \top$ , is induced by a face  $\mathfrak{C}'$  of a chamber  $\mathfrak{C}$ , inducing  $\mathfrak{a}' \subseteq \mathfrak{a}$ , then we have*

$$\begin{array}{ccc} H_{\top}^*(X^A) & \xrightarrow{\text{Stab}_{\mathfrak{C}}} & H_{\top}^*(X) \\ & \searrow \text{Stab}_{\mathfrak{C}/\mathfrak{C}'} \quad \nearrow \text{Stab}'_{\mathfrak{C}'} & \\ & H_{\top}^*(X^{A'}) & \end{array}$$

and  $[\text{Stab}_{\mathfrak{C}}] = [\text{Stab}_{\mathfrak{C}/\mathfrak{C}'}] \cdot [\text{Stab}_{\mathfrak{C}'}]$ .

Combining this with Theorem 4.4.2 we get

**Corollary 4.8.4.** *There is an algebra  $\mathsf{Y}$  with modules  $\{H^*(\mathcal{X}_w)\}$  and an element*

$$R \in \mathsf{Y} \otimes \mathsf{Y}(a)$$

such that  $\rho_{V_i} \otimes \rho_{V_j}(R) = R_{ij}(a)$  (etc.)

*Proof.* To begin with,  $\text{Stab}_{\mathfrak{C}}$  is invertible up to localisation and  $H_{\top}^*(X)$  is free as a  $H_{\top}^*(\text{pt})$ -module, so we have

$$\text{Stab}_{\mathfrak{C}} : \bigoplus_{w=w_1+\dots+w_n} H^*(\mathcal{X}_{w_1}) \otimes \dots \otimes H^*(\mathcal{X}_{w_n}) \otimes \mathbf{Q}(t) \xrightarrow{\sim} H^*(X) \otimes \mathbf{Q}(t)$$

and its inverse is defined. When  $n = 2$  so that  $\mathbf{Q}(\mathfrak{t}) = \mathbf{Q}(a)$  has rank one, we set

$$R_{\mathfrak{C}}(a) = \text{Stab}_{-\mathfrak{C}}^{-1} \cdot \text{Stab}_{\mathfrak{C}}.$$

In particular, we have its matrix coefficients

$$R_{\mathfrak{C}}(a)_{w_1, w_2}^{w'_1, w'_2} : H^\bullet(\mathcal{X}_{w_1}) \otimes H^\bullet(\mathcal{X}_{w_2}) \otimes \mathbf{Q}(a) \xrightarrow{\sim} H^\bullet(\mathcal{X}_{w'_1}) \otimes H^\bullet(\mathcal{X}_{w'_2}) \otimes \mathbf{Q}(a)$$

for any  $w_1 + w_2 = w = w'_1 + w'_2$ .

(conditions needed above for RTT).

□

4.8.5. *Cohomological stable envelope.* Now construct the stable envelopes.

4.8.6. *Proof of Theorem 4.8.2: uniqueness.* First, the way we construct the Lagrangian cycle  $[\text{Stab}_{\mathfrak{C}}]$  below will see that it is unique.

Second, if  $\text{Stab}_{\mathfrak{C}}, \text{Stab}'_{\mathfrak{C}}$  are two such maps as above, then for any  $\gamma \in H_T^\bullet(X^A)$  the difference

$$\delta = \text{Stab}_{\mathfrak{C}}(\gamma) - \text{Stab}'_{\mathfrak{C}}(\gamma) \in H_T^\bullet(X)$$

is supported on a union of attracting sets and

$$\deg_A \iota^* \delta < \frac{1}{2} \text{codim} Z$$

for  $\iota : Z \hookrightarrow X$  any component of the fixed locus.

**Lemma 4.8.7.** *Any such class  $\delta$  is zero.*

*Proof from [MO12].* Pick a component  $Z$  so that  $\delta$  is supported on  $\text{Attr}_{\mathfrak{C}}^f Z$ . Factor

$$Z \xrightarrow{\iota_0} \text{Attr}_{\sigma} Z \xrightarrow{j} \text{Attr}_{\sigma}^f Z \xrightarrow{\iota_1} X$$

as in Lemma 4.6.3. This implies that

$$\delta \cap [X] = \iota_{1*} \alpha$$

for a Borel–Moore class  $\alpha$ . Therefore,

$$\iota^* \delta \cap [Z] = e(N_-) \cap \iota_0^* j^* \alpha$$

where we recall that multiplication by  $e(N_-)$  is injective and  $\deg_A e(N_-) = \frac{1}{2} \text{codim} Z$ . Therefore,  $e(N_-) \cap \iota_0^* j^* \alpha$ , and as  $\iota_0^*$  is an isomorphism this forces that  $j^* \alpha = 0$ , so  $\delta$  is supported on a smaller union of strata. Inducting gives  $\delta = 0$ . □

4.8.8. *Proof of Theorem 4.8.2: existence.* We begin by constructing repeated approximations to the stable envelope, let us call them  $S_0, S_1, \dots$ . To begin with, pick a component  $Z \subseteq X^A$  of the fixed locus, and we set

$$S_0 = \overline{\Delta \cap_{Z \times Z} (\text{Attr}_{\mathfrak{e}} Z \times Z)},$$

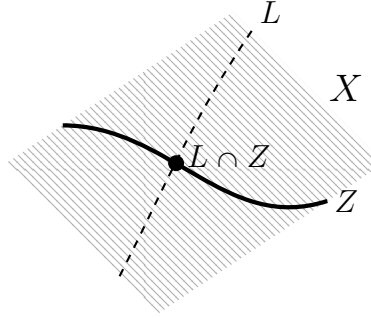
which satisfies (the first two conditions). To construct  $S_1, S_2, \dots$ , we use

**Lemma 4.8.9.** (Gram–Schmidt for Lagrangian cycles) (*Refer to conditions on  $X$* ) and let  $\iota : Z \simeq X_i^A \hookrightarrow X$  be a component of the fixed locus and let  $L \subseteq X$  be an  $A$ -invariant Lagrangian. Note that  $\iota^* \omega$  is symplectic.

- (1)  $L \cap Z$  is isotropic.
- (2) There is a unique Lagrangian cycle  $\text{Res}_{\mathfrak{e}} L$  in  $Z$  supported on  $L \cap Z$  such that in  $H_{\bullet, A}^{\text{BM}}(Z)$ ,

$$\iota^* L = \epsilon[\text{Res}_Z L] + \dots$$

where  $\dots$  consists of terms of smaller  $A$ -degree.



*Proof.* Either skip or write right at the end of prep. □

Specifically, noting that  $S_0$  is a Lagrangian, we set

$$S_1 = S_0 - \overline{\text{Attr}_{\mathfrak{e}}(\epsilon[\text{Res}_Z S_0])}$$

as we note that  $\iota^* \overline{\text{Attr}_{\mathfrak{e}} L} = L$  (check this). Repeating this, this process terminates (dimension reasons) at  $S_n = S_Z$ , and so constructs our candidate  $S = \sum_{Z \in \pi_0(X^A)} S_Z$  for the stable envelope. It satisfies (condition i-iii) because (easy reason), so we are left with showing that it (i.e. its support?) is proper over  $X$ .

We deduce this from the fact that  $\pi : X \rightarrow X_0$  is proper. First, we reduce to the case that  $X_0$  is a vector space with an action of  $A$ . (MO just state this in 3.2.7) Then, (we reason about vector spaces).

4.8.10. *K theoretic stable envelope.* Apparently it's in [OS16], p12-19.

4.8.11. *Elliptic stable envelope.* Only reference Agnagic–Okounkov.

4.9. **Example: Yangians from quiver varieties.**

4.9.1. In the above, we consider a quiver  $Q$  with generic stability condition, e.g.  $\theta = \theta^+$ , and we set  $V_i = H^\bullet(\mathcal{M}(w))$  so that  $V_i(a) = H_{\mathbb{C}^\times}^\bullet(\mathcal{M}(w))$ , and

$$H_A^\bullet(\mathcal{M}(w)^A) = \bigoplus_{w=w_1+\dots+w_n} V_{i_1}(a_1) \otimes \dots \otimes V_{i_n}(a_n)$$

where  $A$  is the maximal torus of  $\mathrm{PGL}(n)$  and  $a_j$  is the equivariant parameter corresponding to the  $j$ -th factor.

Example of the above:  $\mathfrak{sl}_2$ , alias  $Q = \bullet$

**Quiver varieties.**

*Computation.* Let  $Q = \bullet$  so that  $\text{Rep}Q = \text{Vect}$ , with its King stability condition

$$\theta^+ : K_0(\text{Vect}) \xrightarrow{\sim} \mathbf{Z}$$

sending  $V \mapsto \dim V$ . The double of its framed quiver is

$$\begin{array}{c} w \\ \blacksquare \\ \begin{array}{c} \uparrow i \\ \downarrow j \end{array} \\ \bullet \\ v \end{array} \quad (Q^f)^{(2)}$$

The moduli stacks of representations of  $Q$  and  $(Q^f)^{(2)}$  are

$$\mathcal{M}_Q = \bigsqcup_{v \geq 0} \text{BGL}_v, \quad \mathcal{M}_{(Q^f)^{(2)}} = \bigsqcup_{v, w \geq 0} \text{Hom}(W, V) \oplus \text{Hom}(V, W) / \text{GL}_v$$

where  $V, W$  have dimensions  $v, w$ . The cohomologies of both of the above components are  $H^\bullet(\text{BGL}_v) \simeq \mathbf{C}[c_1, \dots, c_v]$  where  $c_i$  has degree  $2i$ .

*Group action.* The action of  $\text{GL}_v$  is by post-composition on  $\text{Hom}(W, V)$  and pre-composition on  $\text{Hom}(V, W)$ , i.e. we have

$$g \cdot (A, B) \mapsto (g \cdot A, B \cdot g^{-1})$$

and by writing  $g = \exp(\epsilon x)$  where  $\epsilon^2 = 0$ , we likewise see that the action of  $\mathfrak{gl}_v$  on the tangent space at  $(A, B)$  is

$$x \cdot (a, b) \mapsto (xa, -bx),$$

where we identify  $T_{(A, B)} T^* \text{Hom}(V, W)$  with  $T^* \text{Hom}(V, W)$ .

*Symplectic reduction.* The symplectic form on  $T^* \text{Hom}(V, W) = \text{Hom}(V, W) \times \text{Hom}(W, V)$  is

$$\omega((a, b), (a', b')) = \text{tr}_V(ba' - b'a)$$

because the evaluation between  $\text{Hom}(V, W)$  and  $\text{Hom}(V, W)^* \simeq \text{Hom}(W, V)$  is given by taking trace.

**Lemma 4.9.2.** *The moment map is given by*

$$\begin{aligned} \mu : \text{Hom}(V, W) \oplus \text{Hom}(W, V) &\rightarrow \mathfrak{gl}_v \xrightarrow{\sim} \mathfrak{gl}_v^* \\ (j, i) &\mapsto ij \mapsto \text{tr}_V(ij(-)), \end{aligned}$$

which is usually just written as  $\mu = ij$ .

*Proof.* We compute the moment map  $\mu : T^*\text{Hom}(V, W) \rightarrow \mathfrak{gl}_v^*$  defined as the unique map so that the composition

$$\mu(x) : T^*\text{Hom}(V, W) \rightarrow \mathfrak{gl}_v^* \xrightarrow{x} \mathbf{C}$$

has exterior derivative at  $(A, B)$  given by  $\omega(x|_{(A, B)}, -)$ . This exterior derivative is given by

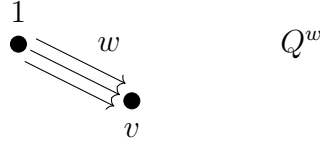
$$(d_{(A, B)}\mu(x))(a, b) = \lim_{t \rightarrow 0} \frac{\mu(A + ta, B + tb)(x) - \mu(A, B)(x)}{t}$$

which as  $x|_{(A, B)} = (xA, -Bx)$  should equal

$$\omega(x|_{(A, B)}, (a, b)) = \text{tr}((-Bx)a - (xA)b) = \text{tr}(-aBx - Abx).$$

This corresponds under  $\mathfrak{gl}_v \simeq \mathfrak{gl}_v^*$  to  $aB + Ab$ , which is the differential of the map  $(A, B) \mapsto AB$ , therefore the exterior derivative of  $\mu(x)$  is  $\text{tr}(-aBx - Abx)$  as required.  $\square$

*Stability.* Recall the Crawley–Boevey quiver of  $Q$



and the stability function  $\hat{\theta} = (\theta, -v\theta)$  attached to stability condition  $\theta \in \mathbf{Z}$  on  $Q$ . Identifying

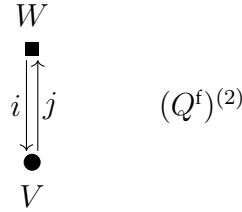
$$\text{Rep}(Q^w, \hat{v}) \simeq \text{Rep}(Q, v, w)$$

gives the stability condition

$$\hat{\theta} : K_0(\text{Rep}(Q^f)^{(2)}, v, w) \simeq K_0(\text{Rep}(Q, v, w)) \simeq K_0(\text{Rep}(Q^w, \hat{v})) \xrightarrow{\hat{\theta}} \mathbf{Z}$$

sending all  $[(V, W)] \mapsto 0$ . **this seems wrong**

**Lemma 4.9.3.** *A representation*



satisfies  $\mu = 0$  if and only if  $ij = 0$ , and is  $\theta^+$ -stable if and only if  $\ker j = 0$ .

*Proof.* **write**

$\square$

Note in particular that the existence of stable points forces that  $w \geq v$ .

**Computing the quiver variety.**

4.9.4. Consider

$$\begin{array}{c} \mathbf{C} \\ \blacksquare \\ \begin{array}{c} \uparrow j \\ \downarrow i \end{array} \\ \bullet \\ \mathbf{C} \end{array} \quad (Q^f)^{(2)}$$

where  $j$  is invertible and  $i = 0$ . The Nakajima quiver variety is then

$$(\mathrm{Hom}(\mathbf{C}(1), \mathbf{C}) \times \mathrm{Hom}(\mathbf{C}, \mathbf{C}(1)))/\mathrm{GL}_1 \hookrightarrow (\mathbf{C}^\times \times 0)/\mathrm{GL}_1 \simeq \mathrm{pt}.$$

Here,  $(n)$  denotes that the  $\mathrm{GL}_1$ -action has weight  $n$ .

4.9.5. Next, consider

$$\begin{array}{c} \mathbf{C}^n \\ \blacksquare \\ \begin{array}{c} \uparrow j \\ \downarrow i \end{array} \\ \bullet \\ \mathbf{C} \end{array} \quad (Q^f)^{(2)}$$

where  $j$  is an embedding and  $i$  induces a map  $\mathbf{C}^n/j(\mathbf{C}) \rightarrow \mathbf{C}$ .

The Nakajima quiver variety is then

$$\begin{array}{ccc} (\mathrm{Hom}(\mathbf{C}(1), \mathbf{C}^n) \times \mathrm{Hom}(\mathbf{C}^n, \mathbf{C}(1)))/\mathrm{GL}_1 & \hookrightarrow & \mu^{-1}(0)^{\mathrm{ss}}/\mathrm{GL}_1 \\ \downarrow \pi & & \downarrow \\ \mathrm{Hom}(\mathbf{C}, \mathbf{C}^n)/\mathrm{GL}_1 & \longleftarrow & \mathbf{P}^{n-1} \end{array}$$

Note that the pullback under

$$\begin{array}{ccc} \mathbf{C}(k)/\mathrm{GL}_1 & \longleftarrow & |\mathcal{O}(-k)| \\ \downarrow & & \downarrow \\ \mathrm{BGL}_1 & \longleftarrow & \mathbf{P}^{n-1} \end{array}$$

and so the pullback to  $\mathbf{P}^{n-1}$  of the bundle  $\pi$  is  $\mathcal{O}(-1)^{\oplus n}$ . If we restrict to the subbundle of  $i \in \mathrm{Hom}(\mathbf{C}^n, \mathbf{C}(1))$  which kills the plane  $\mathbf{C}(1) \subseteq \mathbf{C}^n$  corresponding to a base point in  $\mathbf{P}^{n-1}$ , its pullback is the cotangent bundle, by the Euler sequence

$$0 \rightarrow \mathcal{T}_{\mathbf{P}^{n-1}}^* \rightarrow \mathcal{O}(-1)^{\oplus n} \rightarrow \mathcal{O} \rightarrow 0.$$

Therefore, the Nakajima quiver variety is

$$T^*\mathbf{P}^{n-1} \simeq \mu^{-1}(0)^{\mathrm{ss}}/\mathrm{GL}_1.$$

This has an action of  $G_w = \mathrm{GL}_n$  by the action on  $\mathbf{P}^{n-1}$ , and its cohomology is

$$H_{G_w}^\bullet(\mathcal{M}_{\theta^+}(1, n)) = H_{\mathrm{GL}_n}^\bullet(\mathbf{P}^{n-1}) = H^\bullet(\mathrm{GL}_n \backslash \mathrm{GL}_n/P_n) = H^\bullet(\mathrm{BP}_n)$$

$$\simeq H^\bullet(\mathrm{BGL}_1 \times \mathrm{BGL}_{n-1}) \simeq \mathbf{C}[c_1] \otimes \mathbf{C}[c_1, \dots, c_{n-1}]$$

where  $P_n \subseteq \mathrm{GL}_n$  is the parabolic subgroup given by the stabiliser of a line  $\mathbf{C} \subseteq \mathbf{C}^n$ , i.e. of matrices

$$\begin{pmatrix} \star & \star & \cdots & \star \\ & \star & \cdots & \star \\ & \vdots & \ddots & \vdots \\ & \star & \cdots & \star \end{pmatrix}$$

which admits a map  $P_n \rightarrow \mathrm{GL}_1 \times \mathrm{GL}_{n-1}$ . The non-equivariant cohomology is

$$H_{\mathrm{G}_w}^\bullet(\mathcal{M}_{\theta^+}(1, n)) = H^\bullet(\mathbf{P}^{n-1}) \simeq \mathbf{C}[x]/x^n$$

and the map  $H_{\mathrm{GL}_n}^\bullet(\mathbf{P}^{n-1}) \rightarrow H^\bullet(\mathbf{P}^{n-1})$  sends  $c_i \mapsto \binom{n}{i} x^i$  by the splitting principle.

4.9.6. Finally, consider

$$\begin{array}{c} \mathbf{C}^n \\ \blacksquare \\ \begin{array}{c} \uparrow i \\ \downarrow j \end{array} \\ \bullet \\ \mathbf{C}^k \end{array} \quad (Q^f)^{(2)}$$

where  $n \geq k$ . We have  $\mathrm{Gr}(k, n)$  in place of  $\mathbf{P}^{n-1}$ , and we have an Euler sequence

$$0 \rightarrow \mathcal{T}_{\mathrm{Gr}(k, n)}^* \rightarrow \mathrm{Hom}(\mathcal{O}^{\oplus n}, \mathbf{V}^{(k)}) \rightarrow \mathrm{Hom}(\mathbf{V}^{(k)}, \mathbf{V}^{(k)}) \rightarrow 0$$

where  $\mathbf{V}^{(k)}$  is the pullback of the tautological bundle on  $\mathrm{BGL}_k$ . Therefore, by the same argument,

$$\mathcal{M}_{\theta^+}(k, n) \simeq T^* \mathrm{Gr}(k, n)$$

with equivariant cohomology the equivariant cohomology of the parabolic  $P_{n,k} \subseteq \mathrm{GL}_n$  stabilising a  $k$ -plane,

$$H_{\mathrm{G}_w}^\bullet(\mathcal{M}_{\theta^+}(k, n)) \simeq H^\bullet(\mathrm{BGL}_k \times \mathrm{BGL}_{n-k}) = \mathbf{C}[c_1, \dots, c_k] \otimes \mathbf{C}[c_1, \dots, c_{n-k}].$$

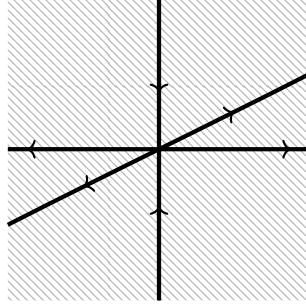
**Torus fixed points.** Consider the action of  $\mathrm{GL}_w$  given by

$$h \cdot (j, i) \mapsto (hj, ih^{-1}).$$

Let us consider a torus  $A \subseteq GL_n$  given by a decomposition into  $w = w_1 + \cdots + w_r$ , i.e.  $n = n_1 + \cdots + n_r$ :

$$A = \left\{ \begin{pmatrix} a_1 & & & \\ & \ddots & & \\ & & a_1 & \\ \hline & & & \ddots \\ & & & & a_r \\ \hline & & & & & \ddots \\ & & & & & & a_r \end{pmatrix} : \det = 1 \right\} \simeq \mathbf{G}_m^{r-1}$$

This splits up  $W = \mathbf{C}^n$  into  $A$ -subrepresentations  $W_1 \oplus \cdots \oplus W_r$ :



and an element of  $T^*\mathrm{Gr}(k, n)$  is fixed by the torus if and only if the associated  $k$ -plane  $\mathbf{C}^k \subseteq W$  is a direct sum of  $k_i$ -planes  $\mathbf{C}^{k_i} \subseteq W_i$ . Thus

$$(T^*\mathrm{Gr}(k, W))^A = \bigsqcup_{k=k_1+\cdots+k_r} T^*\mathrm{Gr}(k_1, W_1) \times \cdots \times T^*\mathrm{Gr}(k_r, W_r).$$

Recall that we set  $G_A \subseteq G_w = GL_n$  the centraliser of  $A$  in  $G_w$ , for instance we write  $\mathbf{G}_{m, \hbar} \subseteq GL_n$  for multiples of the identity. This fixes each  $k$ -hyperplane and acts by scaling the fibres of  $T^*\mathrm{Gr}(k, n) \rightarrow \mathrm{Gr}(k, n)$ .

4.9.7. *Example.* For instance, if  $A$  acts on  $\mathbf{C}^n = \mathbf{C}^{n_1} \times \cdots \times \mathbf{C}^{n_r}$  with different weights, then the fixed locus is

$$(T^*\mathbf{P}^{n-1})^A = T^*\mathbf{P}^{n_1-1} \sqcup \cdots \sqcup T^*\mathbf{P}^{n_r-1}.$$

where each connected component is secretly a product of two factors, but we do not see the  $\mathcal{M}_{\theta^+}(0, n_i) \simeq \mathrm{pt}$  factor.

4.9.8. *Cohomology.* We now consider the maps

$$\tilde{\iota}_{i,*} : H^*(T^*\mathbf{P}^{n_i-1}) \hookrightarrow H^*(T^*\mathbf{P}^{n-1}) : \tilde{\iota}_i^*$$

induced by the inclusion  $\tilde{\iota}_i : T^*\mathbf{P}^{n_i-1} \hookrightarrow T^*\mathbf{P}^{n-1}$ . Note that the pushforward increases cohomological degree by twice the codimension, i.e. by  $4(n - n_i)$ . We also have

$$\tilde{\iota}_i^* \tilde{\iota}_{i,*} = e(N_{\tilde{\iota}_i}) \cap (-)$$

where  $N_{\tilde{\ell}_i}$  is the normal bundle to  $\tilde{\ell}_i$ , and by Atiyah–Bott torus localisation

$$\tilde{\ell}_{i,*} : \bigoplus_i H_{\mathbf{A}}^{\bullet}(T^*\mathbf{P}^{n_i-1}) \otimes_{H^{\bullet}(\mathbf{BA})} \text{Frac} H^{\bullet}(\mathbf{BA}) \xrightarrow{\sim} H_{\mathbf{A}}^{\bullet}(T^*\mathbf{P}^{n-1}) \otimes_{H^{\bullet}(\mathbf{BA})} \text{Frac} H^{\bullet}(\mathbf{BA}) : \tilde{\ell}_i^*$$

is an isomorphism.

To be explicit, we have in the Grothendieck group of vector bundles on  $\mathbf{P}^{n-1}$  that

$$[T(T^*\mathbf{P}^{n-1})] = [T_{\mathbf{P}^{n-1}}] + [T_{\mathbf{P}^{n-1}}^*] = n[\mathcal{O}(1)] + n[\mathcal{O}(-1)] - 2[\mathcal{O}]$$

whose Euler class is thus

$$e(T(T^*\mathbf{P}^{n-1})) = (x)^n(-x)^n = 0 \in H^{\bullet}(\mathbf{P}^{n-1}) \simeq H^{\bullet}(T^*\mathbf{P}^{n-1}).$$

However, in the equivariant Grothendieck group  $K_0^{\mathbf{A}}(\mathbf{P}^{n-1})$ , we have

$$[T(T^*\mathbf{P}^{n-1})] = \sum_{i=1}^r n_i[\mathcal{O}(1) \otimes \chi_i] + n_i[\mathcal{O}(-1) \otimes \chi_i^{-1}] - 2[\mathcal{O}]$$

where  $\chi_i$  is the character of  $\mathbf{A}$  corresponding to the  $i$ -th factor, and so its Euler class is

$$e_{\mathbf{A}}(T(T^*\mathbf{P}^{n-1})) = \prod_{i=1}^r (x + a_i)^n (-x - a_i)^n \in H_{\mathbf{A}}^{\bullet}(\mathbf{P}^{n-1}) \simeq H_{\mathbf{A}}^{\bullet}(T^*\mathbf{P}^{n-1}) \simeq \mathbf{C}[a_1, \dots, a_r, x]/x^n.$$

Likewise, the normal bundle is

$$e_{\mathbf{A}}(N_{T^*\mathbf{P}^{n_i-1} \hookrightarrow T^*\mathbf{P}^{n-1}}) = \prod_{j \neq i} (x + a_j)^n (-x - a_j)^n \in H_{\mathbf{A}}^{\bullet}(\mathbf{P}^{n_i-1}) \simeq H_{\mathbf{A}}^{\bullet}(T^*\mathbf{P}^{n_i-1}) \simeq \mathbf{C}[a_1, \dots, a_r, x]/x^{n_i}.$$

Note that elements of

$$H_{\mathbf{A}}^{\bullet}(T^*\mathbf{P}^{n-1}) \otimes_{H^{\bullet}(\mathbf{BA})} \text{Frac} H^{\bullet}(\mathbf{BA}) = \mathbf{C}(a_1, \dots, a_r)[x]/x^n$$

are rational functions

$$\frac{f(a_1, \dots, a_r, x)}{g(a_1, \dots, a_r, x)}$$

where the leading term of  $g$  is not nilpotent, i.e. contains no  $x$ 's.

what are the restriction maps  $i^*$ ?

**4.9.9. General Grassmannian case.** Note that  $\text{Gr}(k, n)$  admits a canonical short exact sequence of vector bundles

$$0 \rightarrow \mathbf{V}^{(k)} \rightarrow \mathcal{O}^{\oplus n} \rightarrow \mathcal{O}^{\oplus n}/\mathbf{V}^{(k)} \rightarrow 0$$

where  $\mathbf{V}^{(k)}$  is the pullback of the tautological bundle on  $\text{BGL}_k$ . The cohomology of  $\text{Gr}(k, n)$  is generated by the Chern classes of  $\mathbf{V}^{(k)}$  and  $\mathcal{O}^{\oplus n}/\mathbf{V}^{(k)}$ , with the relations given by the coefficients of  $c(\mathbf{V}^{(k)})c(\mathcal{O}^{\oplus n}/\mathbf{V}^{(k)}) = 1$ .

**Lemma 4.9.10.** *The tangent bundle of the Grassmannian is computed in terms of*

$$\mathrm{Gr}(k, n) \rightarrow \mathrm{BGL}_k$$

as  $\mathrm{Hom}(\mathbf{V}^{(k)}, \mathcal{O}^{\oplus n}/\mathbf{V}^{(k)})$ , where  $\mathbf{V}^{(k)} \hookrightarrow \mathcal{O}^{\oplus n}$  is the pullback of the tautological bundle on  $\mathrm{BGL}_k$  and its tautological inclusion into the rank  $n$  trivial bundle. The normal bundle to

$$\iota_i : \mathrm{Gr}(k, W_i) \hookrightarrow \mathrm{Gr}(k, W)$$

is  $N_{\iota_i} = \bigoplus_{j \neq i} \mathrm{Hom}(\mathbf{V}^{(k)}, W_j \otimes \mathcal{O})$ .

*Proof.* We have an Euler sequence

$$0 \rightarrow \mathrm{Hom}(\mathbf{V}^{(k)}, \mathbf{V}^{(k)}) \rightarrow \mathrm{Hom}(\mathbf{V}^{(k)}, W \otimes \mathcal{O}) \rightarrow T_{\mathrm{Gr}(k, n)} \rightarrow 0$$

which both implies the first statement and may be used in conjunction with the same sequence for  $\mathrm{Gr}(k, W_i)$  to compute the normal bundle using

$$0 \rightarrow N_{\iota_i} \rightarrow T_{\mathrm{Gr}(k, W)}|_{\mathrm{Gr}(k, W_i)} \rightarrow T_{\mathrm{Gr}(k, W_i)} \rightarrow 0.$$

□

**Corollary 4.9.11.** *The Euler class of the normal bundle to  $N_{\iota_i}$  is*

$$e(N_{\iota_i}) = (-e(\mathbf{V}^{(k)}))^{n-n_i}.$$

*The equivariant Euler class is*

$$e_{\mathrm{GL}_W}(N_{\iota_i}) = \prod_{j \neq i} (-e(\mathbf{V}^{(k)})x_1^{(j)} \cdots x_{n_j}^{(j)})$$

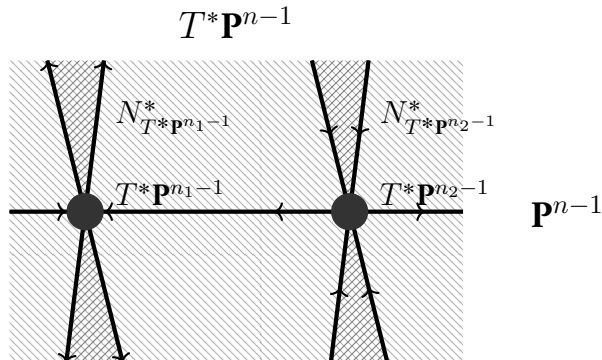
where  $x_1^{(j)}, \dots, x_{n_j}^{(j)}$  are the Chern roots of the maximal torus of  $\mathrm{GL}_{n_j}$  acting on  $W_j$ .

### Attracting sets.

. Consider the action of  $A = \mathbf{G}_m$  on  $\mathbf{C}^n = \mathbf{C}^{n_1} \oplus \mathbf{C}^{n_2}$  given by

$$a \cdot (v_1, v_2) = (av_1, a^{-1}v_2).$$

Then unless a  $k$ -plane intersects  $\mathbf{C}^{n_1}$ , its  $a \rightarrow 0$  limit will be a  $k$ -plane in  $\mathbf{C}^{n_2}$ . Note that if a fixed point  $p \in \mathbf{P}^{n-1}$  is attracting, then it will be repelling in its cotangent fibre  $T_p^* \mathbf{P}^{n-1}$ , and vice versa. Thus



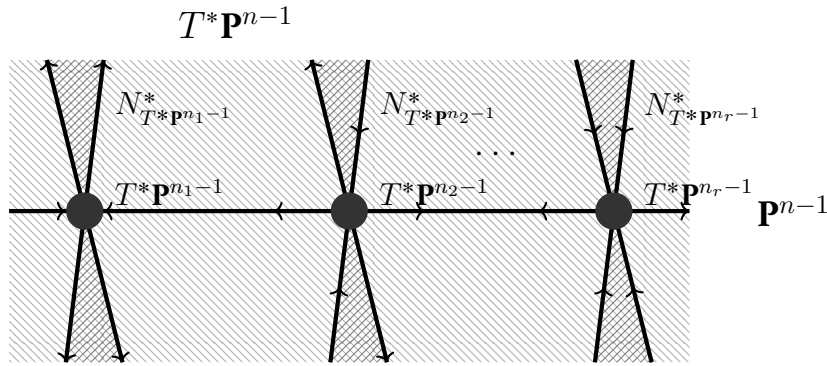
Thus

$$\begin{aligned} [\text{Attr}^f T^* \mathbf{P}^{n_1-1}] &= [\mathbf{P}^{n-1}] + [N_{\mathbf{P}^{n_2-1}}^*] \\ [\text{Attr}^f T^* \mathbf{P}^{n_2-1}] &= [N_{\mathbf{P}^{n_2-1}}^*]. \end{aligned}$$

Likewise, consider the action of  $A = \mathbf{G}_m$  on  $\mathbf{C}^n = \mathbf{C}^{n_1} \oplus \cdots \oplus \mathbf{C}^{n_r}$  given by

$$a \cdot (v_1, \dots, v_r) = (av_1, a^2 v_2, \dots, a^r v_r)$$

or more generally any sequence of increasing weights.



and so

$$\begin{aligned} [\text{Attr}^f T^* \mathbf{P}^{n_1-1}] &= [\mathbf{P}^{n-1}] + [N_{\mathbf{P}^{n_2-1}}^{*, -}] + [N_{\mathbf{P}^{n_3-1}}^{*, -}] + \cdots + [N_{\mathbf{P}^{n_r-1}}^{*, -}] \\ [\text{Attr}^f T^* \mathbf{P}^{n_2-1}] &= [\mathbf{P}^{n-n_1-1}] + [N_{\mathbf{P}^{n_3-1}}^{*, -}] + \cdots + [N_{\mathbf{P}^{n_r-1}}^{*, -}] \\ &\vdots \\ [\text{Attr}^f T^* \mathbf{P}^{n_r-1}] &= [N_{\mathbf{P}^{n_r-1}}^{*, -}] \end{aligned}$$

where  $N^{*, -}$  is the  $A$ -attracting part of the conormal bundle.

4.9.12. We now consider the *pair* attracting space inside  $(T^* \mathbf{P}^{n-1})^A \times T^* \mathbf{P}^{n-1}$ .

4.9.13. Inside  $(T^* \mathbf{P}^1)^{\mathbf{G}_m} \times T^* \mathbf{P}^1$ , the pair attracting space coincides with a disjoint union of the attracting spaces

$$0 \times \text{Attr}^f 0 \sqcup \infty \times \text{Attr}^f \infty \simeq (\mathbf{P}^1 \cup N_\infty^*) \sqcup N_\infty^*$$

since the fixed points are isolated, but in general they differ.

**Stable envelopes.**

*Cohomology.* Let us take  $A = \mathbf{G}_m$  acting on  $T^*\mathbf{P}^1$  as before, and take

$$G_A = C_A(\mathrm{GL}_2) = \mathbf{G}_{m,\hbar} \times A = \mathbb{T}$$

be the maximal torus in  $\mathrm{GL}_2$ . Note that

$$H_{G_A}^\bullet(T^*\mathbf{P}^1) = H_A^\bullet(\mathbf{P}^1) \otimes H_{\mathbf{G}_{m,\hbar}}^\bullet(\mathrm{pt}) = \mathbb{C}[a, x]/(x+a)(x-a) \otimes \mathbb{C}[\hbar].$$

To see the second equality, note that

$$\mathbf{P}^1/A = (\mathbf{A}^2 \setminus 0)/\mathbf{G}_{m,1} \times \mathbf{A}_{1,-1}$$

characterises pairs of line bundles  $\mathcal{L}, \mathcal{L}_A$  with a nowhere vanishing section of  $\mathcal{L} \otimes \mathcal{L}_A \oplus \mathcal{L} \otimes \mathcal{L}_A^{-1}$ . In particular, the Euler class of this bundle is zero, so writing  $x = c_1(\mathcal{L})$  and  $a = c_1(\mathcal{L}_A)$ , we have  $(x+a)(x-a) = 0$ .

Finally, the fully equivariant Euler classes of these line bundles are

$$e_{G_A}(\mathcal{L}) = x + \hbar, \quad e_{G_A}(\mathcal{L}_A) = a + \hbar.$$

Likewise of course,

$$H_{G_A}^\bullet((T^*\mathbf{P}^1)^{\mathbf{G}_m}) = \mathbb{C}[a, \hbar] \oplus \mathbb{C}[a, \hbar]$$

corresponding to the point  $0, \infty$ . Note that the class  $x$  pulls back as

$$x \mapsto (a, -a).$$

4.9.14. Let us now compute the stable envelope inside

$$H_{G_A}^\bullet((T^*\mathbf{P}^1)^{\mathbf{G}_m} \times T^*\mathbf{P}^1) = (\mathbb{C}[a, \hbar, x]/(x+a)(x-a)) \oplus (\mathbb{C}[a, \hbar, x]/(x+a)(x-a))$$

where the first summand corresponds to the  $A$ -fixed point  $0$  and the second to  $\infty$ .

Recall that it is uniquely defined by a map

$$\mathrm{Stab}_{\mathfrak{C}} : H_{\mathbb{T}}^\bullet((T^*\mathbf{P}^1)^A) \rightarrow H_{\mathbb{T}}^\bullet(T^*\mathbf{P}^1),$$

such that for any component  $\{z\} \subseteq (T^*\mathbf{P}^1)^A$  and  $\gamma \in H_{\mathbb{T}}^\bullet(\{z\})$ ,

- (1)  $\mathrm{Stab}_{\mathfrak{C}}(\gamma)$  is supported on  $\mathrm{Attr}_{\mathfrak{C}}^f\{z\}$ ,
- (2)  $\mathrm{Stab}_{\mathfrak{C}}(\gamma)|_{\{z\}} = \pm e(N_{z,\mathfrak{C}}^-) \cup \gamma$ ,
- (3)  $\deg_A \mathrm{Stab}_{\mathfrak{C}}(\gamma)|_{z'} < \frac{1}{2} \mathrm{codim}\{z'\}$  for all downstream  $\{z'\} < \{z\}$ .

To explain the above, if  $i : \mathrm{Attr}^f\{z\} \hookrightarrow T^*\mathbf{P}^1$  is the inclusion, a cohomology class is *supported on*  $\mathrm{Attr}^f\{z\}$  if it is in the image of

$$i_* : \Gamma(i_* i^! k) \rightarrow \Gamma(k).$$

If  $\text{Attr}^f\{z\} = \sqcup Z_k$  splits into smooth irreducible closed components with regular closed embeddings  $i_k$  of codimension  $c_k$ , this agrees with the image of

$$(i_{k*}) : \bigoplus_k \Gamma(i_{k,*} i_k^! k) \simeq \bigoplus_k \Gamma^{\bullet+2c_k}(Z_k, k) \rightarrow \Gamma(T^*\mathbf{P}^1, k)$$

if in the distinguished triangle

$$i_{12*} i_{12}^! k \rightarrow i_{1*} i_1^! k \oplus i_{2*} i_2^! k \rightarrow i_* i^! k \rightarrow$$

where  $i_{12}$  is the inclusion of the intersection of the two components, the first map is injective on cohomology, which is the case for us. Recall that  $i_k^* i_{k*}(1) = e(N_{i_k})$  is the Euler class of the normal bundle.

For instance,  $\text{Attr}^f 0 = \mathbf{P}^1 \cup N_\infty^*$  has

$$H_{\mathbf{G}_A}^*(\text{Attr}^f 0) = H_{\mathbf{G}_A}^*(\mathbf{P}^1) \oplus_{H_{\mathbf{G}_A}^*(\text{pt})} H_{\mathbf{G}_A}^*(N_\infty^*) \simeq k[a, x, \hbar]/(x+a)(x-a) \oplus_{k[a, \hbar]} k[a, \hbar].$$

The normal bundle to  $\mathbf{P}^1 \hookrightarrow T^*\mathbf{P}^1 \simeq \mathcal{O}(1)$  is the cotangent bundle, which has Euler class  $e(T^*\mathbf{P}^1) = a - \hbar$ , and the normal bundle to  $N_\infty^* \hookrightarrow T^*\mathbf{P}^1$  is trivial with  $\mathbf{G}_{m, \hbar}$ -weight  $-\hbar$ , so  $e(N_{N_\infty^* \hookrightarrow T^*\mathbf{P}^1}) = -\hbar$ .

Next,  $\text{Attr}^f \infty = N_\infty^*$  which as before has  $e(N_{N_\infty^* \hookrightarrow T^*\mathbf{P}^1}) = -\hbar$ .

**Lemma 4.9.15.**

The only such one is for  $z_i, z_j \in \{0, \infty\}$  that

$$\text{Stab}_{\pm \mathfrak{c}}(z_i)|_{z_j} \in \left\{ \begin{pmatrix} -a - \hbar & \\ -\hbar & a \end{pmatrix}, \begin{pmatrix} -a & -\hbar \\ & a - \hbar \end{pmatrix} \right\}$$

so that  $R(a) = \frac{1 - \frac{\hbar}{a} \sigma_{0, \infty}}{1 - \frac{\hbar}{a}}$  acting on  $H_{\mathbf{A}}^\bullet(X^A) = \mathbf{C} \oplus \mathbf{C}(u) = H^\bullet(0) \oplus H^\bullet(\infty)(u)$ .

$$[\text{Stab}_{\mathfrak{c}}(0)] = [\mathbf{P}^1] + [T_\infty^* \mathbf{P}^1], \quad [\text{Stab}_{\mathfrak{c}}(\infty)] = -[T_\infty^* \mathbf{P}^1].$$

and for the other chamber,

$$[\text{Stab}_{-\mathfrak{c}}(0)] = -[T_0^* \mathbf{P}^1], \quad [\text{Stab}_{-\mathfrak{c}}(\infty)] = [\mathbf{P}^1] + [T_0^* \mathbf{P}^1].$$

## 5. Discussion on the $q$ KZ equations and quasimaps

### 5.1. Differential and difference equations.

5.1.1. Recall the *mapping stack* of maps  $f : X \rightarrow Y$  over a base scheme  $B$  is the stack  $\mathrm{Maps}_B(X, Y)$  such that for any test scheme  $S$ ,

$$\mathrm{Maps}_B(X, Y)(S) = \mathrm{Maps}_B(X \times S, Y).$$

Likewise, the *tangent bundle* of a smooth scheme  $X$  is

$$T_X = \mathrm{Maps}_X(\mathrm{Spec} \mathbb{C}[\epsilon]/\epsilon^2 \times X, X)$$

which has a natural map  $T_X \rightarrow X$  and action  $T_X \times_X X \rightarrow X$  given by evaluating at the ideals  $(0)$  and  $(\epsilon)$  respectively.

**Lemma 5.1.2.**  $T_{\mathrm{id}} \mathrm{Maps}_X(X, X) = T_{\mathrm{id}} \mathrm{Aut}(X) = T_X$ .

*Proof.* For the first statement involving  $\mathrm{Maps}(X, X)$  this is tautological, and for the second we use that  $\mathrm{Aut}(X)(S) = \mathrm{Maps}(X, X)(S) \times_{\mathrm{Maps}(X, X)(k)} \mathrm{Aut}(X)(k)$ .  $\square$

Recall that the *stack quotient* of a stack  $X$  by the action of a group  $G$ , both defined over a base  $B$ , is the homotopy pushout

$$\begin{array}{ccc} G \times_B X & \xrightarrow{\pi_2} & X \\ \downarrow \mathrm{act} & & \downarrow \\ X & \longrightarrow & X/G \end{array}$$

and a  $G$ -equivariant sheaf on  $X$  is a sheaf on the stack quotient  $X/G$ . In other words, it is a sheaf  $\mathcal{F}$  on  $X$  along with an isomorphism  $\varphi_{g,x} : \mathcal{F}_x \xrightarrow{\sim} \mathcal{F}_{g \cdot x}$  for any  $g \in G$  and  $x \in X$ , satisfying the cocycle condition

$$\varphi_{g_2, g_1 \cdot x} \cdot \varphi_{g_1, x} = \varphi_{g_2 g_1, x}.$$

5.1.3. *Grothendieck's crystals.* Let  $X$  be a smooth scheme, and consider the formal group  $G = \exp(T_X)$  of **jets** on  $X$ . This is a formal group scheme over  $B = X$  acting on  $X$ , and we can take the quotient

$$\begin{array}{ccc} e^{T_X} \times_X X & \simeq e^{T_X} \xrightarrow{\pi_2} & X & (e^{\epsilon\xi}, x) \longmapsto x \\ \downarrow \mathrm{act} & & \downarrow & \downarrow \\ X & \longrightarrow & X/e^{T_X} & e^{\epsilon\xi} \cdot x \end{array}$$

**Lemma 5.1.4.** Let  $V$  be a vector bundle over  $X$ . A  $G$ -equivariant structure, i.e. a lift of the action on  $X$  to  $V$ , is equivalent to a flat connection  $\nabla$  on  $V$ .

In this case the  $G$ -equivariant structure becomes the parallel transport isomorphism

$$\varphi_{v,x} : V_x \xrightarrow{\sim} V_{x+\epsilon v},$$

corresponding to the action of  $v \in T_{X,x}$  satisfying the cocycle condition. A flat section  $f \in \Gamma(X, V)$  is then equivalent to a  $G$ -equivariant section.

5.1.5. *Example.* For instance, when  $X = \mathbf{A}^1$  and  $V = \mathcal{O}$ ,

$$e^{\epsilon \partial_z} \cdot f(z) = f(z + \epsilon)$$

by Taylor's Theorem. [continue on example](#)

**Definition 5.1.6.** Let  $V$  be a vector bundle over a smooth scheme  $X$ .

5.1.7. *Example: automorphisms.* Let  $X$  be a scheme acted on by a group  $G$  and consider  $\mathbf{Z} \simeq g^{\mathbf{Z}}$  acting on  $X$ . Then a sheaf on  $X/g^{\mathbf{Z}}$  is a sheaf  $\mathcal{F}$  with compatible isomorphisms

$$\mathcal{F} \simeq (g^n)^* \mathcal{F} \tag{15}$$

for every  $n \in \mathbf{Z}$ .

5.1.8. *Example.* We can consider  $\mathcal{G} = \mathbf{Z}$  acting on  $X$  generated by automorphism  $q$ , in which case a  $\mathcal{G}$ -differential equation is just a quasicoherent sheaf  $M$  along with compatible automorphisms

$$q^* M \simeq M.$$

Examples of this are when  $X$  is itself a group and the automorphism is action by a point  $q \in X$ .

For instance, if  $\mathcal{V} = V \otimes \mathcal{O}_X$  is the trivial vector bundle, then the sections consist of functions  $f(x)$  with

$$q \cdot f(x) = f(qx) = f(x).$$

5.1.9. *Example: groups.* We want to globalise the above construction, making  $g$  vary over  $G = \text{Aut}(X)$ . We form

$$(X \times G)/\mathbf{Z} \rightarrow G$$

where  $n \in \mathbf{Z}$  acts as  $n \cdot (x, g) = (g^n \cdot x, g)$ . A sheaf on the quotient coming from

We form

$$(X \times G)/G$$

Let  $X = G$  be a group, with an action of itself  $\underline{G} = G$ . Then the group scheme over  $X$  given by

$$\underline{G}_G = \underline{G} \times G$$

5.1.10. *Example.* We can construct “mixed” examples as follows. Say a two dimensional torus  $T \simeq \mathbf{G}_m \times \mathbf{G}_m$  acts on  $X$ , and  $v = (1, 0) \in \mathfrak{t}$  and  $q = (1, t) \in T$ . Then we can take

$$\mathcal{G} = \exp(\mathcal{O} \cdot v) \times \mathbf{Z} \cdot q.$$

Loosely speaking, a  $\mathcal{G}$ -differential equation is a connection along the flowlines of action of the first  $\mathbf{G}_m$ , and a difference equation along the second.

For instance, take the universal curve

$$\pi : \mathcal{E} \rightarrow \mathcal{M}_{1,1}$$

and consider both:

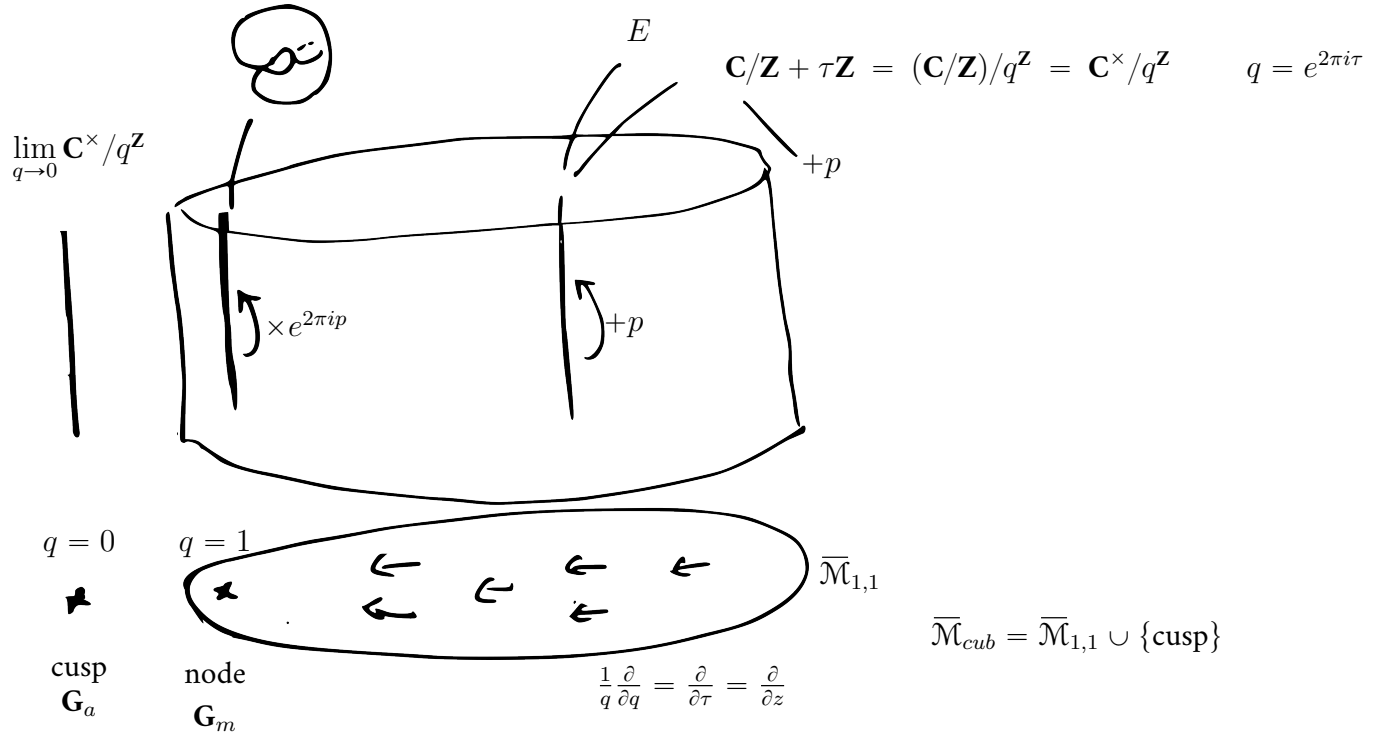
- an automorphism  $+p$  on  $\mathcal{E}_\tau$  given by adding a point, (we need to specify a point, i.e. work with  $\mathcal{M}_{1,2}$ , or quotient by all of  $\text{Aut}(\mathcal{E}_\tau)$ ),
- vector fields on the base,  $\mathcal{T}_{\mathcal{M}_{1,1}}$ .

In particular, we have that

$$\mathcal{E}_{dR} = \mathcal{E} / (p^{\mathbf{Z}} \times \exp(\pi^* \mathcal{T}_{\mathcal{M}_{1,1}}))$$

and so  $M \in \text{QCoh}(\mathcal{E}_{dR})$  corresponds to a quasicoherent sheaf with an action of differential operators on the base and an automorphism of the fibre. In an important example of  $M$  in [FTV3], these are called the *heat equation* and the *qKZB equation*, respectively.

5.1.11. *The picture of all cubic curves.*



$$\mathbf{P}(4, 6)_z = \overline{\mathcal{M}}_{1,1} = (\mathbf{H}_{\tau} \cup \mathbf{P}(\mathbf{Q}))/\mathrm{PSL}(2, \mathbf{Z})$$

The variants of the  $q$ KZ equations are all equivariance with respect to various automorphisms of the universal curve  $\mathcal{E}$ . The vector fields  $\partial_z$  (should be  $\bar{\partial}_z$ ?) are *gauge fields* coming from the fact that the HT theory depends holomorphically on the curve's moduli. The discrete automorphism by  $\mathbf{Z}$  is given by the fact that we are evaluating the HT theory on the *quotient* of the HT manifold

$$(\mathbf{R} \times \mathcal{E})/\mathbf{Z} \rightarrow \overline{\mathcal{M}}_{cub}.$$

The  $q = 1$  limit gives the multiplicative KZ equation, and the exponential map

$$\exp : \mathbf{C} \rightarrow \mathbf{C}^{\times} \subseteq E_{\text{node}}$$

gives the additive  $q$ KZ equation.

In physics,  $(p, e^{2\pi i p})$  is usually written  $(\hbar, q)$ , but this is not the same as the  $q$  above so we do not to avoid confusion. In particular, there are *four* limits one can consider

|                            | $q \rightarrow 0$ | $q \rightarrow 1$ |
|----------------------------|-------------------|-------------------|
| $\hbar \rightarrow 0$      |                   |                   |
| $\hbar \rightarrow \infty$ |                   |                   |

where the  $q$  limits correspond to  $\mathbf{G}_a$  and  $\mathbf{G}_m$  formal group laws;<sup>6</sup> note that we have a family of formal group laws over  $\overline{\mathcal{M}}_{cub}$ , induced by the relative pointed curve  $\mathcal{E}$ .

## 5.2. Digression: loop spaces and HKR.

5.2.1. The *derived loop space* of a scheme  $X$  is

$$LX = X \times_{X \times X} X$$

whose algebra of functions is Hochschild chains  $\mathcal{O}(X) \otimes_{\mathcal{O}(X \times X)}^L \mathcal{O}(X)$ .

5.2.2. *Example.* For instance, if  $X = \text{Spec} \mathbf{C}[x]$ , then we compute  $\mathcal{O}(LX) = \mathbf{C}[x] \otimes_{\mathbf{C}[x,y]}^L \mathbf{C}[x]$  using the Koszul resolution of  $\mathbf{C}[x]$  as a  $\mathbf{C}[x, y]$ -module,

$$\mathcal{O}(LX) = (\mathbf{C}[x, y] \xrightarrow{y} \mathbf{C}[x, y]) \otimes_{\mathbf{C}[x,y]} \mathbf{C}[x] \simeq \mathbf{C}[x] \oplus \epsilon \cdot \mathbf{C}[x]$$

where the first summand is generated by  $1 \otimes 1$  and the second by the degree 1 element  $\epsilon = y \otimes 1$ . As an  $\mathbf{C}[x]$ -algebra, this satisfies  $\epsilon^2 = 0$ , and so

$$\mathcal{O}(LX) \simeq \wedge_{\mathbf{C}[x]}^\bullet T_{\mathbf{C}[x]}$$

is the algebra of polyvector fields on  $X$ .

5.2.3. Note that we have

$$\mathcal{O}(L_{\mathbf{G}_a} X) = \text{Maps}(\mathbf{B}\mathbf{G}_a, X) = \wedge_X^\bullet \mathbf{L}_X$$

is the de Rham complex without the differential, and by [BN12, Thm. 1.3] the differential is encoded by the translation action of  $\mathbf{B}\mathbf{G}_a$ . In particular, we have by [BN12, Thm. 1.5] that if  $X$  is a smooth underived scheme, an equivalence to the category of equivariant sheaves

$$(\wedge_X^\bullet \mathbf{T}_X^*, d_{dR})\text{-Mod} \simeq \text{QCoh}_{S^1}(L_{\mathbf{G}_a} X) \simeq \text{QCoh}_{\mathbf{B}\mathbf{G}_a}(L_{\mathbf{G}_a} X). \quad (16)$$

Given this, we use the Koszul duality

$$(\wedge_X^\bullet \mathbf{T}_X^*, d_{dR}) \simeq \mathcal{D}(X)$$

to conclude that all of (16) is equivalent to the category  $\mathcal{D}\text{-Mod}(X)$ .

To understand where the  $\text{Vect}(X)$ -action comes from, note

$$(\text{Sym}^\bullet \mathbf{T}_X, 0)\text{-Mod} \stackrel{\text{KD}}{\simeq} (\wedge_X^\bullet \mathbf{T}_X^*, 0)\text{-Mod} \simeq \text{QCoh}(L_{\mathbf{G}_a} X) \quad (17)$$

and the  $\mathbf{T}_X$ -action on the right turns into the vector field action after we turn on the de Rham differential.

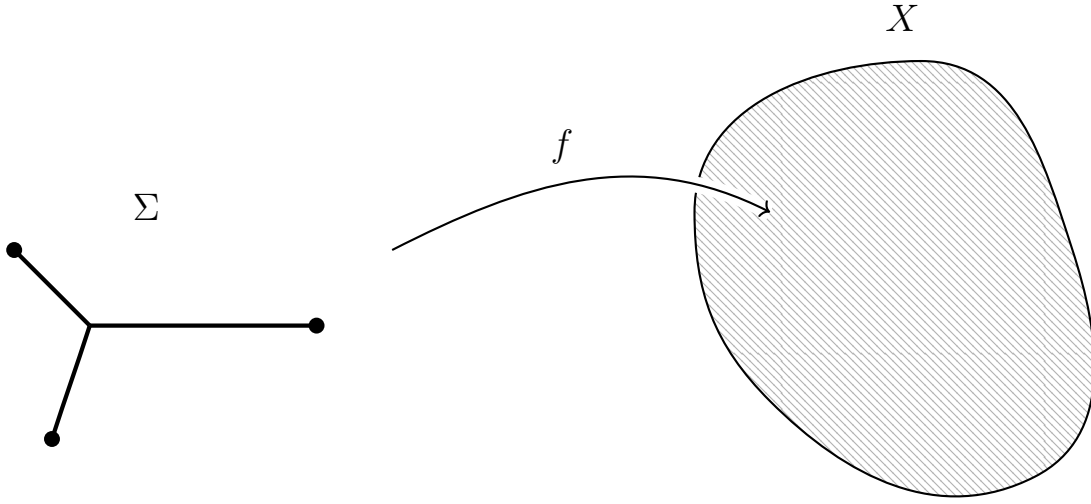
Thus, if we want to consider *difference*, *multiplicative difference* or *kZB heat* equations instead of differential equations, we need to first generalise (17). The right side has been generalised to arbitrary (algebraic) curves  $C$  replacing  $\mathbf{G}_a$ , by [BK25]. However the definition is the same for  $\mathbf{G}_a, \mathbf{G}_m, E$  if  $X$  is

<sup>6</sup>See <https://www.math.wustl.edu/~matkerr/436/ch17.pdf>

a scheme, so it is expected that we need to consider the analytic curves  $\mathbf{G}_a/\mathbf{Z}\hbar$ ,  $\mathbf{G}_m/q^{\mathbf{Z}}$ ,  $\mathcal{E}/\mathbf{Z} \times \mathbf{Z}$  where  $\mathcal{E}$  is the universal elliptic curve.

### 5.3. Background: quantum cohomology and quasimaps.

5.3.1. *Cohomology.* Let  $X$  be a topological space. We can consider the cup product on  $H^*(X)$  as arising from the following singular cobordism  $\Sigma \simeq \text{pt}$  between  $\text{pt} \sqcup \text{pt}$  and  $\text{pt}$ . Namely, consider maps from  $\Sigma$  into  $X$ :



This induces for us a correspondence

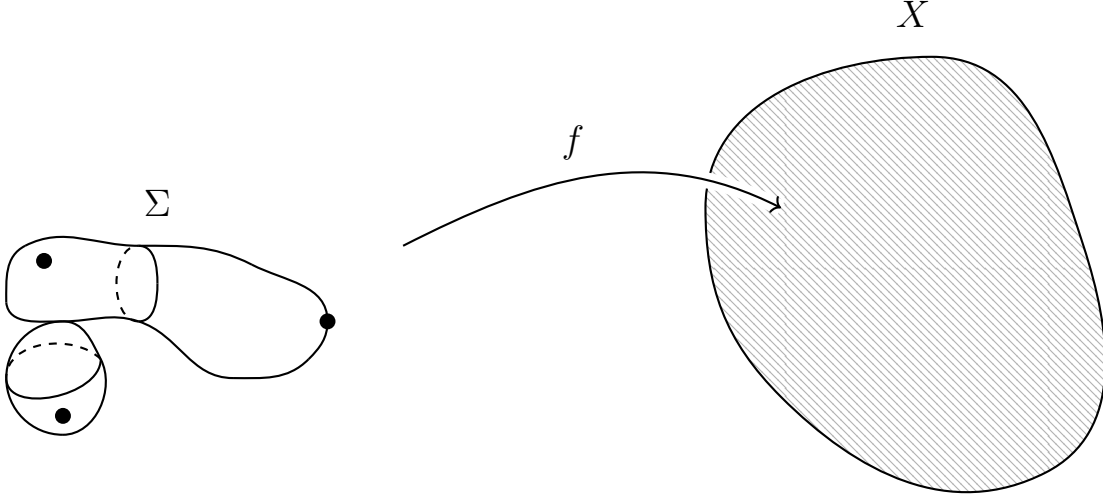
$$\begin{array}{ccc} & \text{Maps}(\Sigma, X) & \\ q \swarrow & & \searrow p \\ X \times X & & X \end{array}$$

which is homotopic to the ordinary diagonal correspondence  $X \times X \leftarrow X \rightarrow X$ , and so the pull-push map

$$p_* q^* : H^*(X) \otimes H^*(X) \rightarrow H^*(X)$$

gives the cup product.

5.3.2. *Quasimaps and Deligne–Mumford stacks.* Let us now consider  $X = Y//_{\theta} G$  a GIT quotient, and let  $\Sigma$  be a *prestable curve*: one with only nodal singularities and a unique connected component, see [St25, 0E6S].



There is now an entire stack  $\text{Maps}(\Sigma, X)$  of such curves and of such maps, moreover the space  $\text{Maps}_\beta(\Sigma, X)$  with a fixed image of  $[\Sigma] \in H_2(\Sigma)$ . We define two substacks

$$\overline{\mathcal{M}}_\Sigma(X, \beta) \hookleftarrow \mathcal{M}_\Sigma(X, \beta) \hookrightarrow \text{Maps}_\beta(\Sigma, \mathcal{X}) \xrightarrow{\circ} \text{QMaps}_\beta(\Sigma, X)$$

where  $\mathcal{X} = Y/G$  is the *stack* quotient.

**Definition 5.3.3.** (Ciocan–Fontanine–Kim–Maulik) Let  $X = Y//_\theta G$  be a GIT quotient. A **quasimap** from a pointed curve  $(\Sigma, p_1, \dots, p_n)$  is a map

$$\bar{f} : \Sigma \rightarrow Y/G$$

such that the inverse image of  $Y^s/G$  is dense. It is called **stable** if all nodes and marked points are mapped to  $Y^s/G$ , and<sup>7</sup> if  $\omega_\Sigma(p_1 + \dots + p_n) \otimes \bar{f}^* \mathcal{L}_\theta^\epsilon$  is ample for every positive rational  $\epsilon$ , where  $\mathcal{L}_\theta$  is the line bundle on  $Y/G$  corresponding to the character  $\theta$ .

If  $Y$  is affine and  $G$  reductive the space of quasimaps is proper.

**Lemma 5.3.4.** If  $X$  is a Nakajima quiver variety, then  $\text{QMaps}_{p_1, \dots, p_n}(\Sigma, X)$  is a scheme.

5.3.5. *Quantum cohomology.* Next, we consider the Deligne–Mumford stack

$$\overline{\mathcal{M}}_{0,3}(X)$$

of stable maps from genus zero curves with three marked points into a smooth proper complex scheme  $X$ . Here, we have a genuine map to  $X$ , which is fixed, and we allow  $\Sigma$  to vary. The *quantum cohomology* is

$$\text{QH}_T(X) = H_T^\bullet(X)[[q^{H^2(X, \mathbb{Z})}]]$$

with product  $\cup_q$  defined by pull-pushing along

<sup>7</sup>This last point ensures separatedness

$$\begin{array}{ccc} & \overline{\mathcal{M}}_{0,3}(X) & \\ q \swarrow & & \searrow p \\ X \times X & & X \end{array}$$

Note that the conditions on  $X$  imply that  $p$  is smooth and proper as a map of Deligne–Mumford stacks, so we can push forward along it. The connected components of  $\overline{\mathcal{M}}_{0,3}(X)$  are labelled by the degree  $\deg f \in H^2(X, \mathbf{Z})$ , and so we set

$$\alpha \cup_z \beta = \sum_{d \in H^2(X, \mathbf{Z}) > 0} z^d p_{d,*}(q_d^*(\alpha \otimes \beta))$$

where  $q_d$  and  $p_d$  are the restrictions of  $q$  and  $p$  to the degree  $d$ th component.

**5.3.6.  $\sigma$ -twisted quasimaps.** Let  $G$  act on  $X$  and let  $\sigma : \mathbf{G}_m \rightarrow G$  be a cocharacter. We define the  $X$ -bundle over  $\mathbf{P}^1$

$$\begin{array}{ccccc} X_\infty & \longrightarrow & X^\sim = ((\mathbf{C}^2 \setminus 0) \times X)/\mathbf{G}_m & \longleftarrow & X_0 \\ \downarrow & & \downarrow & & \downarrow \\ \infty & \longrightarrow & \mathbf{P}^1 & \longleftarrow & 0 \end{array}$$

Note that its total space inherits an action of  $G^\sim = G \times \mathbf{C}_\epsilon^\times$  on  $X^\sim$ ,  $\mathbf{C}_\epsilon^\times$  acts by scaling the second factor of  $\mathbf{C}^2$ .

The **shift operator** is

$$S(\sigma) : H_{G^\sim}^\bullet(X_0) \rightarrow H_{G^\sim}^\bullet(X_\infty)[[q^{H_2(X^\sim; \mathbf{Z})}]]$$

given by pull-pushing along the roof of

$$\begin{array}{ccccc} & M^\sim(X) & & & \\ & \swarrow \downarrow \searrow & & & \\ X_0 & & & & X_\infty \\ \downarrow & & \downarrow & & \downarrow \\ & \overline{\mathcal{M}}_{0,2}(\mathbf{P}^1, X^\sim) & & & \\ \downarrow \swarrow \searrow & & & & \\ X^\sim & & & & X^\sim \end{array}$$

summing over classes in  $H_2(X^\sim; \mathbf{Z})$  pushing down to  $[\mathbf{P}^1] \in H_2(\mathbf{P}^1; \mathbf{Z})$ .

#### 5.4. The $q$ KZ equation (Frenkel–Reshetikhin).

**5.4.1.** Let  $V_1, \dots, V_n$  be finite-dimensional representations of  $\mathfrak{g}$ , or equivalently of  $U_h(\mathfrak{g})$ .<sup>8</sup> We define the differential and difference equations on the trivial vector bundle  $\mathbf{V}$  with fibre  $V_1 \otimes \dots \otimes V_n$ .

<sup>8</sup>In the non-formal case we need to begin with representations of  $U_q(\mathfrak{g})$ .

5.4.2. *Warning.* Although we use the same vector bundle on the same base, they play very different roles in the two settings. In the KZ equations, it is the *fibre*

$$\mathbf{V}_{x_1, \dots, x_n} = V_1 \otimes \cdots \otimes V_n$$

which is a product of  $U_h(\mathfrak{g})$ -modules, whereas in the  $q$ KZ equations, it is (for  $\Sigma = \mathbf{C}, \mathbf{C}^\times$ ) the *space of sections*

$$\Gamma((\Sigma^n)_\circ, \mathbf{V}) = V_1(z_1) \otimes \cdots \otimes V_n(z_n) [(z_i - z_j)^{-1}]_{i \neq j}$$

which will be a product of  $Y_h(\mathfrak{g}), U_h(\widehat{\mathfrak{g}})$ -modules formed by restriction along the family of evaluation homomorphisms

$$\text{ev}_z : Y_h(\mathfrak{g}), U_h(\widehat{\mathfrak{g}}) \rightarrow U_h(\mathfrak{g})$$

depending on  $z$ , so that  $V(z) = V \otimes \Gamma(\Sigma, \mathcal{O})$  and we localise away from the diagonals in  $\Sigma^n$ . When  $\Sigma = E$  we need to consider the sheaf  $\mathbf{V}$  itself.

5.4.3. *Warning.* There is a *Kazhdan–Lusztig* equivalence of braided monoidal categories

$$\text{Rep}^{f.d.} U_q(\mathfrak{g}) \simeq \text{Rep}_\kappa \widehat{\mathfrak{g}}$$

where  $q = e^{\pi i/(\kappa + h^\vee)}$ , and the right hand side is identified with certain (smooth) modules for  $U_\kappa(\widehat{\mathfrak{g}})$ . Under this equivalence, the KZ equations for  $U_q(\mathfrak{g})$ -modules are identified with equations on the space of  $\widehat{\mathfrak{g}}$ -intertwiners between tensor products of  $U_\kappa(\widehat{\mathfrak{g}})$ -modules. It is thus a priori not obvious which side one should generalise from. [FR92] takes the view of considering intertwiners of  $U_{q', \kappa}(\widehat{\mathfrak{g}})$ -modules, i.e. modules of level  $\kappa$  over the  $q'$ -deformed algebra.

5.4.4. *Differential equations.* Let  $\Sigma$  be a Riemann surface and

$$r(z_1, z_2) : (\Sigma \times \Sigma) \setminus \Sigma \rightarrow \mathfrak{g} \otimes \mathfrak{g}, \text{ i.e. } r(z_1, z_2) \in \mathfrak{g} \otimes \mathfrak{g}(z_1, z_2)$$

is an antisymmetric function  $r(z_2, z_1) = -r(z_1, z_2)$  satisfying the *classical Yang–Baxter equation*:

$$[r_{12}(z_1, z_2), r_{13}(z_1, z_3)] + [r_{12}(z_1, z_2), r_{23}(z_2, z_3)] + [r_{13}(z_1, z_3), r_{23}(z_2, z_3)] = 0,$$

Here  $(z_1, z_1) = \mathcal{O}(\Sigma \times \Sigma \setminus \Sigma)$ . We then we define the associated **KZ equation** on  $(\Sigma^n)_\circ$  as the vanishing of the set of differential operators on  $\mathbf{V}$ :

$$\nabla_i = \hbar \sum_{i > j} r_{ij}(z_i, z_j) + \partial_{z_i} - \hbar \sum_{i < j} r_{ij}(z_j, z_i)$$

where  $\{\partial_{z_i}\}$  is a commuting set of vector fields on  $(\Sigma^n)_\circ$ . We have used antisymmetry of  $r$  to write the above in a form which will generalise better.

**Lemma 5.4.5.** (Belavin–Drinfeld) *The above implies that  $\Sigma \in \{\mathbf{C}, \mathbf{C}^\times, E\}$  is a one-dimensional algebraic group, with*

$$r(z_1, z_2) = \frac{\Omega}{z_1 - z_2}, \quad r(z_1, z_2) = \frac{\tilde{r}z_1 + \tilde{r}^{21}z_2}{z_1 - z_2}, \quad r(z_1, z_2) = r^E(z_1, z_2)$$

respectively, where  $r^E(z_1, z_2)$  is given in terms of elliptic functions on  $E$  as in [ES].

Any such  $\Sigma$  admit a nowhere vanishing vector field, given by the translation action of the algebraic group. The KZ equations in these cases are called *additive*, *multiplicative* and *elliptic* KZ equations, respectively.

5.4.6. *Difference equations.* If  $\Sigma$  is a Riemann surface and

$$R(z_1, z_2) : (\Sigma \times \Sigma) \setminus \Sigma \rightarrow U_{\hbar}(\mathfrak{g}) \otimes_{\hbar} U_{\hbar}(\mathfrak{g}),$$

$$\text{i.e. for } \Sigma = \mathbf{C}, \mathbf{C}^{\times}, \text{ an element } R(z_1, z_2) \in (Y_{\hbar}(\mathfrak{g}) \otimes_{\hbar} Y_{\hbar}(\mathfrak{g}))_{\text{loc}}, (U_{\hbar}(\widehat{\mathfrak{g}}) \otimes_{\hbar} U_{\hbar}(\widehat{\mathfrak{g}}))_{\text{loc}}$$

is a function with leading term  $1 \otimes 1 + \mathcal{O}(\hbar)$ , and satisfying the *spectral Yang–Baxter equation*:

$$R_{12}(z_1, z_2) R_{13}(z_1, z_3) R_{23}(z_2, z_3) = R_{23}(z_2, z_3) R_{13}(z_1, z_3) R_{12}(z_1, z_2).$$

**Lemma 5.4.7.**  $R(z_1, z_2) = 1 \otimes 1 + \hbar r(z_1, z_2) + \mathcal{O}(\hbar^2)$  where  $r(z_1, z_2)$  is a solution to the classical Yang–Baxter equation.

We define the associated  **$q$ KZ difference equation** on  $(\Sigma^n)_{\circ}$  by the set of difference operators

$$\varphi_i = \prod_{i>j} R_{ij}(\kappa_i(z_i), z_j) \cdot d_i \cdot \prod_{i<j} R_{ji}(z_j, z_i)^{-1}$$

where  $\{\kappa_i\}$  are a set of commuting automorphisms of  $\Sigma$  and  $\{d_i\}$  is a commuting set of automorphisms on  $(\Sigma^n)_{\circ}$  such that each  $d_i \otimes d_j$  commutes with  $R_{ij}$ . We suggest  $d_i = \kappa_i$ , assuming that all  $\kappa_i = \kappa$  agree and  $R_{ij}$  is automorphism-invariant:  $R_{ij}(\kappa(z_i), \kappa(z_j)) = R_{ij}(z_i, z_j)$ .

5.4.8. *Remark.* Thus, the  $q$ KZ limits to the KZ equation. This should follow from the Baker–Campbell–Hausdorff formula, but we will not write out the details here.

5.4.9. Thus, a solution to the  $q$ KZ equations is a  $V_1 \otimes \cdots \otimes V_n$ -valued function  $f(z_1, \dots, z_n)$  such that

$$f(z_1, \dots, \kappa_i(z_i), \dots, z_n) = \varphi_i(f(z_1, \dots, z_n))$$

for each  $i$ .

## 5.5. Construction of solutions to $q$ KZ using quasimaps.

5.5.1. *Computation of the  $q$ KZ equation in terms of quiver varieties [MO12, 1.4.5].* Let  $X^{\sigma}$  be the fixed points of a quiver variety with respect to a character  $\sigma : \mathbf{C}^{\times} \rightarrow \mathbf{G}_w$ .

**Proposition 5.5.2.** [MO12, 1.4.5] Then the  $q$ KZ equation on  $(\mathbf{C}^{\times})_{\circ}^2$  on two tensor factors is

$$\Psi(t + \sigma) = (-1)^{\frac{1}{2} \text{codim} X^{\sigma}} (q^v \otimes 1) R_{\sigma} \cdot \Psi(t)$$

as a  $H_{\top}^*(X^{\sigma}) \otimes \mathbf{Q}(q^{\beta})$ -valued function of  $t \in \mathfrak{t}$ , where  $R_{\sigma}$  is the  $R$ -matrix of the chamber  $\sigma$ .

5.5.3. *Computing a solution.* We pick an explicit element, called the *capping operator*. First we consider

$$\begin{array}{ccc} & \mathrm{QMap}_{C,p_1,p_2}^d(X) & \\ \mathrm{ev}_2 \swarrow & & \searrow \mathrm{ev}_1 \\ X & & X \end{array}$$

for a fixed curve  $C$  with marked points  $p_1, p_2$  and  $d \in H_2(X, \mathbf{Z})$ . When  $C = \mathbf{P}^1$  and  $X$  are acted on by  $\mathbf{C}_q^\times = \mathrm{Aut}(\mathbf{P}^1, 0, \infty)$ , we define the **capping operator**

$$J(u, z) = \sum_d z^d \cdot (\mathrm{ev}_1 \times \mathrm{ev}_2)_*(\hat{\mathcal{O}}_{\mathrm{vir}}) \in K_{G \times \mathbf{C}_q^\times}(X)_{\mathrm{loc}}^{\otimes 2} \otimes \mathbf{Q}[[z]]$$

where  $z$  are the Kahler variables,  $u \in G$  are equivariant variables, and we *define* the pushforward using torus localisation.

Let  $X$  be a quiver variety and  $\sigma : \mathbf{C}^\times \rightarrow G_w$  be a cocharacter.

**Theorem 5.5.4.** [Ok15, Thm. 8.2.20] *The capping operator satisfies*

$$J(q^\sigma u) = \frac{S(\sigma)}{E(\sigma)} \cdot J(u)$$

where  $E(\sigma)$  is some torus localisation term, and moreover,  $S(\sigma)/E(\sigma)$  is the  $qKZ$  difference operator.

*Proof.* Write

$$E(\sigma) = \frac{\text{edge term for } f_\sigma}{(\text{edge term for } f_0)|_{u \rightarrow q^\sigma u}}$$

where  $f_\sigma : \Sigma \rightarrow X^\sigma$  is a constant map to some  $x \in X^\sigma$ .

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□

## 5.6. A Drinfeld–Kohno theorem for $qKZ$ ?

5.6.1. A Drinfeld–Kohno Theorem for  $U_q(\hat{\mathfrak{g}})$  was announced but not proven by Gautam–Toledano-Laredo.

5.6.2. *What is the structure on  $\mathrm{Rep} U_q(\hat{\mathfrak{g}})$ ?* Remark that we want the limit as  $z \rightarrow 0, \infty$  to define an ordinary braided monoidal category, [FR92, p.49].

## 5.7. Casimir connections.

5.7.1. If  $\mathfrak{g}$  is a finite-dimensional simple Lie algebra over the complex numbers, we have *two* natural vector spaces on which things are defined:

- (1)  $\mathbf{C}^n$ , the base of the KZ equations,
- (2)  $\mathfrak{t}$ , the Cartan.

Likewise, if  $\mathcal{M}(v, w)$  is a quiver variety, we have two natural groups:

- (1) the gauge group  $G_w$ ,
- (2) the picard group  $\text{Pic}(\mathcal{M}(v, w))$ .

Note that these are related by  $\mathbf{C}^n = \text{Spec} H_{G_w}^\bullet(\text{pt})$  and  $\mathfrak{t} = H_2(\mathcal{M}(v, w), \mathbf{C})$ .

It turns out that we have a flat connection on the *product*  $\mathbf{C}^n \times \mathfrak{t}$ . In other words, we have a connection on  $\mathfrak{t}$  which *commutes* with the relevant KZ connections on  $\mathbf{C}^n$ .

5.7.2. *Differential equations case.* The **Casimir connection** over  $\mathfrak{t}_{reg} = \mathfrak{t} \setminus \cup_{\alpha \in \Delta} H_\alpha$  is a flat connection on the vector bundle  $\mathbf{V} = V \otimes \mathcal{O}$  whose fibre is a *single* representation of  $U_h(\mathfrak{g})$ , namely

$$\nabla = d - \hbar \sum_{\alpha \in \Delta_+} \frac{d\alpha}{\alpha} \Omega_\alpha$$

where  $\Omega_\alpha \in \text{Sym}^2 \mathfrak{g}$  is the Casimir element for the  ${}_2$ -triple corresponding to  $\alpha$ . It is flat and commutes with the KZ connection on  $\mathbf{C}^n$ .

This gives an action of the quantum Weyl group on the fibre  $V$ . One can consider the KZ-Casimir connection on  $(\mathbf{C}^n)_\circ \times \mathfrak{t}_{reg}$ , and the monodromy of this gives an action of the affine braid group on  $V$ .

5.7.3. Consider the **Casimir connection** (alias **Dubrovin connection** or **quantum differential equation**) given by

$$\nabla_\lambda = \frac{d}{d\lambda} - \lambda \cup_h (-) : H_G^\bullet(X) \rightarrow H_G^\bullet(X) \otimes \mathbf{Q}(q^\beta)$$

where  $\frac{d}{d\lambda} q^\beta = \langle \lambda, \beta \rangle q^\beta$  and  $\cup_h$  is the quantum cup product, and  $\lambda \in H_G^\bullet(X)$ . We likewise define

$$\nabla_\lambda^\sigma = \text{Stab}_+^{-1} \nabla_\lambda \text{Stab}_+ : H_T^\bullet(X^\sigma) \rightarrow H_T^\bullet(X^\sigma) \otimes \mathbf{Q}(q^\beta)$$

as opposed to  $R_\sigma = \text{Stab}_-^{-1} \text{Stab}_+$ .

**Theorem 5.7.4.** [MO12, Thm. 9.3.1] *Let  $X$  be a symplectic resolution such that  $\sigma$  preserves  $\omega$  and is miniscule, i.e.  $\Gamma(X, \mathcal{O}_X)$  is generated by functions of  $\sigma$ -weight in  $\{-1, 0, 1\}$ . Then we have*

$$\text{Stab}_+^{-1} \underbrace{S(\sigma)}_{\text{defined using quantum cohomology}} \text{Stab}_+ = \underbrace{(-1)^{\frac{1}{2} \text{codim} X^\sigma} (q^{v_1} \otimes \text{id}) R_{\sigma, \infty}}_{q\text{KZ difference operator}}$$

where  $R_{\sigma, \infty} \in \text{End} H_{G^\sim}^\bullet(X_\infty^\sigma) \otimes \mathbf{Q}(\mathfrak{g}^\sim)$  is the  $R$ -matrix for the action of  $\sigma$  on  $X_\infty$ .

This implies [MO12, Thm. 1.4.1] that

**Corollary 5.7.5.**  $\text{Stab}_+^{-1}S(\sigma)\text{Stab}_+$  commutes with the  $q\text{KZ}$  difference connection

$$\Psi(t + \sigma) = \pm q^\zeta R_\sigma \Psi(t)$$

where  $R_\sigma$  is the  $R$ -matrix for the action of  $\sigma$  on  $X$ .

#### 5.7.6. Shift operators and quantum multiplication.

**Theorem 5.7.7.** [MO12, §8.3] The  $\epsilon = 0$  specialisation of  $S(\sigma)$  can be identified with quantum multiplication by

$$S(\sigma)|_{\epsilon=0}(1) \in H_{G^\sigma}^\bullet(X)[[q^{H_2(X;\mathbb{Z})}]]$$

5.7.8. *Example.* For quiver varieties,  $X^\sigma = \bigsqcup \mathcal{M}(v, w) \times \mathcal{M}(v', w')$  and  $q^\zeta = q^v \otimes 1$ .

From the Theorem above about stable envelopes and quantum multiplication commuting, the above gives:

**Corollary 5.7.9.**  $(q^v \otimes 1)R(u)$  coincide with quantum multiplication.

5.7.10. *Danilenko's Theorem.* Pick a Weyl chamber  $\mathfrak{C} \subseteq \mathfrak{h} \subseteq \mathfrak{g}$ . This gives a choice of Casimir  $\Omega$ , hence of a trigonometric KZ connection

$$\nabla_i^{\text{KZ}} = z_i \partial_{z_i} + h_i + \hbar \sum_{i \neq j} \frac{z_i \Omega_{ij, \mathfrak{C}} - z_j \Omega_{ji, -\mathfrak{C}}}{z_i - z_j}$$

on  $\underbrace{\mathbb{C}^n}_{H_T^2(\text{pt})} \times \mathfrak{h} \times \mathbb{C}_h$ , where  $h_i \in \mathfrak{h}$ .

**Theorem 5.7.11.** [Da20] For  $G$  simply laced the map

$$(V_{\lambda_1} \otimes \cdots \otimes V_{\lambda_n})_\mu \xrightarrow{\sim} H_{T, \bullet}^{\text{BM}}(\mathcal{M}_C^T) \xrightarrow{\text{Stab}_{\mathfrak{C}}} H_{T, \bullet}^{\text{BM}}(\mathcal{M}_C)_{T-\text{loc}}$$

of vector bundles over  $\mathbb{C}^n \times \mathfrak{h}$  intertwines

$$\begin{array}{ccc} \nabla_i^{\text{KZ}}, & \text{and} & c_1(\mathcal{E}_i)^* \\ z_i, & \text{and} & q^{e_i} \end{array}$$

where  $V_\lambda$  is irreducible finite dimensional with highest weight  $\lambda$ .

Here,  $T$  is the framing torus and  $H$  is the internal torus.

**General principle.** We have taken cohomology twice,

$$H_{T, \bullet}^{\text{BM}}(\mathcal{M}_C) = H_{T, \bullet}^{\text{BM}}(\text{Spec} H_{\bullet}^{\text{BM}}(\mathcal{B}_z, m))$$

and therefore we can try to apply any pair of cohomology theories:

$$A_T^\bullet(B_H^\bullet(\mathcal{B}_z))$$

and recall that cohomology theories are labelled by one-dimensional formal groups:

|                      |          |                      |          |              |
|----------------------|----------|----------------------|----------|--------------|
|                      | <b>C</b> | <b>C<sup>×</sup></b> | <i>E</i> | ( $\Sigma$ ) |
| <b>C</b>             |          |                      |          |              |
| <b>C<sup>×</sup></b> |          |                      |          |              |
| <i>E</i>             |          |                      |          |              |

KZ  
*q*KZ  
*q*KZB heat

There is also *critical cohomology* versions.

PART II

ENUMERATIVE GEOMETRY AND  
WALL-CROSSING

## 6. Plan

**6.1. Lecture 1 and 2: Hall algebras and DT invariants.** Go back and edit this as necessary to respond to background needed for Lectures 3-6.

6.1.1. *Big picture.* Explain what structure we expect attached to a CY3.

6.1.2. *Pre-CoHA.* Motivic Hall algebras attached to  $(Q, W)$ , relation to Joyce's motivic Hall algebra, Reinecke's Poisson algebra, finite Hall algebra over  $\mathbf{F}_q$ .

$$\begin{array}{ccc}
 K^\bullet(\mathcal{X}) & \xrightarrow{ch} & H_\bullet^{BM}(\mathcal{X}) \\
 & \nwarrow \scriptstyle f_* (\mathcal{O}_{\mathcal{Y}}) & \nearrow \scriptstyle \text{Trace of } f_!(1_{\mathcal{Y}}) \\
 \mathbf{Q}^{|\# \mathcal{X}(\mathbf{F}_q)|} & \xleftarrow{|\# \mathcal{Y}(\mathbf{F}_q)|} & K_0(\text{Stk}/\mathcal{X})
 \end{array}$$

Computation in preprojective case.

Motivic Milnor function.

Behrend function, example  $\text{Hilb}_n(\mathbf{C}^3)$  and MacMahon

Integration maps (e.g. Joyce's to  $K_0(\text{Coh}(\mathcal{M}))$ ).

6.1.3. CoHAs and KHAs. Decomposition as a product over the Harder–Narasimhan strata, and the characters as relation to the DT invariants.

BPS Lie algebra, PBW and cohomological integrality.

**6.2. Lecture 3 and 4: Kontsevich–Soibelman wall-crossing and cluster varieties.**

6.2.1. Collect the different definitions of wall crossing structure and stability data.

Define the motivic Hall algebra.

**Definition 6.2.2.** (*Definition of stability data*)

**Definition 6.2.3.** (*Definition of wall-crossing structure*)

**Theorem 6.2.4.** (Existence of motivic DT invariants) *For any stability condition  $Z$  and sector  $V$ , there are invariants  $A_V^{Hall}$  and  $A_V^{mot}$ , and a homomorphism called the integration map*

$$\Phi : \hat{H}(\mathcal{C}) \rightarrow \hat{R}(\mathcal{C})$$

*such that  $\Phi(A_V^{Hall}) = A_V^{mot}$ .*

**Theorem 6.2.5.** (Relation to cluster coordinates)

**6.3. Lecture 5 and 6: Gromov–Witten invariants, scattering diagrams, Bridgeland–Joyce conjectures.**

## 7. Calabi–Yau categories

Let  $\mathcal{C}$  be a monoidal Calabi–Yau-three dg category.

- (1) Due to Costello <https://arxiv.org/pdf/math/0412149>, there is a two-dimensional TQFT—the *B model of strings inside*  $\mathcal{C}$ —whose value on the circle is the Hochschild homology

$$\mathcal{Z}(S^1) = \mathrm{HH}_\bullet(\mathcal{C}).$$

The product structure on this is given by the tensor product  $\otimes$  on  $\mathcal{C}$ . This theory is meant to be mirror to the Gromov–Witten *A* model, and works for any Calabi–Yau- $n$  category.

- (2) When  $n = 3$ , coming from short exact sequences in  $\mathcal{C}$

$$\begin{array}{ccc} & \mathrm{Ext}_{\mathcal{C}} & \\ \nearrow & & \nwarrow \\ \mathcal{C} \times \mathcal{C} & & \mathcal{C} \end{array}$$

we get algebra structures on many vector spaces attached to  $\mathcal{C}$ , such as  $\mathrm{HH}_\bullet(\mathcal{C})$ .

- (3) There are notions of *stable* objects in [extemporise](#)
- (4) Explain why things are critical loci.

<https://arxiv.org/pdf/0804.1274> is an excellent review

### 7.1. Calabi–Yau categories.

**Definition 7.1.1.** A **Calabi–Yau  $n$  category** is a dg category  $\mathcal{C}$  with a symmetric non-degenerate pairing of degree  $n$  on the morphism dg vector spaces:

$$\mathrm{RHom}_{\mathcal{C}}(c, d) \simeq \mathrm{RHom}_{\mathcal{C}}(d, c)^\vee[n].$$

7.1.2. *Example.* If  $X$  is a smooth proper CY3, then its dualising sheaf  $\omega_X \simeq \mathcal{O}_X$  is trivial, and Serre duality gives a pairing of the form above on  $D^b\mathrm{Coh}(X)$ , so  $D^b\mathrm{Coh}(X)$  is a CY3 category.

7.1.3. *Example.* Matrix factorisations are the categorified version of vanishing cycles for a function  $W : Y \rightarrow k$  on a separated scheme of finite type  $Y$  over  $k$ .

**Definition 7.1.4.** The category of **matrix factorisations**  $\mathrm{MF}(Y, W)$  has objects

$$E_0 \xrightarrow{\delta_0} E_1 \xrightarrow{\delta_1} E_0$$

a  $\mathbf{Z}/2$ -periodic complex of coherent sheaves on  $Y$  such that  $\delta^2 = W \cdot \mathrm{id}$ , modulo the subcategory of exact triples, and morphisms are maps of  $\mathbf{Z}/2$ -periodic complexes.

**Theorem 7.1.5.** <https://arxiv.org/pdf/1101.5834>, 8.3.4. If  $Y$  is a smooth scheme of dimension  $n$  with a nonvanishing top dimensional volume form, then  $\mathrm{MF}(Y, W)$  is a Calabi–Yau- $n$  category.

7.1.6. *Remark.* In physics <https://arxiv.org/pdf/1604.05365> the Calabi–Yau structure on matrix factorisations was introduced by Kasputin–Li. Let  $f$  be a function on affine open subset  $X \subseteq \mathbf{C}^n$  with isolated critical points whose only critical value is zero. Then consider a trivial  $\mathbf{Z}/2$ -graded vector bundle with an odd endomorphism  $D$  satisfying  $D^2 = f \cdot \text{id}$ . Then the Calabi–Yau- $n$  structure can be viewed as a pairing

$$\theta : \text{Ext}_{\text{MF}(X,f)}^n(x, x) \rightarrow \mathbf{C}$$

which is equivalent to the above by applying  $x = c + d[\bullet]$ . The Kasputin–Li formula is

$$\theta(\Phi) = \frac{1}{n!} \sum_{p \in \text{Crit} f} \text{Res}_p \frac{\text{tr}(dD)^n \Phi}{\partial_1 f \cdots \partial_n f}$$

7.1.7. *Example.* Let  $Q$  be a quiver and let  $\overline{W} \in \mathbf{C}Q/[\mathbf{C}Q, \mathbf{C}Q]$ , which we call a **potential**. There is a unique representative  $W \in \mathbf{C}Q$  which is *cyclic*, i.e. a linear combination of cyclic paths in  $Q$ , e.g.

$$W = e_0 e_1 e_2 + e_1 e_2 e_0 + e_2 e_0 e_1$$

for a cycle of length three in  $Q$  with edges  $e_i$ . We can then define the **Jacobian algebra**

$$\text{Jac}(Q, W) = \frac{\mathbf{C}Q}{\langle \partial W / \partial e : e \in Q_1 \rangle}$$

where  $\partial W / \partial e$  is formed by formally removing  $e$  from each word.

## 7.2. Various operations on categories.

### 7.2.1. Explain the philosophy

7.2.2. *Moduli stack of objects.* If  $\mathcal{C}$  is a dg category, there is an Artin stack (locally of finite type if  $\mathcal{C}$  is finite type)  $\mathcal{M}_{\mathcal{C}}$  whose  $k$ -points are identified with objects in  $\mathcal{C}$ :

$$\mathcal{M}_{\mathcal{C}}(k) = \text{Ob}(\mathcal{C})$$

and more generally,  $\mathcal{M}_{\mathcal{C}}(R) = \text{Hom}_{\text{dgCat}}(\mathcal{C}^{\text{op}}, \text{Perf} R)$  on a general  $k$ -algebra  $R$ . The assignment  $\mathcal{C} \mapsto \mathcal{M}_{\mathcal{C}}$  is visibly contravariant, and has a left adjoint given by taking  $\text{Perf}(-)$ . Prop 3.4 <https://arxiv.org/pdf/math/0503269>

**Theorem 7.2.3** ([PTVV; BBBJ]). *If  $\mathcal{C}$  is a CY3 dg-category (with appropriate finiteness, so that  $\mathcal{M}_{\mathcal{C}}$  exists as a derived Artin stack), then etale-locally there is an equivalence of  $d$ -critical stacks*

$$\mathcal{M}_{\mathcal{C}} \simeq \text{Crit}(W)$$

for some regular function  $W : \mathcal{Y} \rightarrow \mathbf{C}$  on a smooth scheme  $\mathcal{Y}$ .

7.2.4. *Example.* The moduli stack of the Jacobi algebra is given as follows. We have the smooth moduli stack  $\mathcal{M}_Q$  of quiver representations, and a function

$$\text{tr} W : \mathcal{M}_Q \rightarrow \mathbf{C}, \quad V \mapsto \text{tr}_V W.$$

Then  $\mathcal{M}_{\text{Jac}(Q,W)\text{-Mod}} \simeq \text{Crit}(\text{tr} W)$ .

7.2.5. (Critical)  $K$  theory.

7.2.6. *Toy model: constructible functions and  $\mathbf{F}_q$ -points.* Given a stack  $\mathcal{X}$  defined over  $\mathbf{Z}$ , we may consider the ring of functions

$$f : \mathcal{X}(k) \rightarrow \Lambda$$

valued in a ring  $\Lambda$ , which is constant on a locally closed stratification of  $\mathcal{X}_k$ . In what follows we will make this notion precise for  $k = \mathbf{C}, \mathbf{F}_q$ . In the latter case, the groupoid  $\mathcal{X}(\mathbf{F}_q)$  is often finite, in which the ring of such functions is simply a function on the  $\mathbf{F}_q$ -points.

7.2.7. *Grothendieck group of stacks.* Assume all stacks are locally of finite type over  $\mathbf{C}$  and with affine stabilisers.

**Definition 7.2.8.** For  $\mathcal{X}$  a stack,  $K(\mathrm{Stk}/\mathcal{X})$  is the abelian group spanned by stacks of finite type over  $\mathcal{X}$  modulo the following relations:

- (1)  $[\mathcal{Y}_1 \sqcup \mathcal{Y}_2] = [\mathcal{Y}_1] + [\mathcal{Y}_2]$ ,
- (2)  $[\mathcal{Y}_1] = [\mathcal{Y}_2]$  if there is a map  $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$  over  $\mathcal{X}$  giving a bijection on  $\mathbf{C}$ -points,
- (3)  $[\mathcal{Y}_1] = [\mathcal{Y}_2]$  whenever  $\mathcal{Y}_i \rightarrow \mathcal{X}$  are Zariski fibrations with the same fibres.

The second condition collapses to equivalence many different stacks, for instance

$$[\mathbf{P}^n] = \mathbf{L}^{n-1} + \mathbf{L}^{n-2} + \cdots + \mathbf{L} + 1$$

where we write  $\mathbf{L} = \mathbf{A}_{\mathcal{X}}^1$  and  $1 = \mathrm{pt}_{\mathcal{X}}$ . In the same way, we have that  $[\mathcal{Y}] = [\mathcal{U}] + [\mathcal{Z}]$  whenever  $\mathcal{Y} = \mathcal{U} \sqcup \mathcal{Z}$  is a union of complementary open and closed substacks, and more  $[\mathcal{Y}] = \sum [\mathcal{Y}_i]$  for any stratification.

The space of *stack functions* on  $\mathcal{Y}$  relative to a ring homomorphism  $K(\mathrm{Stk}/\mathbf{C}) \rightarrow \Lambda$  is

$$K(\mathrm{Stk}/\mathcal{Y}) \otimes_{K(\mathrm{Stk}/\mathbf{C})} \Lambda,$$

and has an interpretation due to Bridgeland and Joyce as the vector space on certain functions  $f : \mathcal{Y} \rightarrow \mathbf{C}$ .

**Lemma 7.2.9.** For any map  $f : \mathcal{X} \rightarrow \mathcal{X}'$ ,

- (1) there is a map  $f_* : K(\mathrm{Stk}/\mathcal{X}) \rightarrow K(\mathrm{Stk}/\mathcal{X}')$  given by composition with  $f$ ,
- (2) if  $f$  is of finite type, there is a map  $f^* : K(\mathrm{Stk}/\mathcal{X}) \rightarrow K(\mathrm{Stk}/\mathcal{X}')$  given by fibre product.

Moreover, these are compatible with composition and base change.

In particular, applying this to the diagonal map we get that

**Corollary 7.2.10.**  $K(\mathrm{Stk}/\mathcal{X})$  is a commutative algebra with respect to  $\times_{\mathcal{X}}$ .

7.2.11. *Hochschild homology.* Consider any **simplicial module**: a sequence of  $k$ -vector spaces  $k$ -vector spaces

$$\cdots \begin{array}{c} \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \end{array} M_2 \begin{array}{c} \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \end{array} M_1 \begin{array}{c} \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \end{array} M_0$$

with face  $d_i : M_n \rightarrow M_{n-1}$  and degeneracy  $s_i : M_{n-1} \rightarrow M_n$  maps satisfying

$$d_i d_j = d_{i-1} d_j, \quad s_i s_j = s_{j+1} s_i \quad d_i s_j = \begin{cases} s_{j-1} d_i & \text{if } i < j \\ \text{id} & \text{if } i = j, j+1 \\ s_j d_{i-1} & \text{if } i > j+1 \end{cases}.$$

The **(Hochschild) homology** of the simplicial module  $M_\bullet$  is the cohomology of

$$\cdots \xrightarrow{d} M_2 \xrightarrow{d} M_1 \xrightarrow{d} M_0$$

where  $d = \sum_{i=1}^n (-1)^i d_i$ .

7.2.12. *Example.* Let  $X$  be a topological space and  $M_n = \mathbf{C} \cdot \{\text{Maps}(\Delta^n, X)\}$  be the vector space spanned by  $n$ -chains in  $X$ . The Hochschild homology agrees with the homology of  $X$ .

7.2.13. *Remark.* In what follows, we will have examples coming from  $M_n = \mathbf{C} \cdot \{\text{Maps}(\Delta^n \times S^1, X)\}$ , where the loop parameter acts by

$$t : (x_0, \dots, x_n, \theta) \mapsto (x_1, \dots, x_n, x_0, \theta + x_0).$$

The Hochschild cohomology is  $H_\bullet(LX)$ , but there will be *three* relevant cohomology theories,

$$\pi_\bullet(\mathbf{C}_\bullet(LX)^{S^1}) \rightarrow \pi_\bullet(\mathbf{C}_\bullet(LX)^{tS^1}) \rightarrow H_\bullet^{S^1}(LX)$$

given by invariants, the Tate construction, and coinvariants (which is equivariant homology).

7.2.14. *Cyclic homology.* The definition of cyclic homology and its variants due to Loday <https://www.math.univ-paris13.fr/~vallette/GdT/Cyclic%20Homology%20-%20Loday.pdf> was made for any **cyclic module**: a sequence of  $k$ -vector spaces

$$\cdots \begin{array}{c} \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \end{array} M_2 \begin{array}{c} \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \end{array} M_1 \begin{array}{c} \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\quad} \end{array} M_0$$

$\begin{array}{ccc} \mathbf{Z}/3=\langle t \rangle & & \mathbf{Z}/2=\langle t \rangle \\ & \cap & \cap \\ & \text{---} & \text{---} \end{array}$

with face  $d_i : M_n \rightarrow M_{n-1}$  and degeneracy  $s_i : M_{n-1} \rightarrow M_n$  maps satisfying

$$d_i d_j = d_{i-1} d_j, \quad s_i s_j = s_{j+1} s_i \quad d_i s_j = \begin{cases} s_{j-1} d_i & \text{if } i < j \\ \text{id} & \text{if } i = j, j+1 \\ s_j d_{i-1} & \text{if } i > j+1 \end{cases}$$

$$d_i t = \begin{cases} -t d_{i-1} & \text{if } i > 0 \\ (-1)^n d_n & \text{if } i = 0 \end{cases}, \quad s_i t = \begin{cases} -t s_{i-1} & \text{if } i > 0 \\ (-1)^n t^2 s_n & \text{if } i = 0 \end{cases}$$

We then form the double complex on  $C_n = M_n$

$$\begin{array}{ccccccc}
 & \vdots & & \vdots & & \vdots & \\
 & \downarrow & & \downarrow^b & & \downarrow^{-b'} & \\
 \cdots & \longleftarrow & C_2 & \xleftarrow{N} & C_2 & \xleftarrow{1-t} & C_2 \longleftarrow \cdots \\
 & \downarrow & & \downarrow & & \downarrow & \\
 \cdots & \longleftarrow & C_1 & \xleftarrow{N} & C_1 & \xleftarrow{1-t} & C_1 \longleftarrow \cdots \\
 & \downarrow & & \downarrow & & \downarrow & \\
 \cdots & \longleftarrow & C_0 & \xleftarrow{N} & C_0 & \xleftarrow{1-t} & C_0 \longleftarrow \cdots \\
 & & -1 & & 0 & & 1
 \end{array}$$

where  $b = \sum_{i=0}^n (-1)^i d_i$ ,  $b' = \sum_{i=0}^{n-1} (-1)^i d_i$ ,  $N = 1 + t + \cdots + t^n$  and  $B = (1 - t)s_n N$  where  $s = (-1)^n t s_n$ .

**Definition 7.2.15.** (*Loday*) The **negative cyclic**, **periodic cyclic** and **cyclic chain complexes**

$$CC^- \rightarrow CP \rightrightarrows CC$$

are the total complexes of the  $\leq 1$ , all of, and  $\geq 0$  parts of the above double complex.

7.2.16. *Example.* Let  $\mathcal{C}$  be any (small) dg category. We set

$$M_k = \bigoplus_{f_0, \dots, f_k \in \text{Ob}(\mathcal{C})} \text{Hom}(f_0, f_k) \otimes \text{Hom}(f_k, f_{k-1}) \otimes \cdots \otimes \text{Hom}(f_1, f_0)$$

where  $d_i$  composes the  $i, i - 1$ st maps, and

$$t(f_0 \otimes \cdots \otimes f_k) = (-1)^k (f_k \otimes f_0 \otimes \cdots \otimes f_{k-1}).$$

<https://arxiv.org/pdf/1211.7360>

## 8. Hall algebras of various sorts

We define Hall algebras of various sorts and their DT invariants. The analogy is:

|                             |                            |                                      |                                                                                                        |
|-----------------------------|----------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------|
| Algebra $U_h(\mathfrak{g})$ | Lie algebra $\mathfrak{g}$ | elements $e_\alpha \in \mathfrak{g}$ | $\mathfrak{g}$ is finite, $\mathfrak{g}$ is affine (over $\mathbf{C}, \mathbf{C}^\times, E$ ), $\dots$ |
| Hall algebra                | BPS Lie algebra            | DT invariants                        | which quantisation we apply                                                                            |

This analogy is more or less precise with finite and cohomological Hall algebras.

There is another picture due to [GGL], to do with *categorification*, formed by taking  $K_0(-)$  (conjecturally, I have not checked the left column):

|                                                            |                                                                |                                                     |                                                         |
|------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------|
| Category $\mathcal{E}_{UV} = U_h(\mathfrak{g})\text{-Mod}$ | Object $U_q(\mathfrak{g})_V = \prod_\mu U_q(\mathfrak{g})_\mu$ | Category $\mathcal{E}_{IR} = \text{Vect}_q^\Lambda$ | functor $\mathcal{E}_{UV} \rightarrow \mathcal{E}_{IR}$ |
| Motivic Hall algebra                                       | Element $A_V^{Hall} = \prod_\ell A_\ell$                       | Quantum torus $\mathbf{T}$                          | Integration map                                         |

Finally, there is another categorification picture, formed by taking  $\text{HP}_\bullet(-)$  (or  $K_0$  or  $\text{Fun}_{\mathbf{F}_q}(\mathcal{M}_{(-)})$  to get KHA or finite Hall algebras):

|                                                         |                                                                |
|---------------------------------------------------------|----------------------------------------------------------------|
| CY3 Category $\mathcal{C}$                              | Another CY3 category $\mathcal{C}_T$ acting on $\mathcal{C}$ ? |
| Cohomological Hall algebra<br>$U_q(\hat{\mathfrak{b}})$ | torus CoHA?<br>$U_q(\hat{\mathfrak{t}})?$                      |

### 8.1. Motivic Hall algebras.

8.1.1. We first consider the zero potential version, due to Joyce. Let  $\mathcal{A}$  be an abelian category of finite global dimension over algebraically closed field  $k$  such that

$$\dim \text{Ext}^n(a, a') < \infty$$

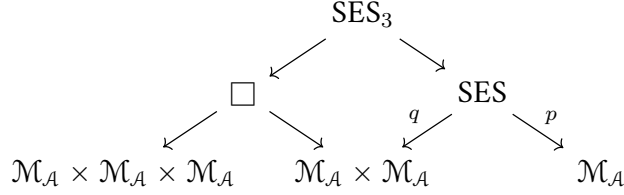
are all finite dimensional vector spaces over  $k$ .

**Theorem 8.1.2.** <https://arxiv.org/pdf/math/0503029> There is an associative product structure on  $K(\text{Stk}/\mathcal{M}_{\mathcal{A}})$  given in terms of

$$\begin{array}{ccc} & \text{SES} & \\ q \swarrow & & \searrow p \\ \mathcal{M}_{\mathcal{A}} \times \mathcal{M}_{\mathcal{A}} & & \mathcal{M}_{\mathcal{A}} \end{array}$$

by  $\star = q^* p_*$ .

*Proof Sketch.* The assumption on  $\mathcal{A}$  ensures that  $p$  is of finite type, since the fibres over a point  $(a, a')$  are given by the vector bundle stack  $\text{Ext}^{\bullet-1}(a, a')$ , which is a finite type derived stack if the above condition on  $\mathcal{A}$  holds. The product is associative by applying base change to the composition



which is the same for both  $\square \in \{\text{SES} \times \mathcal{M}_{\mathcal{A}}, \mathcal{M}_{\mathcal{A}} \times \text{SES}\}$ . □

Note that for delta functions  $\nu_a$  corresponding to maps  $a : \text{pt} \rightarrow \mathcal{M}_{\mathcal{A}}$ , we have that

$$\nu_a \star \nu_{a'} = [\text{Ext}^{\bullet-1}(a, a')]$$

as  $\text{Ext}^{\bullet-1}(a, a') = \text{SES} \times_{\mathcal{M}_{\mathcal{A}} \times \mathcal{M}_{\mathcal{A}}} \{(a, a')\}$ , with the map to  $\mathcal{M}_{\mathcal{A}}$  induced by the map of stacks  $\text{Ext}^1(a, a') \rightarrow \mathcal{M}_{\mathcal{A}}$  sending a class to its associated extension.

**8.1.3. Kontsevich–Soibelman’s formalism.** Kontsevich–Soibelman [KS08, §6] consider the setting of ind-constructible categories.

**Definition 8.1.4.** [KS08, §3] An **ind-constructible category**  $\mathcal{C}$  is one for which:

- its object set  $\text{Ob}(\mathcal{C})$  is a countable union of  $\bar{k}$ -points of locally closed subschemes of a  $k$ -variety,
- its morphism set  $\text{Mor}(\mathcal{C})$  is the  $\bar{k}$ -points of a complex of finite dimensional vector bundles  $(\text{Mor}^n(\mathcal{C}))$  over  $\text{Ob}(\mathcal{C}) \times \text{Ob}(\mathcal{C})$ ,
- $\text{Mor}^n(\mathcal{C})$  vanishes on each connected component of  $\text{Ob}(\mathcal{C}) \times \text{Ob}(\mathcal{C})$  for  $n \ll 0$ .

Kontsevich–Soibelman also define the notion of  $A_{\infty}$  structure on an ind-constructible category, and prove [KS08, Prop. 10] that  $\text{K}(\text{Stk}/\mathcal{M}_{\mathcal{C}})$  has an associative product, defined similarly to the above, except that they compute  $[\text{Ext}^{\bullet-1}(a, a')]$  explicitly in terms of  $[\text{Ext}^1(a, a)]$  and  $\mathbf{L}$ , then make that their definition.

**8.1.5. Critical version?** As far as I am aware there is no version of  $\text{K}(\text{Stk}/\mathcal{M}_{\mathcal{A}})$  in the case where  $\mathcal{M}_{\mathcal{A}}$  is a critical locus.

## 8.2. Finite Hall algebras.

**8.2.1.** Finite Hall algebras have been defined by Ringel and Green for any abelian category  $\mathcal{A}$  defined over  $\mathbf{F}_q$  such that

$$\dim(\text{Ext}^n(a, a')) < \infty,$$

or equivalently all  $\mathbf{F}_q$ -vector spaces  $\text{Ext}^n(a, a')$  are all finite sets. Morally we would now like to take  $\text{SF}(\mathcal{X}_{\mathbf{F}_q})$ , except that I do not know a good definition of this that recovers the below. Instead, we define

$$H_q(\mathcal{A}) = \bigoplus_{a \in \pi_0(\text{Ob}(\mathcal{A}))} \mathbf{C}[a]$$

which has the same functoriality properties of  $K(\text{Stk}/\mathcal{X})$  for complex stacks. It is called the **finite Hall bialgebra** in view of the following Theorem due to Ringel and Green.

**Theorem 8.2.2.** [Sc06, §1]  $H_q(\mathcal{A})$  carries an associative product  $\star = p_*q^*$  and (graded-completed) coassociative coproduct  $\Delta = q_*p^*$ , which form a bialgebra inside the category

$$\text{Vect}_q^\Lambda$$

of vector spaces graded by  $\Lambda = K_0(\mathcal{A})$ , and with braiding given by symmetrised multiplicative Euler form  $\beta(a, a') = \langle a, a' \rangle \cdot \langle a', a \rangle$ .

*Proof Sketch.* Explicitly, as  $[a]$  is the function which takes value  $1/|\text{Aut}(a)|$  on the  $\mathbf{F}_q$ -point  $a$  and zero elsewhere, we have

$$\begin{aligned} \text{Aut}(a)[a] \cdot \text{Aut}(a')[a'] &= p_* \sum_{a \rightarrow e \rightarrow a'} \# \mathbf{F}_q(q^{-1}(a, a')) \cdot [a \rightarrow e \rightarrow e'] \\ &= p_* \sum_{a \rightarrow e \rightarrow a'} \underbrace{\prod_i q^{(-1)^i \# \text{Ext}^{i-1}(a, a')}}_{\langle a, a' \rangle} \cdot [a \rightarrow e \rightarrow e'] \\ &= \langle a, a' \rangle \sum_{a \rightarrow e \rightarrow a'} [e]. \end{aligned}$$

The coproduct

$$\begin{aligned} \Delta(\text{Aut}(e)[e]) &= q_* \sum_{a \rightarrow e \rightarrow a'} \# \mathbf{F}_q(p^{-1}(e)) \cdot [a \rightarrow e \rightarrow a'] \\ &= \sum_{a \rightarrow e \rightarrow a'} \langle a, a' \rangle [a] \otimes [a'] \end{aligned}$$

is only valued in a completion  $H_q(\mathcal{A}) \hat{\otimes} H_q(\mathcal{A})$  since a priori there are infinitely many pairs of elements  $a, a'$  which  $e$  is an extension of.

□

8.2.3. *Example.* For  $\mathcal{A} = \text{Vect}$ , we have that

$$H_q(\text{Vect}) = \mathbf{C}[S^0] \oplus \mathbf{C}[S] \oplus \mathbf{C}[S^{\oplus 2}] \oplus \dots$$

where  $S$  is the one-dimensional vector space, with  $\text{Hom}(S, S) = \mathbf{F}_q$  so that  $\langle S, S \rangle = q$ . Therefore,

$$\begin{aligned} [S^{\oplus n}] \cdot [S^{\oplus m}] &= q^{nm} \# \mathbf{F}_q(\text{Gr}(n, n+m)) [S^{\oplus n+m}], \\ \Delta[S^{\oplus n}] &= \sum_{n=n_1+n_2} q^{n_1 n_2} \# \mathbf{F}_q(\text{Gr}(n_1, n)) [S^{\oplus n_1}] \otimes [S^{\oplus n_2}] \end{aligned}$$

where the size of  $\text{Gr}(k, n)$  can be easily computed as a quotient of  $\text{GL}(n)$  by a parabolic subgroup.

**Lemma 8.2.4.**  $[S]^{\star n} = q^{n(n-1)} [S^{\oplus n}]$ .

8.2.5. *Relation to quantum groups.* Let us consider the abelian category

$$\mathcal{A} = \text{Rep}^{\text{nil}} Q$$

of representations of  $Q$  such that any path of edges greater than some integer  $N \gg 0$  is zero.

Let  $Q$  be any quiver without vertex self-loops and  $\mathfrak{g}$  be the Lie algebra on the Cartan matrix given in terms of the adjacency matrix  $B$  of  $Q$  by  $A = 2\text{id} - B - B^\top$ . Recall that  $U_q(\mathfrak{n})$  is the associative algebra generated by  $e_\alpha$  for  $\alpha$  a node in the quiver, modulo the quantum Serre relations

$$\sum_{k=0}^{1-a_{\alpha\beta}} (-1)^k \binom{1-a_{\alpha\beta}}{k}_q e_\alpha^k e_\beta e_\alpha^{1-a_{\alpha\beta}-k} = 0$$

and with the coproduct

$$\Delta(e_\alpha) = e_\alpha \otimes 1 + 1 \otimes e_\alpha.$$

This defines a bialgebra inside  $\text{Vect}_q^\Lambda$ , with braiding given by

$$\mathbf{C}_\lambda \otimes \mathbf{C}_\mu \xrightarrow[\sim]{q^{\kappa(\lambda, \mu)}} \mathbf{C}_\lambda \otimes \mathbf{C}_\mu \xrightarrow{\sim} \mathbf{C}_\mu \otimes \mathbf{C}_\lambda$$

where  $\kappa$  is the associated Killing form. We have the following *bosonisation* result:

**Proposition 8.2.6.** *We have that*

$$U_q(\mathfrak{n})\text{-Mod}(\text{Vect}_q^\Lambda) \simeq U_q(\mathfrak{b})\text{-Mod}$$

where  $U_q(\mathfrak{b}) \simeq U_q(\mathfrak{n}) \rtimes \mathbf{C}[\Lambda]$  has subalgebras generated by  $U_q(\mathfrak{n})$  and grouplike invertible elements  $\{K_\alpha\}$  labelled by vertices of the quiver, with modified coproduct

$$\Delta(e_\alpha) = e_\alpha \otimes 1 + K_\alpha \otimes e_\alpha.$$

Combining this with the following result allows us to obtain  $U_q(\mathfrak{b})$  from finite Hall algebras:

**Theorem 8.2.7.** [Sc06, Thm. 3.16] *There is an embedding of bialgebras*

$$\begin{aligned} U_q(\mathfrak{n}) &\hookrightarrow H_q(\text{Rep}^{\text{nil}} Q) \\ e_\alpha &\mapsto [S_\alpha] \end{aligned}$$

which is an isomorphism if  $Q$  is of ADE type.

*Proof.* It is easy to see that  $\Delta([S_\alpha]) = [S_\alpha] \otimes 1 + 1 \otimes [S_\alpha]$ , so for this map of bialgebras in  $\text{Vect}_q^\Lambda$  to be well-defined it remains to show that the quantum Serre relations are satisfied, i.e.

$$\sum_{k=0}^{1-a_{\alpha\beta}} (-1)^k [S_\alpha]^{\star k} [S_\beta] [S_\alpha]^{\star(1-a_{\alpha\beta}-k)} = 0. \quad (18)$$

It is enough to restrict to the case where the quiver  $Q$  has two vertices,  $\alpha$  and  $\beta$ , with  $n$  arrows in one direction and  $m$  in the other:

$$\begin{array}{ccc} \alpha & \xrightarrow{\times n} & \beta \\ \bullet & & \bullet \\ & \xleftarrow{\times m} & \end{array}$$

so that  $n + m = -a_{ij}$ .

To do this, note that the left hand side of (18) is a linear combination over  $[E]$  for  $E$  a nilpotent representation of dimension  $(n + m + 1)\alpha + \beta$  which admitting a two-step extension

$$E_0 \hookrightarrow E_1 \hookrightarrow E$$

with subquotients  $S_\alpha^{\oplus k}, S_\beta, S_\alpha^{1+n+m-k}$ . In other words,  $E_0 = S_\alpha^{\oplus k}$ , the representation  $E_1$  is

$$\begin{array}{ccc} \mathbf{C}^k & \xrightarrow{0} & \mathbf{C} \\ & \xleftarrow{v_1, \dots, v_m} & \end{array}$$

and so the representation  $E$  is

$$\begin{array}{ccc} & (0, f_1), \dots, (0, f_n) & \\ \mathbf{C}^k \oplus \mathbf{C}^{1+n+m-k} & \xrightarrow{\quad} & \mathbf{C} \\ & \xleftarrow{(v_1, 0), \dots, (v_m, 0)} & \end{array}$$

Let  $N$  be the joint kernel and  $M$  the joint image of those  $n$  and  $m$  arrows, respectively. Nilpotency implies that  $M \subseteq N$ ,<sup>9</sup> and

$$\begin{aligned} [S_\alpha]^{\star k} [S_\beta] [S_\alpha]^{\star \ell} &= q^{-rn-\ell m+k\ell+k(k-1)+\ell(\ell-1)} \sum_{[E]} \# \mathbf{F}_q(\mathrm{Gr}(\mathbf{C}^\ell/M, N/M)) \cdot [E] \\ &= q^{-rn-\ell m+k\ell+k(k-1)+\ell(\ell-1)} \sum_{[E]} q^{(N-\ell)(\ell-M)} \binom{N-M}{\ell-M}_q \cdot [E] \end{aligned}$$

where we have conflated  $N, M$  with their dimensions. We finish as in [Sc06] by the following which follows for any  $1 - r \leq d \leq r - 1$  with  $d = r - 1 \pmod{2}$ :

$$\sum_{n=0}^r (-1)^n q^{-dn} \binom{r}{n}_q = 0.$$

□

8.2.8. *Question.* Is there a cleverer way to prove the quantum Serre relations?

### 8.3. Cohomological Hall algebras.

<sup>9</sup>This is the first and only time we use nilpotency!

8.3.1. *Vanishing cycles approximate construction.* Let  $\mathcal{X}$  be a space lying within a category of spaces admitting a sheaf theory  $\mathrm{Sh}(-)$  with the six functors. Fix a function  $W : \mathcal{X} \rightarrow \mathbf{C}$ .

Whereever it is defined, the *nearby cycles* endofunctor  $\psi_W$  on  $\mathrm{Sh}(\mathcal{X})$  is defined something like follows. Consider the universal cover  $\widetilde{\mathbf{C}^\times}$ —this is the part which requires careful defining based on the category of spaces we are working in—and take the pullback

$$\begin{array}{ccccccc} \mathcal{X}_0 & \xrightarrow{i} & \mathcal{X} & \xleftarrow{j} & \mathcal{X}^\times & \xleftarrow{p} & \widetilde{\mathcal{X}} \\ \downarrow & & \downarrow W & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbf{C} & \xleftarrow{} & \mathbf{C}^\times & \xleftarrow{} & \widetilde{\mathbf{C}^\times} \end{array}$$

The nearby cycles functor is then defined as

$$\psi_W = i_* i^* j_* \underbrace{p_* p^* j^*}_{\text{monodromy}},$$

that is pull-pushing along  $\widetilde{\mathcal{X}} \rightarrow \mathcal{X}$  then along  $\mathcal{X}_0 \rightarrow \mathcal{X}$ . The *vanishing cycle functor*  $\varphi_W$  is defined as the cofibre of the counit map

$$i_* i^* \rightarrow \psi_W \rightarrow \varphi_W \xrightarrow{+1}$$

All sheaves are clearly supported on  $W^{-1}(0) = \mathcal{X}_0$ .

8.3.2. *Remark.* Notice that if  $\mathcal{L}$  is a local system on  $S^1$  with monodromy  $T : L \rightarrow L$  acting on the fibre, we have that

$$H^0(U^\times, \mathcal{L}) = L^T, \quad H^1(U^\times, \mathcal{L}) = L_T \quad (19)$$

are monodromy invariants and coinvariants.

8.3.3. *Example.* We can use the above Remark to compute nearby and vanishing cycles when  $\mathcal{X} = \mathbf{C}$ , valued in the pushforward of a local system  $\mathcal{L}^\times$  on  $\mathcal{X}^\times = \mathbf{C}^\times$  given by monodromy  $T : L \xrightarrow{\sim} L$  on the fibre. In this case,  $p_* p^* \mathcal{L}^\times$  is the local system with fibre and monodromy

$$T[1] : \bigoplus_{n \in \mathbf{Z}} L \xrightarrow{\sim} \bigoplus_{n \in \mathbf{Z}} L.$$

In this case,

$$\begin{aligned} \psi_W j_* \mathcal{L}^\times &= H^\bullet(\mathbf{C}^\times, p_* p^* \mathcal{L}^\times) = L \oplus L[-1] \\ i_* i^* j_* \mathcal{L}^\times &= H^\bullet(\mathbf{C}^\times, \mathcal{L}^\times) = L^T \oplus L_T[-1]. \end{aligned}$$

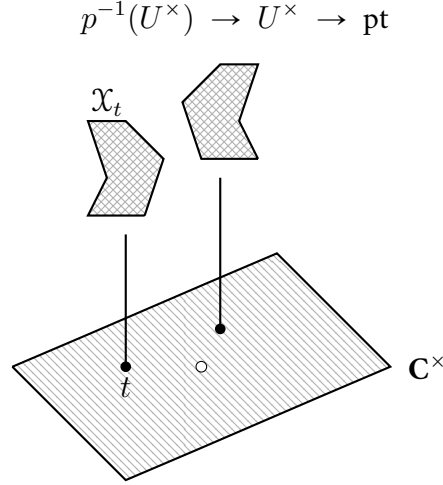
Thus at least in degree zero  $L/L^T$ —those germs at  $t$  which vanish upon parallel transporting to the origin—includes into the degree zero part of vanishing cycles.

8.3.4. *Question.* What about in higher degrees?

8.3.5. *Example.* In the case where  $\mathcal{X}$  is a complex manifold and we consider constructible sheaves, let us examine the fibre of  $(\psi_W \mathcal{F})_x$  at a point:

$$(\psi_W \mathcal{F})_x = \lim_{x \in U} H^\bullet(p^{-1}(U^\times), p^* \mathcal{F}|_{\mathcal{X}^\times}).$$

We compute this as the pushforward along



We may pick  $U^\times \sim S^1$  contractible. We can use (19) to try to compute the cohomology of the constructible sheaf  $p_* p^* \mathcal{F}|_{\mathcal{X}^\times}$ .

8.3.6. *Example.* If  $\mathcal{F} = \mathbf{C}_\mathcal{X}$  is the constant sheaf the situation simplifies a lot. Note that since  $\mathcal{X}$  is a complex manifold, in a small enough contractible neighbourhood we may assume  $\mathcal{X}^\times \rightarrow \mathbf{C}^\times$  is a fibre bundle, and so  $p_* p^* \mathcal{F}|_{\mathcal{X}^\times}$  is the Gauss–Manin connection. The nearby cycles fibre can then be computed using (19) and a spectral sequence argument.

Thus its  $H^0$  should be thought of as being a family of sections  $f_t$  of  $\mathcal{F}|_{\mathcal{X}_t}$  which have trivial monodromy as  $t \in \mathring{\mathcal{X}}$  moves in a path lifting the loop in  $\mathbf{C}^\times$ . Its  $H^n$  should be thought of as a combination of a family of monodromy-invariant sections of  $H^n(\mathcal{F}|_{\mathcal{X}_t})$  and

Moreover,

**Theorem 8.3.7.** *For Artin stacks over  $\mathbf{C}$ , the vanishing cycles functor satisfies:*

- If all critical values of  $W$  are 0, it is supported over  $\text{Crit}(W)$
- there is a map  $f^* \varphi_W \rightarrow \varphi_{f^* W} f^*$  and  $f_! \varphi_{f^* W} f^! \rightarrow \varphi_W$  for every map  $f : \mathcal{Y} \rightarrow \mathcal{X}$ ,
- the vanishing cycle functor is Verdier self-dual:  $\mathbf{D} \varphi_\mathcal{X} = \varphi_\mathcal{X} \mathbf{D}$ ,
- (Thom–Sebastiani isomorphism)  $\varphi_{W_1 \boxplus W_2} \simeq \varphi_{W_1} \boxtimes \varphi_{W_2}$  for a product  $\mathcal{X}_1 \times \mathcal{X}_2$  with a direct sum of two functions  $W_1 \boxplus W_2$ .

8.3.8. We can now define the cohomological Hall algebra.

**Theorem 8.3.9.** *If  $\mathcal{A}$  is a CY3 abelian category, whose moduli stack is a global critical locus, then*

$$\mathbf{H}^\bullet(\mathcal{M}_{\mathcal{A}}, \varphi_W)$$

*has an associative product  $\star = p_* q^*$ .*

*Proof.*

□

**Question 8.3.10.** Compute the CoHA of  $\mathcal{C} = \text{Vect}$ .

8.3.11. Let  $\mathcal{C}$  be a CY3 category. We can form the following (at least locally, i.e. for  $\mathcal{C} \simeq \text{MF}(\mathcal{D}, W)$ ).

$$\begin{array}{c}
 \mathcal{C} \\
 \swarrow \quad \searrow \quad \downarrow \text{HP} \quad \downarrow K_0 \\
 \text{Fun}(\mathcal{M}_{\mathcal{C}}(\mathbf{F}_q)) \xleftarrow{?} K^{(\text{crit})}(\text{Stk}/\mathcal{M}_{\mathcal{C}}) \xrightarrow{f_! 1_{\mathcal{Y}}} H_{\text{crit}}^\bullet(\mathcal{M}_{\mathcal{C}}) \xleftarrow{ch} K_{\text{crit}}^\bullet(\mathcal{M}_{\mathcal{C}}) \\
 \quad \quad \quad \searrow f_* \mathcal{O}_{\mathcal{Y}} \quad \quad \quad \downarrow
 \end{array} \tag{20}$$

We first note that under conditions on  $\mathcal{C}$ , the maps (20) send  $\mathcal{C}$  to the denoted vector spaces:

- Theorem 1.1 <https://arxiv.org/pdf/1212.2859> If  $Y$  is a smooth and quasiprojective scheme over  $k = \mathbf{C}$ , then

$$\text{HP}_\bullet(\text{MF}(Y, W)) \simeq \mathbf{H}^{\bullet-1}(Y, \varphi_W \mathbf{C}_Y).$$

- Theorem 8.2.6 <https://arxiv.org/pdf/1101.5834> The Hochschild cohomology of  $\text{MF}(Y, W)$  is equivalent to the vanishing cycle chains  $R\Gamma(\Omega_Y^\bullet((\beta)), \beta \cdot -dW \wedge (-))$ .
- [BBDJS] Étale-locally on a critical chart  $\mathcal{M}_{\mathcal{C}}|_U \simeq \text{Crit}(W) \subset \mathcal{Y}$ , the perverse sheaf  $\mathcal{P}_{\mathcal{M}_{\mathcal{C}}}$  computing critical cohomology restricts to  $\varphi_W \mathbf{Q}_{\mathcal{Y}}[\dim \mathcal{Y} - 1]$  (up to an orientation twist), so

$$H_{\text{crit}}^\bullet(\mathcal{M}_{\mathcal{C}}|_U) \simeq \mathbf{H}^\bullet(\mathcal{Y}, \varphi_W \mathbf{Q}_{\mathcal{Y}})[\dim \mathcal{Y} - 1].$$

## 9. Stability conditions

### 9.1. Toy model: vector bundles on $\mathbf{P}^1$ .

9.1.1. We nnte the Theorem, working over the complex numbers:

**Theorem 9.1.2.** *Every finite dimensional vector bundle over  $\mathbf{P}^1$  is isomorphic to*

$$\mathcal{O}(n_1) \oplus \cdots \oplus \mathcal{O}(n_k)$$

for some unique tuple of integers  $n_1, \dots, n_k$ , where  $\mathcal{O}(n) = \mathcal{O}(1)^{\otimes n}$ . Every coherent sheaf is uniquely of the form

$$\mathcal{O}(n_1) \oplus \cdots \oplus \mathcal{O}(n_k) \oplus \mathbf{C}_{(m_1), z_1} \oplus \cdots \oplus \mathbf{C}_{(m_\ell), z_\ell}$$

where  $\mathbf{C}_{(m), z}$  is the order  $m$  skyscraper sheaf at point  $z \in \mathbf{P}^1$ .

*Proof.* Follows from the Jordan normal form of complex matrices. □

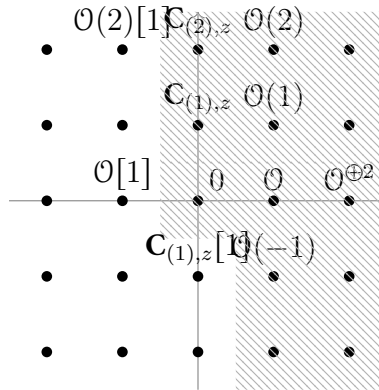
**Question 9.1.3.** How do we count the number of sheaves on  $\mathbf{P}^1$ ? For a general variety  $X$ , how do we get as close to decomposing a coherent sheaf into summands as in the  $\mathbf{P}^1$  case?

To begin with, note that we have extensions

$$0 \rightarrow \mathbf{C}_{(m), z} \rightarrow \mathbf{C}_{(m+m'), z} \rightarrow \mathbf{C}_{(m'), z} \rightarrow 0$$

$$0 \rightarrow \mathbf{C}_{(m), z} \rightarrow \mathcal{O}(n+m) \rightarrow \mathcal{O}(n) \rightarrow 0$$

which gives that the Grothendieck group has rank two generated by  $[\mathcal{O}(1)]$  and  $[\mathbf{C}_{(1), z}]$  for any point  $z \in \mathbf{P}^1$ .



with the cone of classes in  $\text{Coh}(\mathbf{P}^1) \subseteq D^b(\text{Coh}(\mathbf{P}^1))$  shaded.

### 9.2. Stability conditions.

9.2.1. We have the following definition due to Bridgeland:

**Definition 9.2.2.** [Br02; KS08] A **stability condition**  $Z$  on a triangulated category  $\mathcal{D}$  is a “**central charge**” homomorphism  $Z : K_0(\mathcal{D}) \rightarrow \mathbf{C}$ , and a collection of full additive subcategories  $\mathcal{P}(\phi) \subseteq \mathcal{D}$  for each  $\phi \in \mathbf{R}$  (called  **slicings**) such that:

- (1) for each object  $A \in \mathcal{P}(\phi)$ ,  $Z(A) = e^{i\pi\phi} m(A)$  for  $m(A) \in \mathbf{R}_{\geq 0}$ ,
- (2)  $\mathcal{P}(\phi + 1) = \mathcal{P}(\phi)[1]$ ,
- (3) if  $\phi_1 > \phi_2$  and  $A_i \in \mathcal{P}(\phi_i)$  then  $\text{Hom}_{\mathcal{D}}(A_1, A_2) = 0$ ,
- (4) for each object  $E \in \mathcal{D}$  there is a finite sequence of real numbers

$$\phi_1 > \phi_2 > \cdots > \phi_n$$

and a collection of triangles

$$\begin{array}{ccccccc} 0 & \xrightarrow{\quad} & E_1 & \xrightarrow{\quad} & E_2 & \xrightarrow{\quad} & \cdots \xrightarrow{\quad} E_n = E \\ & \swarrow +1 & \nwarrow & \swarrow +1 & \nwarrow & \swarrow +1 & \nwarrow \\ & A_1 & & A_2 & & \cdots & A_n \end{array}$$

with subquotients  $A_i \in \mathcal{P}(\phi_i)$  for each  $i$  (we may assume it is unique up to isomorphism).

Write  $\phi_{\mathcal{P}}^-(A) \leq \phi_{\mathcal{P}}^+(A)$  for the minimum and maximum angles appearing in the above decomposition.

An object  $E$  is *semistable* if every nonzero subobject  $A \subseteq E$  has  $\phi(A) \leq \phi(E)$ .

We write  $\mathcal{P}(a, b)$  for the quasiabelian subcategory generated by  $\mathcal{P}(\phi)$  for  $\phi \in (a, b)$ . A quasiabelian category is an additive category with kernels and cokernels such that every pullback of a strict epimorphism is a strict epimorphism, and every pushout of a strict monomorphism is a strict monomorphism.

A slicing is *locally finite* if there is a fixed real number so that each quasiabelian category  $\mathcal{P}(t - \eta, t + \eta) \subseteq \mathcal{D}$  is of finite length. include?

**Corollary 9.2.3.** For any  $\phi \in \mathbf{R}$  and any  $E \in \mathcal{D}$  there is a distinguished triangle

$$\begin{array}{ccc} A_{>\phi} & \xrightarrow{\quad} & E \\ & \swarrow +1 & \nwarrow \\ & B_{\leq \phi} & \end{array}$$

where  $A_{>\phi} \in \mathcal{P}(\infty, \phi)$  and  $B_{\leq \phi} \in \mathcal{P}(-\infty, \phi]$ .

**Lemma 9.2.4.** If  $I$  is an interval of length  $< 1$  and

$$\begin{array}{ccc} A & \xrightarrow{\quad} & E \\ & \swarrow +1 & \nwarrow \\ & B & \end{array}$$

is a triangle in  $\mathcal{D}$  whose vertices are in  $\mathcal{P}(I)$ . Then  $\phi^+(A) \leq \phi^+(E)$  and  $\phi^-(E) \geq \phi^-(B)$ .

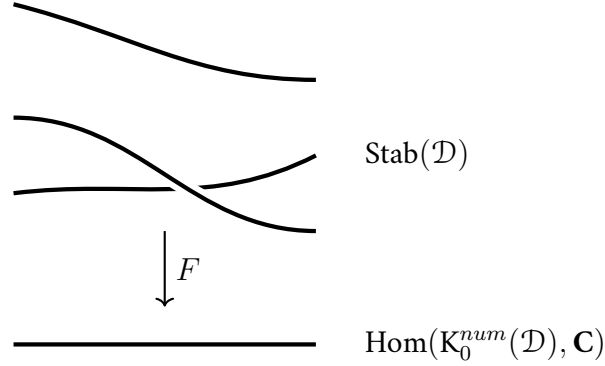
**Lemma 9.2.5.**  $\mathcal{P}( > t ), \mathcal{P}( \leq t )$  forms a  $t$ -structure on  $\mathcal{D}$ .

As an example, you should compute:

**Question 9.2.6.** Compute the space of stability conditions for  $\text{Coh}(\mathbf{P}^1)$ .

**Question 9.2.7.** What about representations for the  $A_2$  quiver  $Q$ ?

We now prove a result about the space of stability conditions



**Theorem 9.2.8.** [Br02] *There is a natural topology on the set  $\text{Stab}(\mathcal{D})$ . If the numerical Grothendieck group is of finite rank, each connected component  $\text{Stab}_\Sigma(\mathcal{D}) \subseteq \text{Stab}(\mathcal{D})$  has the structure of a complex manifold, such that the forgetful map*

$$\begin{aligned} \text{Stab}_\Sigma(\mathcal{D}) &\rightarrow \text{Hom}_{\mathbf{Z}}(\mathbf{K}_0^{\text{num}}(\mathcal{D}), \mathbf{C}) \\ (Z, \mathcal{P}) &\mapsto Z \end{aligned}$$

*is etale (a local homeomorphism) onto a linear subspace  $V(\Sigma) \subseteq \text{Hom}_{\mathbf{Z}}(\mathbf{K}_0^{\text{num}}(\mathcal{D}), \mathbf{C})$ .*

*Proof as in [Br02], assuming  $\mathbf{K}_0(\mathcal{D})$  is finite rank. We define the minimal topology so that*

$$F : \text{Stab}(\mathcal{D}) \rightarrow \text{Hom}_{\mathbf{Z}}(\mathbf{K}_0(\mathcal{D}), \mathbf{C})$$

*and the angle functions*

$$\phi^\pm(A) : \text{Stab}(\mathcal{D}) \rightarrow \mathbf{C} \quad (Z, \mathcal{P}) \mapsto \phi_{\mathcal{P}}^\pm(A)$$

*are continuous, i.e. induce the topology via the embedding  $\text{Stab}(\mathcal{D}) \subseteq \text{Slicing}(\mathcal{D}) \times \text{Hom}_{\mathbf{Z}}(\mathbf{K}_0(\mathcal{D}), \mathbf{C})$  where the set of slices has topology induced by the angle functions above.*<sup>10</sup>

### Local injectivity.

To begin,  $F$  is locally injective: if  $\sigma = (Z, \mathcal{P}), \tau = (Z, \mathcal{Q})$  are stability conditions with the same central charge and

$$|\phi_{\mathcal{P}}^\pm(A) - \phi_{\mathcal{Q}}^\pm(A)| \leq d < 1$$

<sup>10</sup>For infinite rank  $\mathbf{K}_0(\mathcal{D})$ , we need to decide which topology on  $\text{Hom}_{\mathbf{Z}}(\mathbf{K}_0(\mathcal{D}), \mathbf{C})$  to use.

then  $\sigma = \tau$ . Indeed, assume that  $E \in \mathcal{P}(\phi)$  (so that  $\phi_{\mathcal{P}}^{\pm}(E) = \phi$ ) lies outside of  $\mathcal{Q}(\phi)$  (so that  $\phi_{\mathcal{Q}}^{\pm}(E) \in (\phi - 1, \phi + 1)$ ) by the above assumption, which implies  $E$  is not  $\mathcal{Q}$ -semistable by charge considerations).

Thus we can decompose

$$\begin{array}{ccc} A_{(\phi, \phi+1)} & \xrightarrow{\quad} & E \\ & \nwarrow \quad \nearrow & \\ & +1 \quad B_{(\phi-1, \phi]} & \end{array}$$

with both factors nonzero, where  $A_{(\phi, \phi+1)} \in \mathcal{Q}(\phi, \phi + 1)$  and  $B_{(\phi-1, \phi]} \in \mathcal{Q}(\phi - 1, \phi]$ . Note that  $A_{(\phi, \phi+1)} \notin \mathcal{P}[-\infty, \phi]$  by a similar argument, which implies that we have

$$\begin{array}{ccc} C_{(\phi, \phi+1)} & \xrightarrow{\quad} & A_{(\phi, \phi+1)} \\ & \nwarrow \quad \nearrow & \\ & +1 \quad D & \end{array}$$

where now  $C \in \mathcal{P}(\psi) \subseteq \mathcal{P}(\phi, \phi + 1)$  and the map to  $A$  is nonzero. This gives a map  $C \rightarrow E$  which is zero by (2), and so factors to a map  $C \rightarrow B[-1]$ , which is zero, so the map  $C \rightarrow A$  is zero, contradicting.

### Local surjectivity.

**Proposition 9.2.9.** *Suppose that  $\mathcal{A}$  is an abelian category with a function  $Z : K_0(\mathcal{A}) \rightarrow \mathbb{C}$  with the **chain condition**: there is no infinite decreasing sequence of subobjects or kernels:*

$$\cdots \subseteq E_2 \subseteq E_1, \quad E_1 \twoheadrightarrow E_2 \twoheadrightarrow \cdots$$

*with  $\cdots > \phi(E_2) > \phi(E_1)$  and  $\phi(E_1) > \phi(E_2) > \cdots$  respectively. Then  $\mathcal{A}$  has the Harder–Narasimhan property.*

*Proof.* Left as an exercise. □

**Lemma 9.2.10.** (Gluing Lemma) *Assume that*

- (1) *for each interval  $I$  of length  $< 1$ ,  $\mathcal{P}(I)$  is quasiabelian and strict short exact sequences in  $\mathcal{P}(I)$  are equivalent to distinguished triangles in  $\mathcal{D}$  whose elements lie in  $\mathcal{P}(I)$*
- (2) *for each interval  $I$  of length  $< 1 - 2\epsilon$ ,  $W|_{\mathcal{P}(I)}$  takes value in some upper-half plane (so that  $\phi_W$  is defined), and  $\mathcal{P}(I)$  has the chain property for  $\phi_W$ .*

*Then*

$$\mathcal{Q}(\phi) = \langle 0, E \in \mathcal{P}(\phi - \epsilon, \phi + \epsilon) : E \text{ is } W\text{-semistable of phase } \phi \rangle$$

*defines a stability condition for  $W$  (note that  $\phi_W$  is defined on  $\mathcal{P}(\phi - \epsilon, \phi + \epsilon)$  as the interval has length  $< 1$ ).*

*Proof.*

0 *Well-definedness.* [Br02, Lem. 7.5] Next, we say that for thin subcategory  $\mathcal{A} = \mathcal{P}(a, b)$ , an object  $E \in \mathcal{A}$  is *enveloped* by  $\mathcal{A}$  if  $a + \epsilon \leq \psi(E) \leq b - \epsilon$ . This thus generalises  $W$ -semistability. We show that this is a property of the object and not the thin subcategory:

**Lemma 9.2.11.** *Let  $E \in \mathcal{D}$  be enveloped by thin subcategories  $\mathcal{B}, \mathcal{C} \subseteq \mathcal{D}$ . Then  $E$  is  $W$ -semistable in  $\mathcal{B}$  if and only if it is  $W$ -semistable in  $\mathcal{C}$ .*

*Proof.* We can take a fixed  $\mathcal{B}$ , for instance  $\mathcal{B} = \mathcal{P}(\psi(E) - \epsilon, \psi(E) + \epsilon)$  in which  $E$  is  $W$ -semistable by the above Lemma. This reduces us to the case  $\mathcal{B} \subseteq \mathcal{C}$  or by symmetry to  $\mathcal{B} = \mathcal{P}(a, b)$  and  $\mathcal{C} = \mathcal{P}(a, c)$ .

Of course  $W$ -semistability in  $\mathcal{C}$  implies  $W$ -semistability in  $\mathcal{B}$ , so it remains to prove the converse. Assume that  $E$  is unstable in  $\mathcal{C}$ :

$$0 \rightarrow A \rightarrow E \rightarrow B \rightarrow 0$$

where  $\psi(A) > \psi(E) > \psi(B)$ . Then  $\phi^+(A) \leq \phi^+(E)$ , which implies that  $A \in \mathcal{B}$  since  $E \in \mathcal{B}$ . By a longer argument,  $B \in \mathcal{B}$  also.  $\square$

- (1)  $W(E) = m(E) \cdot e^{i\pi\phi}$  for any  $E \in \mathcal{Q}(\phi)$  by definition
- (2) *Shift condition.*  $\mathcal{Q}(\phi + 1) = \mathcal{Q}(\phi)[1]$  as  $W(E[1]) = -W(E)$  and  $\mathcal{P}(\phi + 1 - \epsilon, \phi + 1 + \epsilon) = \mathcal{P}(\phi - \epsilon, \phi + \epsilon)[1]$ .
- (3) *Hom-vanishing.* [Br02, Lem. 7.6] Assume that there is a nonzero map  $f : E_1 \rightarrow E_2$  with  $E_i \in \mathcal{Q}(\phi_i)$  with  $\phi_1 > \phi_2$ . We first prove

**Lemma 9.2.12.** *If  $E$  is  $W$ -semistable in some thin  $\mathcal{A} \subseteq \mathcal{D}$ , then  $E \in \mathcal{P}(\psi(E) - \epsilon, \psi(E) + \epsilon)$ .*

*Proof.* For any exact sequence  $A_\phi \rightarrow E \rightarrow B_{>\phi}$  we have  $\psi(A) \leq \psi(E)$  by  $W$ -semistability, therefore  $\phi = \psi_{\mathcal{P}}^+(E) < \psi(E) + \epsilon$ . Likewise for  $\psi_{\mathcal{P}}^-(E)$ .  $\square$

Therefore it follows that  $\psi_1 - \psi_2 < 2\epsilon$ , and we consider  $\mathcal{A} = \mathcal{P}(\frac{\psi_1 + \psi_2}{2} - \frac{1}{2}, \frac{\psi_1 + \psi_2}{2} + \frac{1}{2})$  which contains  $E_1, E_2$ . Consider the short exact sequences in  $\mathcal{A}$ :

$$0 \rightarrow \ker f \rightarrow E_1 \rightarrow \operatorname{im} f \rightarrow 0, \quad 0 \rightarrow \operatorname{im} f \rightarrow E_2 \rightarrow \operatorname{coker} f \rightarrow 0.$$

continue... therefore  $\psi_1 \leq \psi(\operatorname{im} f) \leq \psi_2$

- (4) *Harder–Narasimhan condition.* [Br02, Lem. 7.7] Let  $E \in \mathcal{D}$ . First we show that for any  $t \in \mathbf{R}$  we can split it into

$$\begin{array}{ccc} A_{\mathcal{Q}( > t)} & \xrightarrow{\quad} & E \\ & \nwarrow \quad \nearrow & \\ & +1 \quad B_{\mathcal{Q}( \leq t)} & \end{array}$$

so that  $\mathcal{Q}( > t)$  forms a  $t$ -structure. This is obvious when  $E \in \mathcal{P}(\geq t + \epsilon), \mathcal{P}(\leq t - \epsilon)$  because these are contained in  $\mathcal{Q}( > t), \mathcal{Q}( \leq t)$ , so enough to consider  $E \in \mathcal{P}(t - \epsilon, t + \epsilon)$ , which follows from the following Lemma.

**Lemma 9.2.13.** *Let  $\mathcal{A} = \mathcal{P}(a, b) \subseteq \mathcal{D}$  be a thin category of finite length. Then every nonzero object in  $\mathcal{P}(a + 2\epsilon, b - 4\epsilon)$  has a finite  $W$ -HN filtrations whose factors are  $W$ -semistable objects enveloped by  $\mathcal{A}$ .*

One can then use that  $E \in \mathcal{P}(\phi_{\mathcal{P}}^-(E), \phi_{\mathcal{P}}^+(E)) = \mathcal{P}(A, B)$ , and so we can repeatedly apply the  $t$ -structure to get

$$\begin{array}{ccccccc}
 0 & \xrightarrow{\quad} & A_{Q(>t_1)} & \xrightarrow{\quad} & \cdots & \xrightarrow{\quad} & A_{Q(>t_{n-1})} & \xrightarrow{\quad} & E \\
 & \swarrow +1 & \swarrow & & \swarrow +1 & \swarrow & \swarrow +1 & \swarrow & \swarrow \\
 & & B_{Q(\leq t_1)} & & B_{Q(\leq t_2)} & & \cdots & & B_{Q(\leq t_{n-1})}
 \end{array}$$

where we pick  $\{t_0, t_1, \dots, t_n\}$  whose convex hull covers  $(A, B)$  and the differences are smaller than those in the above Lemma. Then each  $A_i, B_i$  above has an HN filtration, which we combine with the above to induce an HN filtration on  $E$ .

□

**Old proof** either keep, delete or merge

Finally, we show local surjectivity: if  $\sigma = (Z, \mathcal{P})$  is a (locally finite) stability condition and  $\delta > 0$  then  $F^{-1}(B_\delta(Z)) \cap B_\epsilon^\times(\sigma)^{(l.f.)} \neq \emptyset$  for every  $\epsilon < \delta$ .

Thus, let  $|W - Z| < \delta$ . Our task is to define a stability condition  $\tau = (W, \mathcal{Q})$  with

$$|\phi_{\mathcal{P}}^\pm(E) - \phi_{\mathcal{Q}}^\pm(E)| < \epsilon \cdot |\phi_{\mathcal{P}}^\pm(E)|$$

for every  $E \in \mathcal{D}$ . For any object  $E \in \mathcal{P}(a, b)$  we write

$$\phi(E) = \arg Z(E) \in (a, b), \quad \psi(E) = \arg W(E) \in (a - \epsilon, b + \epsilon)$$

which is well-defined for  $0 < b - a < 1 - 2\epsilon$ . We call such a category  $\mathcal{P}(a, b) \subseteq \mathcal{D}$  **thin**. We call objects  $E$  in a thin category  $\mathcal{A}$   $W$ -semistable (in  $\mathcal{A}$ ) if they are semistable with respect to  $W$  ( $\psi(A) \leq \psi(E)$  for every nonzero subobject  $A \subseteq E$ ); note that we need to be working in a thin subcategory for the  $W$ -phase  $\psi(E)$  to be well-defined.

**Lemma 9.2.14.** *If  $E$  is  $W$ -semistable in some thin  $\mathcal{A} \subseteq \mathcal{D}$ , then  $E \in \mathcal{P}(\psi(E) - \epsilon, \psi(E) + \epsilon)$ .*

*Proof.* For any exact sequence  $A_\phi \rightarrow E \rightarrow B_{>\phi}$  we have  $\psi(A) \leq \psi(E)$  by  $W$ -semistability, therefore  $\phi = \psi_{\mathcal{P}}^+(E) < \psi(E) + \epsilon$ . Likewise for  $\psi_{\mathcal{P}}^-(E)$ . □

Next, we say that for thin subcategory  $\mathcal{A} = \mathcal{P}(a, b)$ , an object  $E \in \mathcal{A}$  is *enveloped* by  $\mathcal{A}$  if  $a + \epsilon \leq \psi(E) \leq b - \epsilon$ . This thus generalises  $W$ -semistability. We show that this is a property of the object and not the thin subcategory:

**Lemma 9.2.15.** *Let  $E \in \mathcal{D}$  be enveloped by thin subcategories  $\mathcal{B}, \mathcal{C} \subseteq \mathcal{D}$ . Then  $E$  is  $W$ -semistable in  $\mathcal{B}$  if and only if it is  $W$ -semistable in  $\mathcal{C}$ .*

*Proof.* We can take a fixed  $\mathcal{B}$ , for instance  $\mathcal{B} = \mathcal{P}(\psi(E) - \epsilon, \psi(E) + \epsilon)$  in which  $E$  is  $W$ -semistable by the above Lemma. This reduces us to the case  $\mathcal{B} \subseteq \mathcal{C}$  or by symmetry to  $\mathcal{B} = \mathcal{P}(a, b)$  and  $\mathcal{C} = \mathcal{P}(a, c)$ .

Of course  $W$ -semistability in  $\mathcal{C}$  implies  $W$ -semistability in  $\mathcal{B}$ , so it remains to prove the converse. Assume that  $E$  is unstable in  $\mathcal{C}$ :

$$0 \rightarrow A \rightarrow E \rightarrow B \rightarrow 0$$

where  $\psi(A) > \psi(E) > \psi(B)$ . Then  $\phi^+(A) \leq \phi^+(E)$ , which implies that  $A \in \mathcal{B}$  since  $E \in \mathcal{B}$ . By a longer argument,  $B \in \mathcal{B}$  also.  $\square$

Let

$$\mathcal{Q}(\psi) := \langle 0, E \text{ is } W\text{-semistable of phase } \psi \text{ in a thin enveloping subcategory } \mathcal{P}(a, b) \rangle$$

which defines a stability condition because:

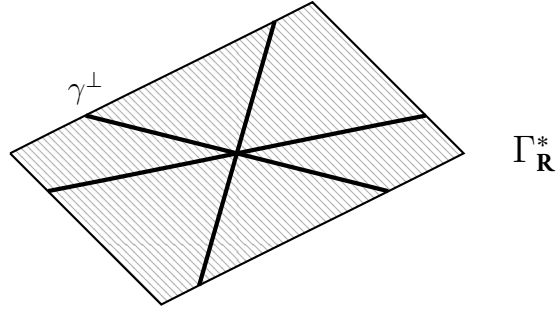
- If  $E \in \mathcal{Q}(\psi_1)$  and  $F \in \mathcal{Q}(\psi_2)$  with  $\psi_1 > \psi_2$  then  $\text{Hom}_{\mathcal{D}}(E, F) = 0$ . To show this,
- Existence of HN filtrations: if  $\mathcal{A} = \mathcal{P}(a, b) \subseteq \mathcal{D}$  is thin of finite length, then every nonzero object in  $\mathcal{P}(a + 2\epsilon, b - 4\epsilon)$  has an HN filtration by  $W$ -semistable objects of  $\mathcal{A}$  enveloped by  $\mathcal{A}$ .

$\square$

## 10. Wall-crossing structures

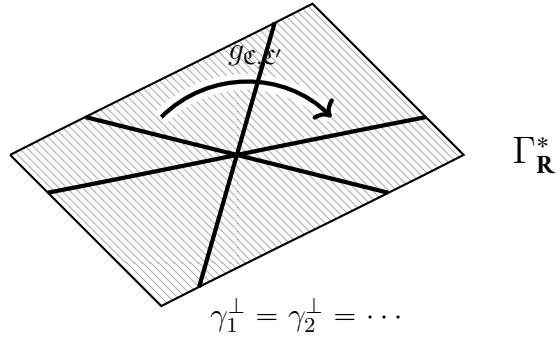
## 10.1. What is a wall-crossing structure?

10.1.1. Let  $\Gamma$  be a finitely generated free abelian group and  $\mathfrak{g} = \bigoplus_{\gamma \in \Gamma} \mathfrak{g}_\gamma$  a  $\Gamma$ -graded Lie algebra. The space of *walls* are the real hyperplanes of the form  $\gamma^\perp$ .



We will later drop this assumption, but for now assume that  $\mathfrak{g}$  is *nilpotent*, i.e.  $\{\gamma : \mathfrak{g}_\gamma \neq 0\}$  is finite and lives in a half-space in  $\Gamma_{\mathbf{R}}$ .

**Definition 10.1.2.** A **wall-crossing structure** is a collection



of elements  $g_{\mathfrak{c}, \mathfrak{c}'} \in G = \exp(\mathfrak{g})$  for any pair of chambers, satisfying the cycle condition, and such that if the chambers meet at a wall  $W$ ,

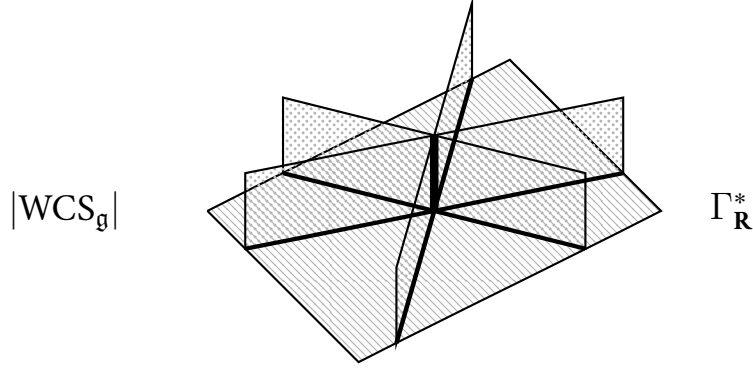
$$g_{\mathfrak{c}, \mathfrak{c}'} \in \exp \left( \bigoplus_{\gamma^\perp = W} \mathfrak{g}_\gamma \right).$$

Any point  $y \in \Gamma_{\mathbf{R}}^*$  induces a triangular decomposition

$$\mathfrak{g} = \mathfrak{g}_-^y \oplus \mathfrak{g}_0^y \oplus \mathfrak{g}_+^y, \quad G = G_-^y G_0^y G_+^y$$

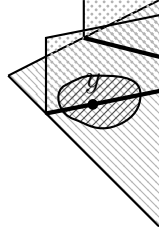
given by the decomposition  $\Gamma = \Gamma_-^y \oplus \gamma^\perp \oplus \Gamma_+^y$  by the sign of  $y$ .

10.1.3. We now define a *wall-crossing sheaf*  $\text{WCS}_{\mathfrak{g}}$  whose total space looks like



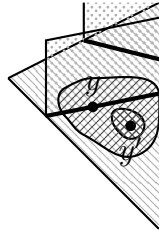
Its fibre at the origin is  $G$  and away from the walls is 1.

**Definition 10.1.4.**  $\text{WCF}_{\mathfrak{g}}$  is the sheaf of sets on  $\Gamma_{\mathbf{R}}^*$  whose value on a small contractible disk around  $y \in \Gamma_{\mathbf{R}}^*$  is



$$\text{WCF}_{\mathfrak{g},y} := G_-^y \backslash G / G_+^y \simeq G_0^y$$

and restriction to a small disk off the wall is given by projection



$$G_-^y \backslash G / G_+^y \rightarrow G_-^{y'} \backslash G / G_+^{y'}$$

or equivalently by the exponential of the projection map  $\mathfrak{g}_0^y \rightarrow \mathfrak{g}_0^{y'}$  killing off the relevant factors.

**Theorem 10.1.5.** [KS13] *There are bijections*

$$G \simeq \Gamma(\Gamma_{\mathbf{R}}^*, \text{WCS}_{\mathfrak{g}}) \simeq \{\text{wall-crossing structures}\}.$$

*Proof.* The first bijection, which takes a section to its fibre at the origin, is a bijection because  $\Gamma(\Gamma_{\mathbf{R}}^*, \text{WCS}_{\mathfrak{g}}) \simeq \Gamma(D_0, \text{WCS}_{\mathfrak{g}}) := G$  for  $D_0$  a disk at the origin, where the first isomorphism follows as  $D_0 \simeq \Gamma_{\mathbf{R}}^*$  are homeomorphic as stratified manifolds, and this homeomorphism pulls back  $\text{WCF}_{\mathfrak{g}}$  to itself.

For the second bijection, let  $s$  be a bijection and  $\mathfrak{C}, \mathfrak{C}'$  two chambers. Pick a generic point  $y \in \bar{\mathfrak{C}} \cap \bar{\mathfrak{C}}'$ , and set

$$g_{\mathfrak{C}, \mathfrak{C}'} = s(y) \in G_y^0.$$

Equivalently, this is the germ  $s|_{(\bar{\mathfrak{C}} \cap \bar{\mathfrak{C}}')^\circ}$  of the section on the generic locus  $(\bar{\mathfrak{C}} \cap \bar{\mathfrak{C}}')^\circ$  of the intersection

$$\Gamma((\bar{\mathfrak{C}} \cap \bar{\mathfrak{C}}')^\circ, \text{WCF}_{\mathfrak{g}}) = \exp \left( \bigoplus_{\gamma \text{ is different signs on } \mathfrak{C}, \mathfrak{C}'} \mathfrak{g}_\gamma \right).$$

Note that  $\bar{\mathfrak{C}} \cap \bar{\mathfrak{C}}'$  is a wall if  $\mathfrak{C}, \mathfrak{C}'$  meet at a wall. This satisfies the cocycle condition by considering the value of the section at a generic point of  $\bar{\mathfrak{C}} \cap \bar{\mathfrak{C}}' \cap \bar{\mathfrak{C}}''$ : here we use that by the nilpotency assumption there is a unique *ordered* factorisation

$$\mathfrak{g}_{\mathfrak{C}, \mathfrak{C}'} := \mathfrak{g}_y^0 = \mathfrak{g}_{y_1}^0 \oplus \cdots \oplus \mathfrak{g}_{y_n}^0$$

where  $y$  is a generic point in  $\bar{\mathfrak{C}} \cap \bar{\mathfrak{C}}'$  and  $\{y_1, \dots, y_n\}$  is the intersection of any straight path from  $\mathfrak{C} \rightarrow \mathfrak{C}'$  with the walls.

Conversely, a wall-crossing structure determines an element  $g_{\mathfrak{C}, -\mathfrak{C}} \in G$  (note that  $\bar{\mathfrak{C}} \cap (-\bar{\mathfrak{C}}) = 0$ ) whose various projections give the other  $g_{\mathfrak{C}', \mathfrak{C}''}$ .  $\square$

10.1.6. *Remark.* Let us now drop the nilpotency assumption. Nevertheless, for any proper half-plane  $C \subseteq \Gamma_{\mathbf{R}}$  (i.e., its closure not containing a line),

$$\mathfrak{g}_C = \prod_{\gamma \in C \setminus 0} \mathfrak{g}_\gamma, \quad G_C = \exp(\mathfrak{g}_C)$$

is a pronilpotent Lie algebra (e.g. ordered by some positive linear function  $\phi : C \rightarrow \mathbf{R}_{\geq 0}$ ) and pronilpotent group. Thus these are limits of nilpotent Lie algebras  $\mathfrak{g}_C^{(k)}$  depending on  $\phi$ , and therefore we can define  $\text{WCS}_{\mathfrak{g}}^{(k)}$  as above.

**Definition 10.1.7.** The **wall crossing sheaf** is the subsheaf of the constant sheaf of sets on  $\Gamma_{\mathbf{R}}^*$

$$\text{WCS}_{\mathfrak{g}} \subseteq \prod_{\gamma \in \Gamma \setminus 0} \mathfrak{g}_\gamma$$

whose fibre at  $y \in \Gamma_{\mathbf{R}}^*$  consists of elements  $\{g(y, \gamma) \in \mathfrak{g}_\gamma\}$  such that:

- (1)  $g(y, \gamma) = 0$  unless  $y \in \gamma^\perp$ ,
- (2) (*Local nilpotency*) every point  $y$  has a neighbourhood  $U_y$  on which the support

$$\text{Supp}g(U_y, -) = \{\gamma \in \Gamma : g(u, \gamma) \neq 0 \text{ for some } u \in U_y\} \subseteq C_{y, U_y}$$

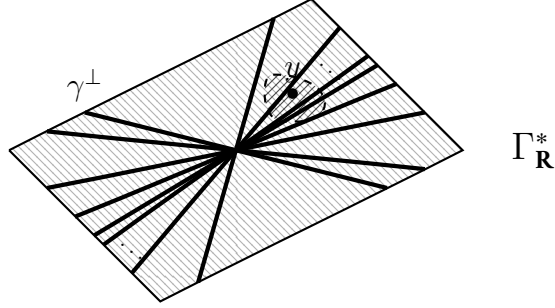
is contained in some proper  $C_{y, U_y} \subseteq \Gamma_{\mathbf{R}}$  is a proper cone,

- (3) (*A wall-crossing structure on each nilpotent piece*) for each function  $\phi \in \Gamma_{\mathbf{R}}^*$  defining a proper map  $\phi : \overline{C_{y, U_y}} \rightarrow \mathbf{R}_{\geq 0}$ ,

$$\exp \left( \sum_{\gamma} g(-, \gamma) \right)$$

defines a section of  $\text{WCS}_{\mathfrak{g}}^{(k)}$  on  $U_y$ .

In plain English, we consider the set of walls (which can now be infinite)



and ask that—even at accumulation points  $y$  of walls—the nearby walls are at least the images of some proper cone  $C_y \subseteq \Gamma$ . We then use the the set of walls it *profinite*—a limit of finite sets of walls—to define the wall crossing sheaf in general.

The actual value on a small open disk  $U_y$  will be a limit

$$\text{WCS}_{\mathfrak{g}}(U_y) = \lim_{k \rightarrow \infty} \text{WCS}_{\mathfrak{g}}^{(k)}(U_y) = \lim_{k \rightarrow \infty} \text{WCS}_{\mathfrak{g}^{(k)}, y_1} \times \cdots \times \text{WCS}_{\mathfrak{g}^{(k)}, y_{n_k}}$$

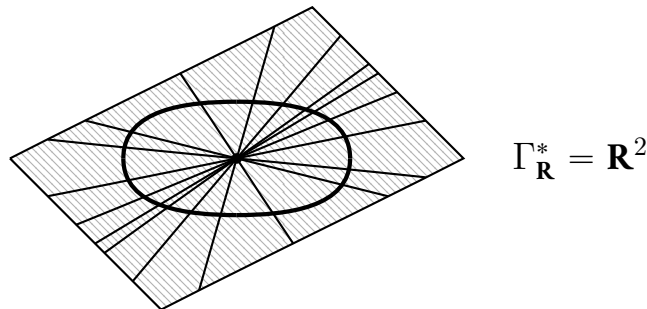
where  $y_i$  lie on the lowest strata of the intersection of  $U_y$  with the walls, and the multiplicands are  $G_{y_i}^{(k), -} \setminus G_{y_i}^{(k)} / G_{y_i}^{(k), +}$ .

For instance, if walls  $\gamma_1^\perp, \gamma_2^\perp, \dots$  are the walls intersecting  $U_y$  nontrivially, and no walls intersect with each other inside  $U_y$ , then

$$\text{WCS}_{\mathfrak{g}}(U_y) = \lim_{k \rightarrow \infty} \prod_{|\gamma_i| \leq k} \underbrace{\exp(\mathbf{C} \cdot \gamma_i)}_{G_{y_i}^0}$$

is the product over all  $\exp(\mathbf{C} \cdot \gamma_i)$ .

10.1.8. *Example.* Let  $\mathfrak{g}_\Gamma = \mathbf{C}[\Gamma]$  be the trivial infinite dimensional Lie algebra on the rank two lattice  $\Gamma = \mathbf{Z}\alpha \oplus \mathbf{Z}\beta$  (contrast this with e.g.  $\mathfrak{g} = \mathfrak{sl}_3$ ). Then we have a wall  $\gamma^\perp \subseteq \Gamma_{\mathbf{R}}^* \simeq \mathbf{R}^2$  for every root  $\gamma = n\alpha + m\beta$ , it is the real line with slope  $n/m$ . We thus identify the walls with  $\mathbf{P}^1(\mathbf{Q})$ :



We have a filtration on the walls, where walls of level  $\leq k$  are those with  $n^2 + m^2 \leq k$ . Every point  $y$  is an accumulation point of infinity many walls, and the walls  $\gamma^\perp$  intersecting with any small disk  $U_y$  avoiding the origin is obviously have  $\gamma$  contained in some proper cone.

10.1.9. *Relation to Bridgeland stability conditions.* We have a map

$$f : S^1 \times \mathrm{Stab}(\mathcal{C}) \rightarrow \Gamma_{\mathbf{R}}^*, \quad (\theta, (Z, \mathcal{P})) \mapsto \mathrm{Im}(e^{-i\theta} Z(-)).$$

A section of  $f^* \mathrm{WCS}_{\mathfrak{g}}$  is called a *global wall-crossing structure*. The induced real codimension one walls in  $\mathrm{Stab}(\mathcal{C})$  correspond to when  $Z(\gamma)$  has argument  $\theta$  for some  $\gamma \in K_0(\mathcal{C})$ .

## 11. DT invariants, the integration map and the quantum torus

So far, for an abelian category  $\mathcal{A}$  we have the motivic Hall algebra  $H(\mathcal{A})$ , the wall-crossing sheaf  $\text{WCS}_{\mathfrak{g}}$  for the torus Lie algebra

$$\mathfrak{g} = \bigoplus_{\gamma \in \Gamma} \mathbf{C} \cdot e_{\gamma}, \quad [e_{\gamma}, e_{\gamma'}] = (-1)^{\chi(\gamma, \gamma')} \chi(\gamma, \gamma') e_{\gamma + \gamma'},$$

graded over  $\Gamma = K_0^{\text{num}}(\mathcal{A})$  the numerical Grothendieck group.

Today we will put this together, defining a wall crossing structure (i.e. a section of  $\text{WCS}_{\mathfrak{g}}$ ) coming from  $H(\mathcal{A})$ . If  $\mathfrak{C}, \mathfrak{C}'$  meet at a single wall  $\gamma^{\perp}$  with  $\gamma$  not a multiple of any other element, these will take form

$$g(\gamma) = \Omega(\gamma) \cdot e_{\gamma} \in \mathfrak{g}_{\gamma}$$

for a number  $\Omega(\gamma) \in \mathbf{C}$  which “counts semistable objects of class  $\gamma$ ” called a *motivic DT invariant*. For more general  $\gamma$ ,  $g(\gamma) = \left( \sum_{k|\gamma} \frac{\Omega(\gamma/k)}{k^2} \right) \cdot e_{\gamma}$ .

We say what the DT invariant is. We will define some kind of generating series

$$A_V = \sum_{\log Z(\gamma) \in V} [\mathcal{M}_{\gamma}^{\text{ss}}]$$

which depends on the choice of a stability condition  $\tau = (Z, \mathcal{P})$  and a sector  $V$  (precise definition given later). This takes values in some pronilpotent Lie algebra  $G_V = \exp(\mathfrak{g}_V)$ , and has leading term 1 so takes the form

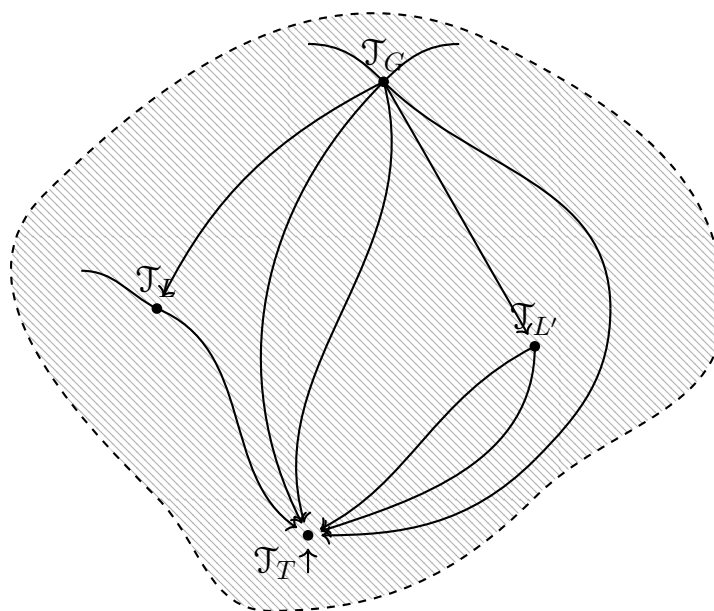
$$A_V = \exp(a_V)$$

where  $a_V \in \mathfrak{g}_V \subseteq \mathfrak{g}$ . A priori it is a sum  $\sum_{\gamma \in V} a_{\gamma} \cdot e_{\gamma}$ , and we Mobius invert to write them in terms of unique *DT invariants*  $\Omega(\gamma)$ :

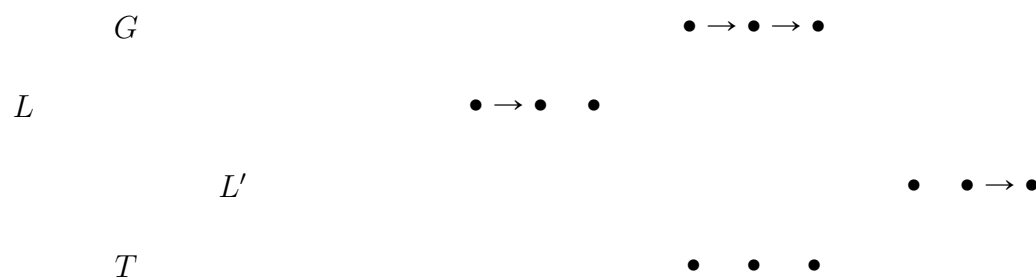
$$A_V = \exp \left( \sum_{\gamma} \left( \sum_{k|\gamma} \frac{\Omega(\gamma/k)}{k^2} \right) \cdot e_{\gamma} \right).$$

### 11.1. Why should this exist?

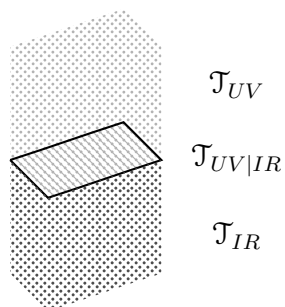
11.1.1. All invariants, quantum groups, etc. we have studied so far are meant to be able to be read off from certain  $4d$  QFTs. There is meant to be a flow on the space of QFTs,



The fixed points and maps between them should be thought of as analogous to either parabolic induction of reductive groups, or a flow in the space of Calabi–Yau threefolds or quivers with potential.



We call QFTs higher upstream “ultraviolet” and lower down “infrared”, and we have maps of QFTs:



## 11.2. Quantum tori.

11.2.1. Recall the quantum affine plane  $\mathbf{A}_q^2$ , generated by two elements  $x, y$  satisfying the relation

$$xy = q^2 yx.$$

If we write  $x^\alpha = x$ ,  $x^\beta = y$  and

$$x^{\alpha+\beta} = q^{-1}xy$$

then the relation becomes

$$x^\alpha \cdot x^\beta = qx^{\alpha+\beta}, \quad x^\beta \cdot x^\alpha = q^{-1}x^{\alpha+\beta}$$

where we package the exponential prefactor in terms of  $\langle \alpha, \beta \rangle = 1$  and  $\langle \beta, \alpha \rangle = -1$ , a skew-symmetric bilinear form on  $\mathbf{Z}\alpha \oplus \mathbf{Z}\beta$ , and we can view above ring of functions as generated by  $x^{n\alpha+m\beta}$  subject to the above relations. Writing  $q = e^{\hbar/2}$ , the leading order term in

$$[x^\alpha, x^\beta] = (q - q^{-1})x^{\alpha+\beta} = \hbar x^{\alpha+\beta} + \mathcal{O}(\hbar^2)$$

defines a Poisson form with  $\{x^\alpha, x^\beta\} = x^{\alpha+\beta}$ .

**Definition 11.2.2.** The **quantum torus** and **Poisson torus** for lattice  $\Gamma$  with antisymmetric form  $\chi$  is the algebra

$$\mathcal{R}_\Gamma = \bigoplus_{\alpha \in \Gamma} \mathbf{Z}[\mathbf{L}^{\pm 1/2}] \cdot x^\alpha, \quad \mathfrak{g}_\Gamma = \bigoplus_{\alpha \in \Gamma} \mathbf{Z} \cdot e_\alpha$$

the first having associative product

$$x^\alpha \cdot x^\beta = \mathbf{L}^{\frac{1}{2}\chi(\alpha, \beta)} x^{\alpha+\beta}.$$

and the second having supercommutative product and Poisson form

$$e_\alpha e_\beta = (-1)^{\chi(\alpha, \beta)} e_{\alpha+\beta}, \quad \{e_\alpha, e_\beta\} = (-1)^{\chi(\alpha, \beta)} \chi(\alpha, \beta) e_{\alpha+\beta}.$$

In particular,  $\mathfrak{g}_\Gamma$  is also a Lie algebra.

11.2.3. *Remark.* In our example,  $\chi$  is the antisymmetrised Euler form on  $\Gamma$  the numerical Grothendieck group.

11.2.4. *Categorification.* Let  $T$  be a torus with character lattice  $\Gamma$ , so that we have

$$K_0(\text{Rep} T) = \bigoplus_{\alpha \in \Gamma} \mathbf{Z} \cdot [\mathbf{C}_\alpha].$$

If we endow  $\text{Rep} T$  with the ordinary monoidal structure  $\mathbf{C}_\alpha \otimes \mathbf{C}_\beta = \mathbf{C}_{\alpha+\beta}$ , this makes the Grothendieck group into a commutative algebra corresponding to  $\chi = 0$ .

To get the quantum torus, we consider  $T \times \mathbf{C}_\hbar^\times$  and

$$K_0(\text{Rep} T \times \mathbf{C}_\hbar^\times) = \bigoplus_{\alpha \in \Gamma} \mathbf{Z}[\mathbf{L}^\pm] \cdot [\mathbf{C}_\alpha].$$

where  $\mathbf{L}$  is a weight one representation of  $\mathbf{C}_\hbar^\times$ , and take the monoidal structure

$$\mathbf{C}_\alpha \otimes \mathbf{C}_\beta = \mathbf{L}^{\chi(\alpha, \beta)} \mathbf{C}_{\alpha+\beta}.$$

This is a slightly unnatural way to get the quantum torus: in examples to do with cohomological Hall algebras, the antisymmetrisation has to do with shifts by cohomological degree, and  $\{-, -\}$  is induced by the classical Yang–Baxter matrix.

### 11.3. DT invariants.

11.3.1. Let  $\mathcal{C}$  be a triangulated category with a homomorphism

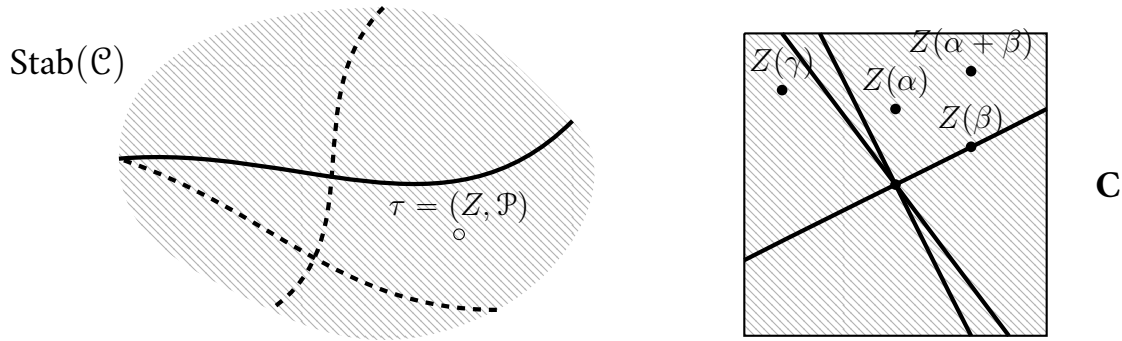
$$\text{ch} : (\mathbf{K}_0(\mathcal{C}), \chi) \rightarrow (\Gamma, \langle -, - \rangle)$$

where  $\Gamma$  is a free finite rank lattice. From now on we will consider Bridgeland stability conditions  $\tau = (Z, \mathcal{P}) \in \text{Stab}(\mathcal{C})$  with the *support property*: that there is a norm on  $\Gamma_{\mathbf{R}}$  such that

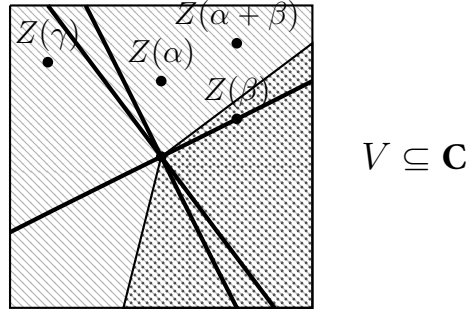
$$\|\text{ch}(E)\| \leq C|Z(E)|$$

for all  $E \in \mathcal{C}^{ss}$ , and  $C > 0$  a universal constant.

11.3.2. Fix a stability condition  $\tau$ , and the image of  $Z : \mathbf{K}_0(\mathcal{C}) \rightarrow \mathbf{C}$ :



(The lines will be the positive rays along  $Z(\gamma)$  with nonzero DT invariant  $\Omega(\gamma) \neq 0$ .)



For any *strict* sector  $V$  (i.e. of angle strictly less than 180), we let

$$\mathcal{C}_V = \langle 0, E : E \in \mathcal{C}^{ss}, \phi_E \in V \rangle = \mathcal{P}(\log(V))$$

be the subcategory generated by semistable objects with  $\log(Z(E)) \in V$ . We likewise define the associative algebra  $\mathcal{R}_V \subseteq \mathcal{R}_{\Gamma}$  and Poisson algebra  $\mathfrak{g}_V \subseteq \mathfrak{g}_{\Gamma}$ , and the moduli stack  $\mathcal{M}_V \subseteq \mathcal{M}_{\mathcal{C}}$  given by the union of components  $\gamma \in \mathbf{K}_0^{num}(\mathcal{C}) = \pi_0(\mathcal{M}_{\mathcal{C}})$  containing an element in  $\mathcal{C}_V$ .

In what follows, we will assume that  $\mathcal{C}$  has countably many objects so that  $[\mathcal{M}_{\mathcal{C}}]$  is a sum over

**Definition 11.3.3.** The *Hall generating series* is

$$A_{Z,V} = [\mathcal{M}_{\mathcal{C}_V}] = \sum_{c \in \text{Ob}(\mathcal{C}_V)} [\text{pt}/\text{Aut}(c) \xrightarrow{c} \mathcal{M}_{\mathcal{C}}] \in H(\mathcal{C}_V)^{\wedge}$$

where we take degree completion in  $\Gamma$ .

Loosely speaking, it is a formal sum of the stacky points in  $\mathcal{M}_{\mathcal{C}}$  corresponding to objects in  $\mathcal{C}_V$ . From the following, it decomposes into an ordered product

**Proposition 11.3.4.** *We have that*

$$A_V = \prod_{\phi \in V}^{\rightarrow} A_{\phi}.$$

*Proof sketch.* As every object in  $\mathcal{C}_{\phi}^{\text{ss}} = \mathcal{P}(\phi)$  is semistable of slope  $\phi$ ,  $A_{\phi} = 1 + \sum_{\arg(Z(\gamma))=\phi} [\mathcal{M}_{\gamma}^{\text{ss}}]$ .

By the Harder–Narasimhan filtration, every object uniquely factors into an  $n$ -step extension of semistable objects with slope  $\phi_1 > \dots > \phi_n$  for some  $n$ . Letting  $\mathcal{M}_V(\phi_1, \dots, \phi_n)$  be the substack of objects with this property, it fits into sequence

$$\begin{array}{ccccc} & & \mathcal{M}_V(\phi_1, \dots, \phi_n) & \longrightarrow & n\text{-SES} \\ & \swarrow & & \searrow & \\ \mathcal{M}_{\phi_1}^{\text{ss}} \times \dots \times \mathcal{M}_{\phi_n}^{\text{ss}} & \longrightarrow & \mathcal{M} \times \dots \times \mathcal{M} & & \mathcal{M}_V(\phi_1, \dots, \phi_n) \end{array}$$

where the last step is a pullback. In other words,

$$[\mathcal{M}_V(\phi_1, \dots, \phi_n)] = [\mathcal{M}_{\phi_1}^{\text{ss}}] \times \dots \times [\mathcal{M}_{\phi_n}^{\text{ss}}]$$

in  $H(\mathcal{C}_V)$ , where on the right we take the Hall product; note that the product is 0 unless  $\phi_1 > \dots > \phi_n$ .

Moreover, as  $[\mathcal{M}_V]$  is a sum over  $[\mathcal{M}_V(\phi_1, \dots, \phi_n)]$ , we have that

$$A_V = [\mathcal{M}_V] = \sum_{n \geq 0} \sum_{\phi_1 > \dots > \phi_n} [\mathcal{M}_{\phi_1}^{\text{ss}}] \times \dots \times [\mathcal{M}_{\phi_n}^{\text{ss}}] = \prod_{\phi \in V}^{\rightarrow} (1 + [\mathcal{M}_{\phi}^{\text{ss}}]) = \prod_{\phi \in V}^{\rightarrow} A_{\phi}.$$

□

**Corollary 11.3.5.** *If  $V_1, V_2$  are disjoint sectors with  $V_1 > V_2$ , then*

$$A_{V_1 \sqcup V_2} = A_{V_1} \cdot A_{V_2}.$$

This thus lets us define the **(Hall) DT invariants**  $\Omega(\gamma) \in \mathbf{C}$  for each  $\gamma \in \Gamma$  by

$$A_V = \exp \left( \sum_{Z(\gamma) \in V} \left( \sum_{k|\gamma} \frac{\Omega(\gamma/k)}{k^2} \right) \cdot e_{\gamma} \right).$$

This formula is independent of  $V$ , indeed it is enough that for each phase  $\phi$  we have

$$A_{\phi} = \exp \left( \sum_{Z(\gamma) \in \mathbf{R}_{\geq 0} e^{i\pi\phi}} \left( \sum_{k|\gamma} \frac{\Omega(\gamma/k)}{k^2} \right) \cdot e_{\gamma} \right).$$

What about if  $\gamma, \gamma'$  both have phase  $\phi$ ?

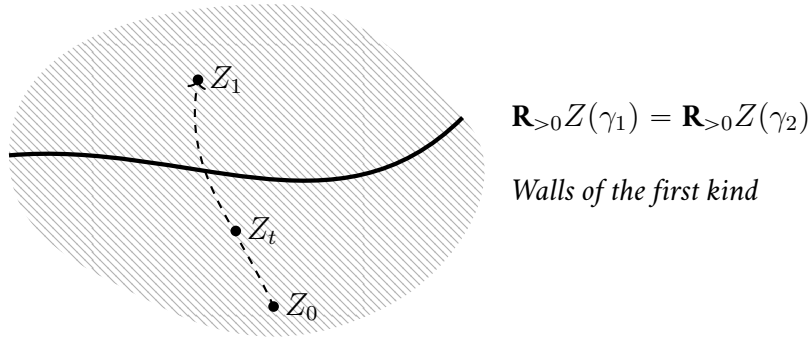
**Conjecture 11.3.6.** *In the quiver with potential case, the DT invariants are integers.*

**11.4. Wall-crossing.** We now vary the stability condition  $Z$ .

**Theorem 11.4.1.** *The above induces symmetric stability data  $a^{\text{mot}}(\gamma), a^{\text{Hall}}(\gamma)$  [KS08, p.24] on the graded Lie algebra  $\mathfrak{g} = \mathfrak{g}_\Gamma = \bigoplus_{\gamma \in \Gamma} R \cdot e_\gamma \simeq H(\mathbb{C}), \mathcal{R}$ , by taking logarithms*

$$A_\ell = \exp \left( \sum_{Z(\gamma) \in \ell, Q(\gamma) \geq 0, \gamma \neq 0} a(\gamma) \right)$$

and DT invariants  $\Omega^{\text{mot}}(\gamma), \Omega^{\text{Hall}}(\gamma)$  by Mobius inversion  $a(\gamma) = -\sum_{n \geq 1} \frac{\Omega(\gamma/n)}{n^2} e_\gamma$ . This satisfies the **wall-crossing formula** as  $Z$  varies over a wall of the first kind:



i.e. for any ray  $\ell_t = \mathbf{R}_{>0}Z_t(\mu)$  for some  $\mu$ ,

$$\prod_{\gamma \in \Gamma^{\text{prim}}, Z_t(\gamma) \in \ell_t} \exp \left( \sum_{n \geq 1} a_t^-(n\gamma) \right) = \prod_{\gamma \in \Gamma^{\text{prim}}, Z_t(\gamma) \in \ell_t} \exp \left( \sum_{n \geq 1} a_t^+(n\gamma) \right)$$

where  $a_t^\pm$  is the limit from the left and right, where the order of the product depends on the argument of  $Z_{t \pm \epsilon}$ .

## 11.5. Integration map.

**11.5.1.** The integration map is Poisson algebra homomorphism from a degeneration of the (motivic) Hall algebra.

The numerical Grothendieck group

$$K_0(\mathcal{C})/K_0(\mathcal{C})^\perp$$

is the quotient by the submonoid generated by objects with Euler pairing  $\chi(-, c) = 0$ . We let  $\Gamma \subseteq K_0(\mathcal{C})/K_0(\mathcal{C})^\perp$  be generated by elements in an abelian heart of  $\mathcal{C}$ .

**Theorem 11.5.2.** (For  $\mathcal{C} = \text{Coh}(X)$ ) Let  $H_{\text{reg}}(\mathcal{C}) \subseteq H(\mathcal{C})$  denote the subspace generated by  $[\mathcal{Y}]$  for  $\mathcal{Y}$  a variety. This is a subalgebra, and the quotient

$$H_{\text{sc}}(\mathcal{C}) = H_{\text{reg}}(\mathcal{C})/(\mathbf{L} - 1)H_{\text{reg}}(\mathcal{C})$$

is a commutative algebra.

**Theorem 11.5.3.** *For a locally constructible function  $\lambda : \mathcal{M}_{\mathcal{C}} \rightarrow \mathbf{Z}$  satisfying certain conditions write which, there is a unique Poisson algebra homomorphism*

$$\int : H_{\text{sc}}(\mathcal{C}) \rightarrow \mathbf{Z}[\Gamma]$$

$$\mathcal{Y} \mapsto \chi(\mathcal{Y}, f^*(\lambda)) \cdot x^\alpha$$

for any map  $\mathcal{Y} \rightarrow \mathcal{M}_{\mathcal{C}}$  that lands in a single connected component labelled by some  $\alpha \in \Gamma$ , and where  $\chi(\mathcal{Y}, f^*(\lambda)) = \sum_{n \in \mathbf{Z}} \chi(\mathcal{Y}, (f\lambda)^{-1}(n))$ .

**Definition 11.5.4.** [KS08, p.95,101] The **motivic DT invariant** is

$$A_V^{\text{mot}} = \sum_{c \in \mathcal{C}_{\log, V}} \frac{w(c)}{[\text{Aut}(c)]} \hat{e}_{\text{cl}(c)} \in \mathcal{R}_V$$

where  $\mathcal{C}_{\log, V}$  is the category of extensions of semistable objects with central charge in  $V$ , and  $w(c) \in A = \mathcal{M}^\mu(\mathcal{C})$  is the motivic weight [KS08, p.93].

**Definition 11.5.5.** [KS08, p.88] The **Hall DT invariant** is

$$A_V^{\text{Hall}} = \sum_{c \in \mathcal{C}_V} \mathbf{1}_c \in \hat{H}(\mathcal{C})$$

where the right side is the set of measures on  $\mathcal{C}$ .

**Theorem 11.5.6.** [KS08, p.96] *There is an “integration map” algebra homomorphism  $\Phi : \hat{H}(\mathcal{C}) \rightarrow \mathcal{R}$  sending  $A_V^{\text{Hall}}$  to  $A_V^{\text{mot}}$ .*

By [KS08, p.88,95] both satisfy  $A_{V_1 \sqcup V_2} = A_{V_1} A_{V_2}$  for disjoint unions of sectors.

## 11.6. Examples.

11.6.1. Let  $\mathcal{C}$  be the CY3 category generated by a single element  $c$  with

$$\text{Ext}^\bullet(c, c) \simeq H^\bullet(S^3).$$

Let us assume that  $c$  is semistable and  $V$  is a sector in which  $Z(c)$  lives. We identify

$$K_0(\mathcal{C}) = \mathbf{Z} \cdot [c].$$

Therefore,  $\mathcal{C}_V = \text{Vect} \cdot c$ . Note that  $c \otimes \mathbf{C}^n$  has group of automorphisms  $\text{GL}(n)$ , and therefore

$$A_V = \sum_{n \geq 0} [\mathcal{M}_{n[c]} \rightarrow \mathcal{M}_{\mathcal{C}}] = \sum_{n \geq 0} \overline{[\text{GL}(n)]} e_{n \cdot [c]}.$$

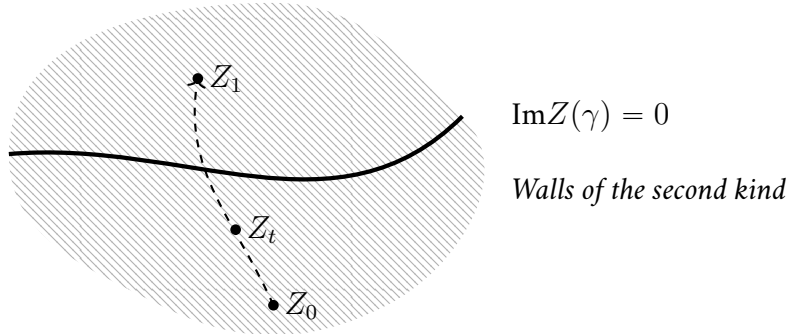
Note that this is identified with the quantum dilogarithm  $\sum_{n \geq 0} \frac{\mathbf{L}^{n^2/2}}{(\mathbf{L}^{n/2}-1) \dots (\mathbf{L}^{n/2}-\mathbf{L}^{(n-1)/2})} e_{nc}$ .

**11.7. To be integrated into the main body.** Note that  $\mathfrak{g}_\Gamma$  with bracket  $[e_\gamma, e_{\gamma'}] = (-1)^{\langle \gamma, \gamma' \rangle} \langle \gamma, \gamma' \rangle e_{\gamma+\gamma'}$  acts as symplectomorphisms of the torus  $\mathbf{T}_\Gamma = \text{Spec} R\langle e_\gamma \rangle / (e_\gamma e_{\gamma'} = (-1)^{\langle \gamma, \gamma' \rangle} e_{\gamma+\gamma'})$ . *Note that  $\langle, \rangle$  needs to be skew-symmetric here!* We define  $G_V = \exp(\mathfrak{g}_V)$ .

**Theorem 11.7.1.** [KS08, §8.3] For a quiver with generic potential  $(Q, W)$ , we consider a cluster collection [KS08, p.129] of objects  $\{E_i\}$  (e.g.  $\mathbf{C}_{\delta_i}$ ) and a sector  $V$  containing all  $Z(E_i)$ . Then  $\mathbf{E}_Q = A_V^{\text{Hall}}$  is independent of  $W$ , and for acyclic  $Q$  is an ordered product of dilogarithms  $\mathbf{E}(i)$ . Conjugation by  $\mathbf{E}_Q$  gives an automorphism of (a double of)  $\mathbf{T}_Q$ , inducing the structure of a cluster variety.

**Conjecture 11.7.2.** (Proven in many cases by now)  $\Omega^{\text{Hall}}(\gamma), \Omega^{\text{mot}}(\gamma)$  are integers.

**Theorem 11.7.3.** [KS08, p.138] For a quiver with generic potential  $(Q, W)$ , as we move over walls of the second kind (so that  $D^b(\text{Rep} J(Q, W))^\heartsuit$  undergoes a mutation of the quiver at vertex  $i$ ),



the  $A_V^{\text{Hall}}$  are related by conjugation by  $\mathbf{E}_Q(\delta_i)$ .

## 11.8. Example: the $A_2$ quiver and the pentagon identity.

11.8.1. Let us consider the Hall algebra for the  $A_2$  quiver.

use the "proposal for parabolic induction of cohas" document here somewhere

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